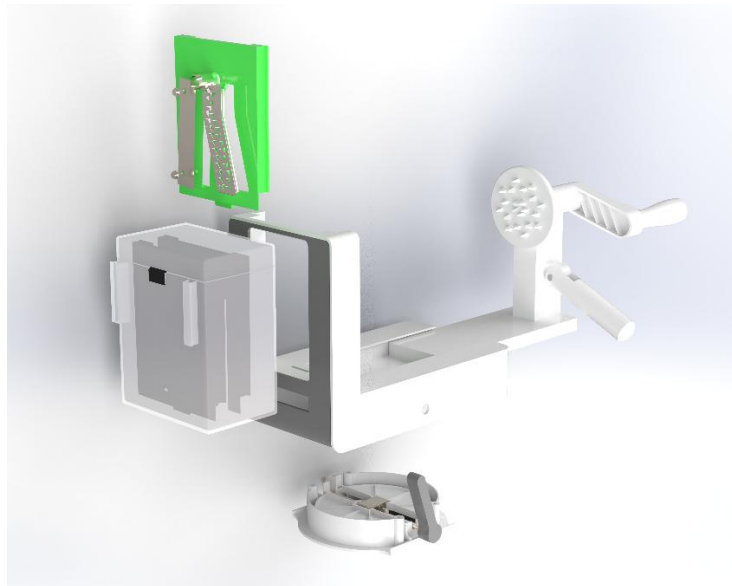


Tabletop Spiralizer



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Executive Summary

This report details the reverse engineering of the OXO Good Grips Tabletop Spiralizer, aimed at gaining an in-depth understanding of the design and manufacturing decisions involved in the product through investigation and analysis, concluding in design improvement proposal. The tabletop spiralizer earns its name from an ability to suction to a smooth tabletop. Vegetables—possessing a noodle-like quality—can be cut into spaghetti, fettuccine, or ribbon ‘noodles’ due to three interchangeable blade options.

The reverse engineering process followed a methodical approach: product decomposition, component modeling, assembly analysis, and redesign suggestions. During disassembly, we paid particular attention to how individual parts contributed to overall functionality. Subsequently, 3D models for all parts were produced and corresponding 2D technical drawings.

Our analysis revealed four distinct subsystems: the main body, suction subsystem, slider subsystem, and cutting subsystem. To better understand the system's operation, we developed black-box and glass-box diagrams, along with Ishikawa diagrams to analyze both overall system functions and individual component relationships. Additionally, patent research provided insight into key design decisions. Finally, we evaluated the entire spiralizer based on design for assembly (DFA) principles. The original design comprises 44 parts with a practical minimum of 25 parts, yielding a part efficiency of 56.8%—below the 60% target.

The assembly analysis illuminated several opportunities for design improvements. Our DFA identified an excessive use of fasteners and the insertion of the suction lever bar as the most difficult assembly process. The findings informed three proposed redesigns—converting closed holes for the suction lever bar into an open slot, integrating intermediate screw attached snap fit into the main body, and eliminating the awkward and underutilized push arm. With the production of 50,000 units annually these improvements would significantly cut costs through assembly time and part reduction.

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Gantt Chart

A Gantt chart was used for our team to track our schedule and key milestones for the reverse engineering project.

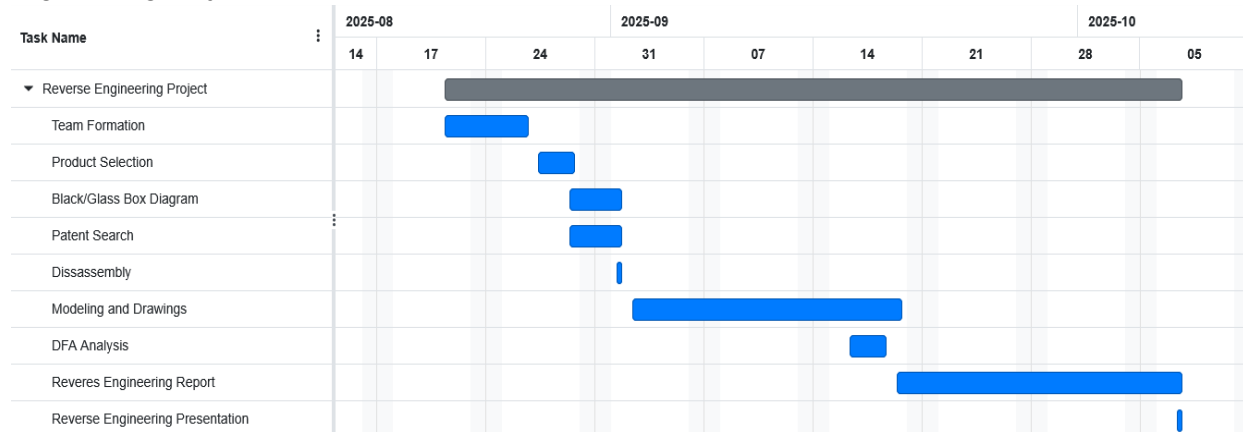


Figure 1: RE Project Gantt Chart

User Experience

When users first handle the spiralizer, they test out all of the functions and the specific interactions used include operating the suction pad, pushing the slider assembly and turning the spiralizer holder. The suction pad requires a smooth, clean surface to function effectively. The push handle is infrequently used during operation. Additional downward force on the cutting plane is needed to supplement the suction cup grip, especially on non-ideal surfaces, and maintain stable position of the spiralizer. The device cores vegetables during processing, prompting manufacturer recommendation of food with a diameter greater than 1.5 inches. These interactions will be taken into consideration during the redesign phase.

Black Box and Glass Box Diagrams

Prior to disassembly, the spiralizer was considered and reduced to its core inputs and outputs. The resulting analysis is present in the black box figure below.



Figure 2: Black Box Diagram

Immediately following the construction of the black box model, a glass box model was constructed to illuminate internal and external components and highlight interactions between inputs, systems, and outputs. This exercise aided in separating and defining subsystems composing the spiralizer.

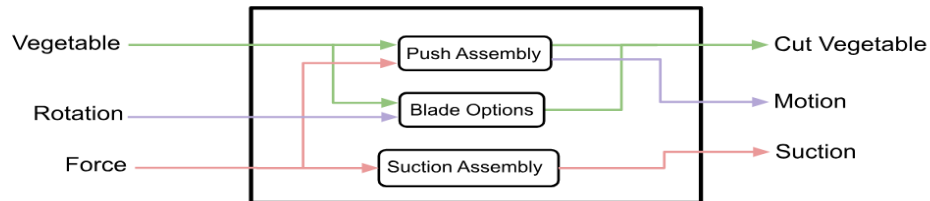


Figure 3: Glass Box Diagram

Ishikawa Diagram

The Ishikawa (Fishbone) Diagram organizes the spiralizer into subassemblies, diagonal lines, and individual components they contain; everything placed along their respective lines. The horizontal line demonstrates that all subassemblies interface with, that therefore come together through, the body.

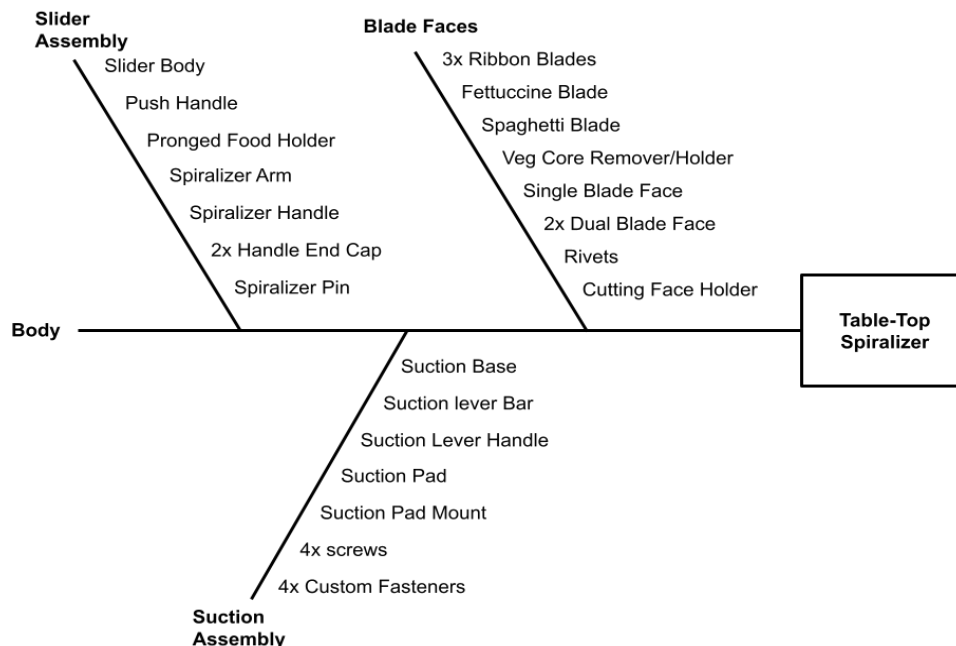


Figure 4: Ishikawa Diagram

Patent Search

A patent search was conducted to better grasp how individual features were innovated individually. This helped inform our understanding of design decisions, as well as inform our design modifications.

Invented by Alfred Börner and Jürgen Börner, patent EP0196550A1 discloses a holder for handling food, specifically fruits and vegetables, for cutting as well as protecting the user's hand. The device is circular to accommodate a majority of raw food shapes and spike to hold said food in place while cutting, as seen in [figure 5](#). This specific patent is part of a system of patents comprising what we would identify as a mandolin.

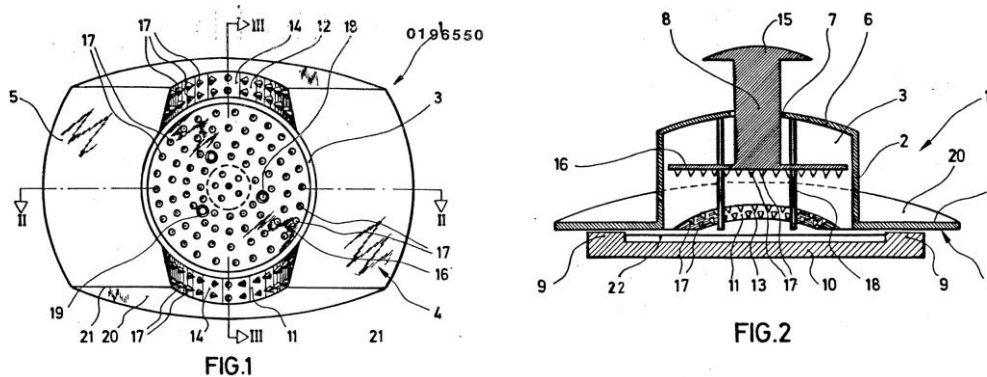


Figure 5: EP0196550A1 - Holder for handling fruit to be prepared in a kitchen utensil - August 10, 1988

Inventor Jocelyn Dube designed this cutting plate specifically for potatoes to be cut into helical strips. The two-piece nature stems from the optionality to cut ribbons or multiple slimmer strips through the addition of a 'slitter blade', seen at the bottom of [figure 6](#).

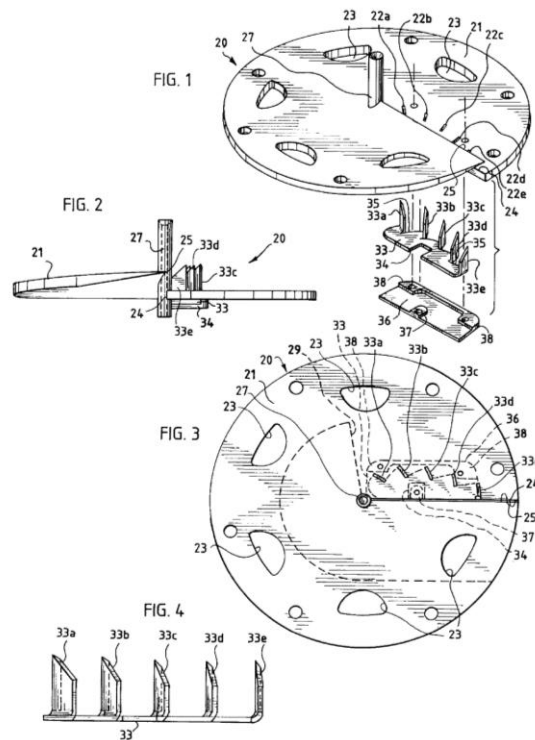


Figure 6: US5992287A - 2-piece cutter blade for producing helical vegetable strips - November 30, 1999

Inventor Cedomir Repac innovated on the prior patents to create a handheld fruit and vegetable spiralizer. With interchangeable cones, seen in [figure 7](#), garlands can be made in various widths. Operation involves the user placing the spiked food holder on the fruit/veg of choice and rotating while pushing the veggie through the selected cutting cone.

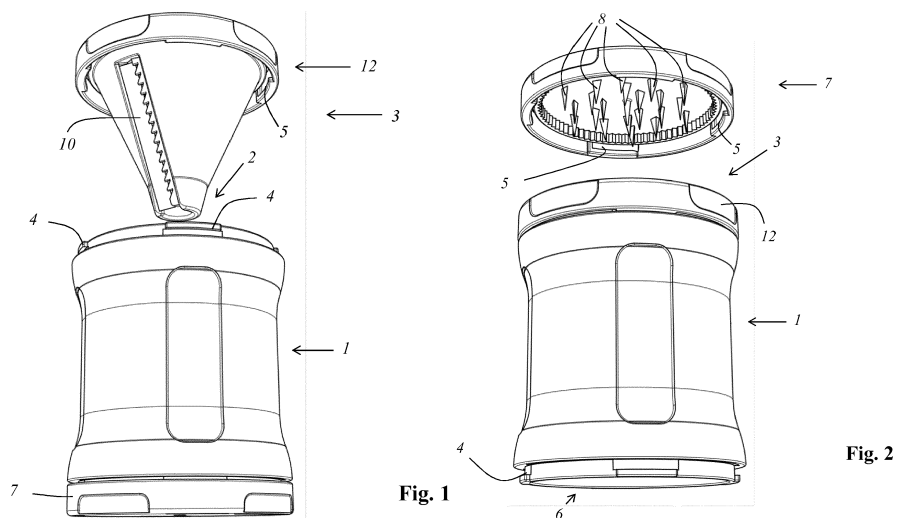


Figure 7: EP3307500B1 - Cutting system for cutting fruit or vegetables in spiral-shaped garlands - August 5, 2020

Product Decomposition

To fully understand the assembly process and part interfaces of each component the spiralizer was diligently deconstructed. As seen in the Ishikawa diagram (figure 4), there are 3 sub-assemblies unified through the main body assembly, comprised of 24 individual parts. All parts and steps of decomposition are cataloged in the product decomposition document in [Appendix A](#).

Material Analysis

After taking the Spiralizer apart and investigating the different components we gained insight into the various kinds of materials being used through the assembly. We landed on 2 primary materials, ABS Plastic and 304 Stainless Steel, which provides the user with a sturdy frame and sharp blades when cutting the vegetables into noodles. There are also 2 secondary materials that we have identified, Silicone Rubber and Polypropylene, which each show up once in the assembly, and provide the user with a more comfortable and aesthetically pleasing design, without significant functionality.

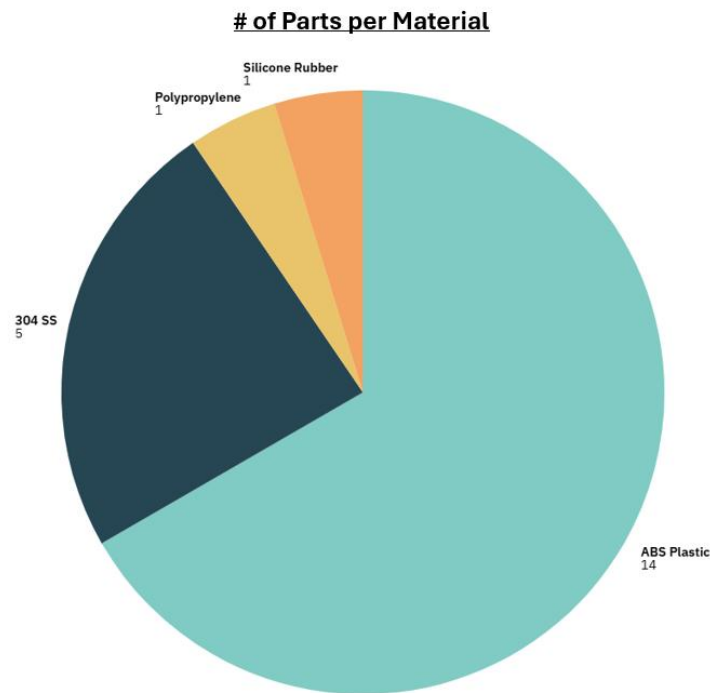


Figure 8: # of Parts per Material

ABS Plastic

Most of our parts are made with ABS plastic, a common material for injection molding. When deciding on a proper material, we need to first determine the wide variety of functionality across the parts.

Design Constraints:

- Needs to withstand cyclical loading (high strength)
- Must transmit torque without excessive twist or deflection
- Somewhat light, not adding additional work (low density)
- Manufacturing method suitable for high-volume production
- Easy to clean, food safe

Given our design constraints, the Ashby chart selected for this investigation is the Young's Modulus-Density chart, with the design guideline of $E^{1/3}/\rho$ to account for the combined loading for torsion and bending that specific piece will be subject to during its lifetime. Materials with a Young's modulus under 1GPa are far too soft and can be bent by hand, making them unfit for the hand-operated part.

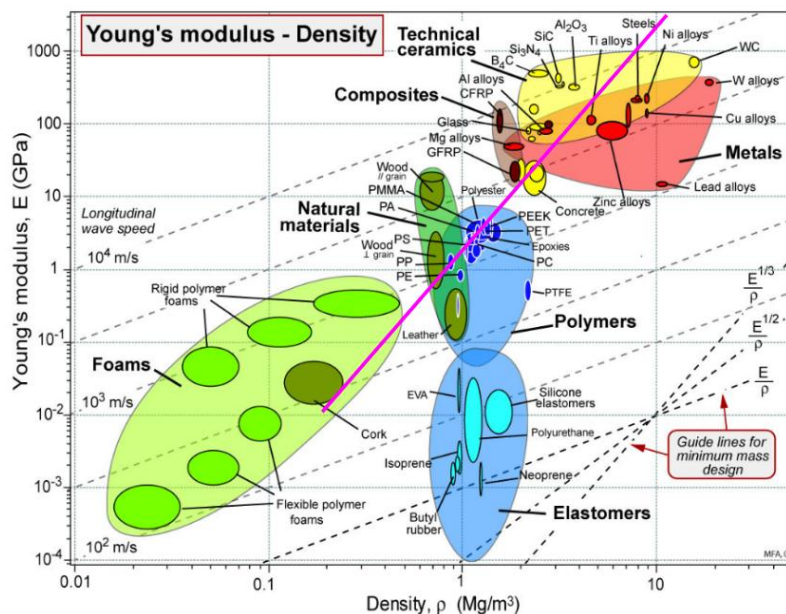


Figure 9: Density vs Strength Ashby Chart (ABS Analysis)

The placement of the selected guideline, in conjunction with a minimum E value, indicates that polymers, technical ceramics, and metals are materials worthy of further investigation.

Beginning with cost, technical ceramics can be eliminated. Cost and mass considerations eliminate most metals—mass driving up cost since the spiralizer arm is one of the larger components. Leaving materials capable of being manufactured, the next highest priority for evaluation is injection moldable materials. The remaining polymers, almost all except PTFE, PE, and PP, are up for consideration. ABS is the most cost-effective option and is capable of being injection molded to fulfill high throughput needs; all other plastics still up for consideration are more expensive.

After thorough consideration, ABS is the most appropriate material for this part.

304 Stainless Steel

All the vegetable cutting blades are made out of stainless steel, to provide a sharp edge that will not degrade over time.

Design Constraints:

- High Stiffness (can't deform while cutting vegetables)
- Low Density (relatively lightweight)
- Low Cost
- Safe for cutting food

Given the design constraints for the blade, we decided to reference a Young's Modulus and Strength Ashby chart to prioritize high-strength and high-stiffness materials. We used the design parameter of $(\sigma_f^{3/2})/E$, which is recommended for elastic constants such as knife edges.

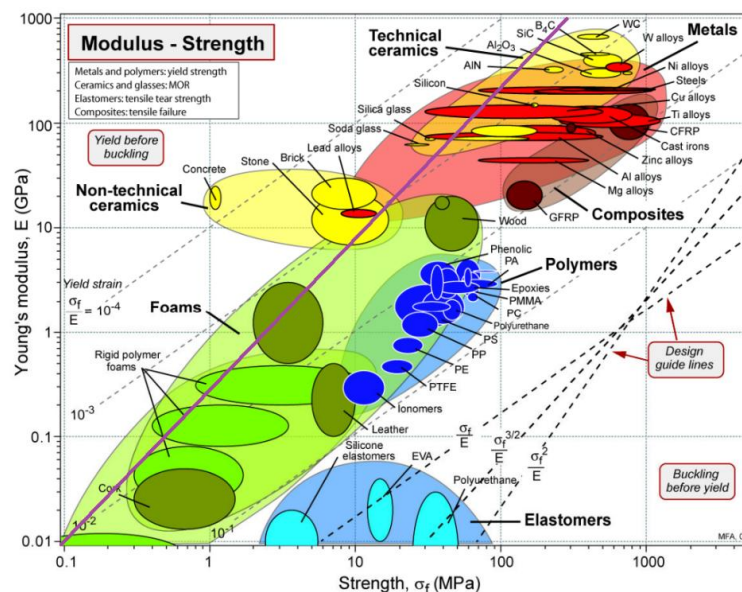


Figure 10: Strength vs Young's Modulus Ashby Chart (SS Analysis)

We land ourselves in the top right corner of the Ashby diagram, which correlates to metals. Titanium, copper, and steel are all viable materials for this application, so we turn to other design constraints to further decide. Of these options, steel is the most cost-effective material for making the blade due to its being a commonly available metal.

In addition to cost, steel is a commonly used material for other kitchen applications, especially in knife blades. Because of this, it is reasonable to assume steel to be a kitchen-safe material to encounter food.

Cost Analysis

To perform cost analysis on our assembly, we needed to set the stage with some assumptions that would be driving key aspects of the unit cost equation (eq. 1).

$$C_U = \frac{(mC_m)}{1-f} + \frac{C_w}{n'} + \frac{C_{tk}}{n} + \left(\frac{1}{n'}\right)\left(\frac{C_e}{Lt_{wo}}\right)(q) + \frac{C_{OH}}{n'} \quad (1)$$

C_m : Material Cost

C_w : Labor Cost

C_t : Tooling Cost

C_e : Capital Cost

C_{OH} : Factory Overhead Cost

C_m : Material Cost

t_{wo} : Capital Write Off Time

L : Load Fraction

f : Fraction Of Scrap

n : Tooling Life

n' : Production Rate

Assumptions

- Made in China → Impacts labor costs (C_w)
- Small sized plant (~25,000 sqft) → Impacts factory expenses and overhead costs (C_{OH})
- All materials are bought in bulk at current market price (C_m)
- For certain injection molded parts, multiple can be made at a time in a single mold
- 1 factory worker per machine (n')

By doing research and making reasonable assumptions to define the key parameters in the cost equation, we were able to produce estimates for each component in our assembly. A reference example of our cost analysis spreadsheet is included in [Appendix B](#). Tables 1-4 are a concise version of some of the notable inputs in our cost equation (mass, material, material costs) and the unit cost and break even cost for 50,000 parts output.

From our calculations we can see that our highest unit cost for any part is our suction pad, at \$3.74, due to the higher material costs of silicone rubber (Part #7). The suction pad increases friction between the spiralizer and the surface it sits on, so that it does not slide while being used to cut vegetables. It is a necessary feature for user experience which is why the higher cost is justified.

The highest break-even cost for 50,000 parts is the main body, white (Part #1). This is due to it having the highest mass out of all the parts and having several complex features embedded in it. Despite having the highest break-even cost, it is still within 20-30¢ of most of the other components in the assembly.

Part #	Description	Material	Material Cost (\$/lb.)	Part Mass (lbs.)	Unit Cost (\$)	Break Even Cost For 50,000 Parts (\$)
Main Body Assembly						
1	Main Body, White	ABS	0.83	0.54810	1.45	0.66
2	Main Body Backing, Gray	ABS	0.83	0.13000	1.07	0.31
3	Main Body Backing Fastener	ABS	0.83	0.00367	0.09	0.20

Table 1: Cost Analysis for the Main Body Assembly

Part #	Description	Material	Material Cost (\$/lb.)	Part Mass (lbs.)	Unit Cost (\$)	Break Even Cost For 50,000 Parts (\$)
Suction Assembly						
4	Suction base, White (screw secured)	ABS	0.83	0.12700	0.20	0.30
5	Suction Lever Bar	304 Stainless Steel	0.8	0.07000	0.14	0.25
6	Suction Lever Handle	ABS	0.83	0.12660	0.20	0.30
7	Suction Pad	Silicon Rubber	6.8	0.05375	3.74	0.56
8	Suction pad mount	ABS	0.83	0.01730	0.57	0.21

Table 2: Cost Analysis for the Suction Assembly

Part #	Description	Material	Material Cost (\$/lb.)	Part Mass (lbs.)	Unit Cost (\$)	Break Even Cost For 50,000 Parts (\$)
Blade Faces Assembly						
10	Ribbon blades	304 Stainless Steel	0.8	0.03230	1.78	0.22
11	Fettuccine blade	304 Stainless Steel	0.8	0.00900	1.76	0.20
12	Spaghetti blade	304 Stainless Steel	0.8	0.00002	1.41	0.20

13	Veg Core Remover/Holder	304 Stainless Steel	0.8	0.00770	1.42	0.21
14	Single Blade Base	ABS	0.83	0.05840	3.46	0.25
15	Dual Blade base	ABS	0.83	0.06090	2.91	0.25
16	Cutting face holder	PP	0.55	0.38350	2.50	0.41

Table 3: Cost Analysis of Blade Faces Assembly

Part #	Description	Material	Material Cost (\$/lb.)	Part Mass (lbs.)	Unit Cost (\$)	Break Even Cost For 50,000 Parts (\$)
Slider Assembly						
18	Slider Body	ABS	0.83	0.19515	2.69	0.36
19	Push handle (dofus arm)	ABS	0.83	0.03450	2.55	0.23
20	Pronged food holder	ABS	0.83	0.11650	2.62	0.30
21	Spiralizer Arm	ABS	0.83	0.05740	2.57	0.25
22	Spiralizer Handle	ABS	0.83	0.02740	2.54	0.22
23	Handle End Cap	ABS	0.83	0.00381	0.98	0.20

Table 4: Cost Analysis of Slider Assembly

In addition to unit cost and the break-even cost for 50,000 parts, OMEs were also calculated for every part. However, we deemed them acceptable when comparing them to our unit costs, and so they are not included in Tables 1-4. Our assembly also included a small number of rivets and fasteners which were not included in our cost analysis as they would not be manufactured in the factory.

Bill of Materials

The parts used in the tabletop spiralizer are listed in the Bill of Materials.

Item #	Part #	Qty	Name	Material
1	1	1	Main Body, White	ABS
2	2	1	Main Body Backing, Gray	ABS
3	3	4	Main Body Backing Fastener	ABS
4	4	1	Suction base, White (screw secured)	ABS
5	5	1	Suction Lever Bar	Stainless Steel
6	6	1	Suction Lever Handle	ABS
7	7	1	Suction Pad	Silicone Rubber

Item #	Part #	Qty	Name	Material
8	8	1	Suction pad mount	ABS
9	9	8	Screws	Stainless Steel
10	10	3	Ribbon blade	Stainless Steel
11	11	1	Fettuccine blade	Stainless Steel
12	12	1	Spaghetti blade	Stainless Steel
13	13	3	Veg Core Remover/Holder	Stainless Steel
14	14	1	Single Blade Base	ABS
15	15	2	Dual Blade base	ABS
16	16	1	Cutting face holder	PP
17	17	6	Rivets	Stainless Steel
18	18	1	Slider Body	ABS
19	19	1	Push handle	ABS
20	20	1	Pronged food holder	ABS
21	21	1	Spiralizer Arm	ABS
22	22	1	Spiralizer Handle	ABS
23	23	2	Handle End Cap	ABS
24	24	1	Spiralizer Pin	Stainless Steel

Table 5: BOM

Design for Assembly Analysis

Design for assembly is a useful tool to analyze the efficiency of a product's assembly process. The goal of designing for assembly is to eliminate assembly steps, decrease difficulty to assemble, and overall decrease the time to assemble. Through this process companies can increase their production output and save time on hourly wages, increasing their cost efficiency.

The spiralizer was analyzed under strict criteria following a DFA (Design for assembly) spreadsheet. In our analysis, we looked closely at opportunities to maintain quality (mistake proofing), handling (consistent grasp and orientation), insertion (locating and securing), and places to reduce secondary operations. Metrics for each criterion were evaluated and discussed to find places to improve current assembly procedures.

All the subsequent column groups to the right of “Functional Analysis” section Table 6 have key metrics in the “Design for Assembly Metrics” row that are a measure of how complex the real assembly process compared to the theoretical minimum number of parts. By creating a dimensionless metric these values can be later compared to updated design iterations.

DFA Analysis Worksheet																					
Assembly Name: <u>Spiralizer</u>				Team: <u>The Noodler</u>												Date: <u>10/2/2025</u>					
If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.																					
Part		DFA Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion			Secondary Operations					
		Number of Parts (Np)	Number of Interfaces (Ni)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
Part Number	Part Name																				
Body and Suction Assembly																					
1	Main Body, White	1	19	1	0	H	1	0	0	0	0	0	0	1	0	0	1	1	0	0	0
2	Main Body Backing, Gray	1	5	0	0	M	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
3	Main Body Backing Fastener	4	3	0	0	L	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0
4	Suction base, White (screw secured)	1	6	0	0	H	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
5	Suction Lever Bar	1	4	0	0	M	1	0	0	0	0	0	1	1	1	1	0	1	0	0	0
6	Suction Lever Handle	1	2	0	0	L	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
7	Suction Pad	1	1	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	Suction pad mount	1	2	0	0	L	1	0	0	0	0	0	1	0	1	1	0	0	0	0	0
9	Screws	8	2	0	0	L	4	0	0	0	0	0	0	1	1	0	0	1	0	0	0
Cutting Faces																					
10	Ribbon blades	3	4	1	0	M	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0
11	Fettuccine blade	1	1	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
12	Spaghetti blade	1	1	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
13	Veg Core Remover/Holder	3	2	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
14	Single Blade Base	1	5	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	Dual Blade base	2	6	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	Cutting face holder	1	4	0	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
17	Rivets	6	2	0	0	L	6	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Slider Assembly																					
18	Slider Body	1	5	1	0	H	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
19	Push handle	1	3	0	0	L	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
20	Pronged food holder	1		1	0	H	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
21	Spiralizer Arm	1	3	0	0	L	0	0	0	1	0	0	0	1	1	0	0	0	0	0	0
22	Spiralizer Handle	1	2	0	0	L	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	Handle End Cap	2	1	0	0	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24	Spiralizer Pin	1	2	0	0	L	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Totals		44	83	7	0	0	25	0	2	2	5	0	2	7	5	3	1	8	0	3	0
Design for Assembly Metrics		60.43177972		15.9%	←Theor. Effy. Pract. Effy.→		56.8%	0.29		1.00			2.43			1.71					
Targets																					

Table 6: Initial DFA Worksheet

Practical Part Count

The first metric that was quantified was the theoretical minimum parts (Table 6, Green Box) and the practical minimum part count (Table 6, Red Box).

This was determined by imagining a spiralizer that only includes parts necessary for its function. The functional requirements we determined for the spiralizer were the ability to cut and hold vegetables and maintain a rotational arm for the rotational cutting feature. We eliminated all aesthetic related parts, user experience related parts, and settled on only 7 required parts to maintain the spiralizer's primary functions. The suction assembly was removed completely, much of the main body was combined into one part, and the slider assembly spiralizer arm was combined into one part that was snap fit onto the slider body.

We then performed an additional analysis to obtain the practical part count by reintroducing parts that were practical to include for the device. For example, the suction functionality was not necessary to perform the cutting process but was deemed practical due to improving the user

experience. After bringing back into the design multiple blades and blade bases to improve user safety and the fasteners that were now relevant, our practical part count was 25. Compared to the number of total parts, the spiralizer was deemed to have a 56%-part efficiency. With 60% being a good guideline for part efficiency, it was necessary to find areas to reduce the part total to increase our part efficiency.

Quality (Error Proofing):

Quality opportunities look at improving parts that can be assembled the wrong way or in an incorrect orientation. When looking at the areas of improvement for ways to avoid assembly mistakes the only places that were identified to have this issue were in the consumer assembled pieces. As seen in the images below, both part #20-pronged food holder and part #18-slider body could be assembled the wrong way.



Figure 11: Improper Assembly of Slider Body

While there is potential for incorrect assembly orientation, the snap fit feature on the pronged food holder and easily removeable slider body made it unnecessary for any design changes that fixed these problems. Our team decided that the 0.29 DFA metric corresponding to quality was low enough and there were more important areas for improvement. If design changes are desired small additions like aligning arrow indent markers could be added to the injection mold of the pronged food holder, main body white, and the slider body.

Handling

Handling opportunities look to analyze problem areas that an assembler could encounter. This includes any interactions with dangerous objects, difficulty grabbing singular parts out of a box due to tangled or nested parts, and the need for extra equipment such as pilers, tweezers, or magnifying glasses to handle the part.

Our initial DFA metric for handling was 1.00 and most of the problem areas came from handling the blades that are used to cut vegetables. The sharpness of the blades does pose a risk to the handlers, but the sharpness cannot be removed without forfeiting the spiralizers' main functionality. Since functionality cannot be lost any safety considerations need to be made in the

assembly line through warnings or training. No formal design changes were made on this subject.

Insertion (Locate and Secure)

The insertion columns address concerns that cause trouble for the assemblers to put parts in place during the assembly process. Examples of issues include small tolerance holes that require precise placements, parts being difficult or strenuous to insert, or lack of vision during insertion.

The DFA metric for this category of 2.43 made it apparent that insertion issues were the most common in the spiralizer. One thing that is not made apparent by the spreadsheet is the severity of the problem for each part that it applies to. During our team's disassembly and reassembly process, part #5-suction lever bar was extremely difficult to align as there were 5 separate alignment holes that were necessary to pass through. These alignment holes can be seen in the images below.

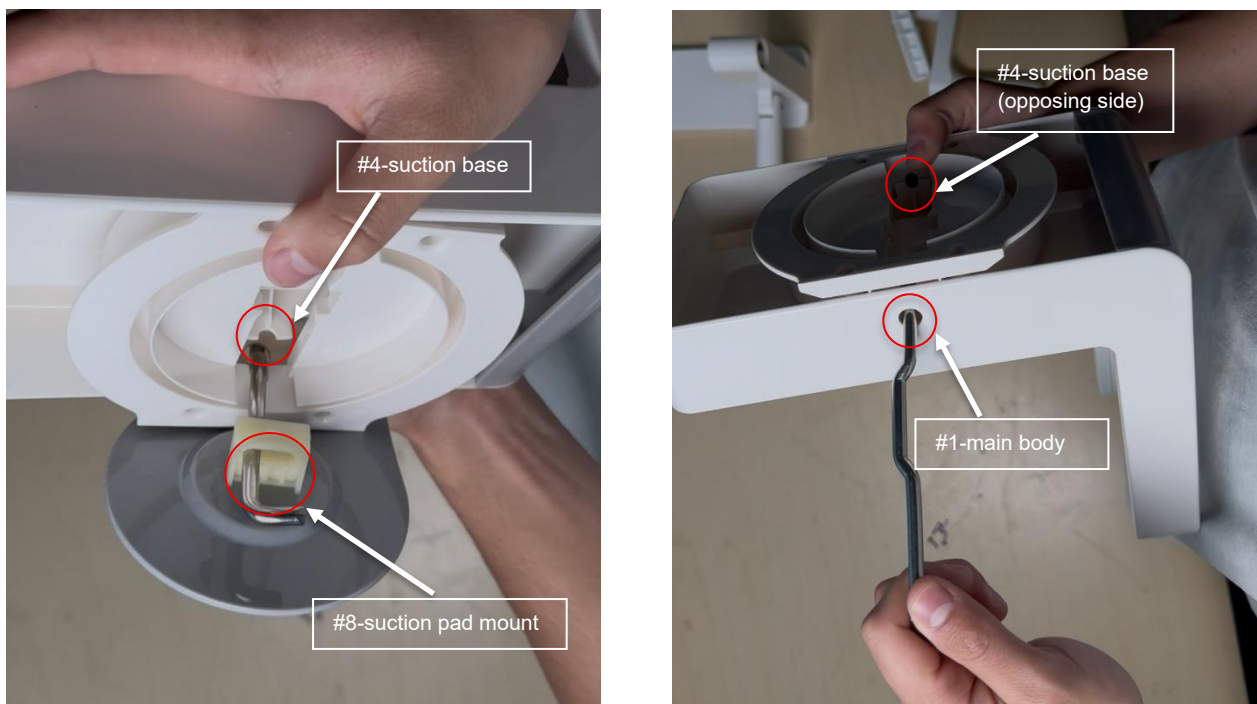


Figure 12: Issues of Suction Assembly

The difficulty of alignment also came from a lack of visibility. Once the lever bar was inserted into the guide hole of part #1-main body white, the alignment gate of part #4-suction base white, and the opening in part #8-suction pad mount, the assembler must lower the bar, and their visibility becomes blocked by the suction pad (Figure 12).

Additionally, the kink in the bar that allows the suction functionality made insertion difficult as many adjustments were necessary to align the bar. Overall, the insertion of each part was straightforward and did not require notable extra time. The lever bar insertion was the one standout process that required significant time to insert and could benefit from a design change.

Secondary Operations

Secondary operations include any process that must be done past putting the part in its place. Screwing, glueing, soldering, welding, and riveting are all examples of secondary operations. Our DFA metric for this section was 1.71, indicating that secondary operations were our second most common assembly instance. As the scoring blades (Parts #11 and #12) each required a rivet, and 8 parts were either screwed in or heat sunk, there were many places where secondary operations are currently necessary. While the rivets are the optimal method for mounting the scoring blades, the inclusion of 8 screws to connect four interfaces seemed excessive. Removing screws could provide long term health benefits to assemblers from a decrease in repetitive motion if the number of screws were reduced.



Figure 13: Secondary Operation (Screw Locations)

Our team's analysis determined that design changes regarding the blades and how they mount would not be overly beneficial to the assembly process. Trying to eliminate fasteners would be targeted in our design changes to help alleviate unnecessary stress on assemblers and simplify the design.

DFA Conclusion

The spiralizer succeeded in many facets of DFA. General metrics, including the practical part efficiency being close to 60% and the error proofing metric being as low as 0.27 gave a strong start to optimizing the part for assembly. Our in-depth analysis provided potential areas for improvement and enlightened areas that would greatly improve the assembly process. Design changes related to DFA are discussed later on in the report.

Proposed Changes

The reverse engineering and DFA analysis conducted on the tabletop spiralizer revealed specific opportunities to reduce part count, simplify assembly operations, and improve manufacturing efficiency. We propose three redesigns targeting the highest-impact DFA metrics: secondary operations (1.71), insertion complexity (2.43), and overall part efficiency (56.8%). Each modification addresses documented assembly difficulties while maintaining product functionality and user experience.

Redesign Change 1: Snap fit body assembly

The main body assembly requires four custom fasteners screwed to the back of the main body, which then serve as attachment points for the main body backing to snap into place. This design necessitates eight parts (four screws plus four custom fasteners) and their associated secondary operations during assembly.

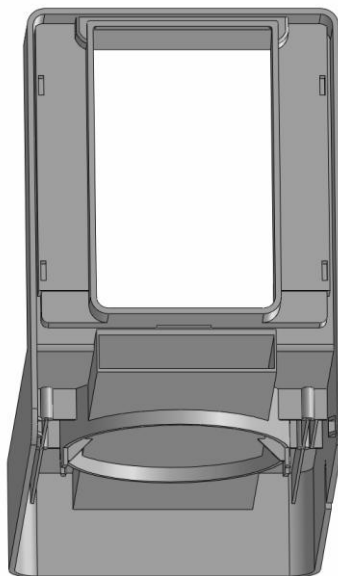


Figure 14: Redesign of Slider Body

A parametric redesign, seen in “figure 14” that integrates the snap-fit features directly into the main body during injection molding, eliminating the need for separate fasteners. The main body backing would snap directly onto these integrated features. This modification improves part count efficiency from 56.8% to 61.5%, exceeding the target benchmark. The secondary operations metric improves from 1.71 to 1.29, representing a 25% reduction in fastening operations.

Redesign Change 2: Open slot for suction lever bar

Assembly of the suction subassembly was identified as the most difficult operation in the DFA analysis (Figure 12). The suction lever bar must be threaded through five separate alignment features: the main body guide hole, the suction base alignment gates (both sides), and the suction pad mount openings. The kinked geometry of the bar compounds the difficulty, and visibility is blocked by the suction pad during final alignment.

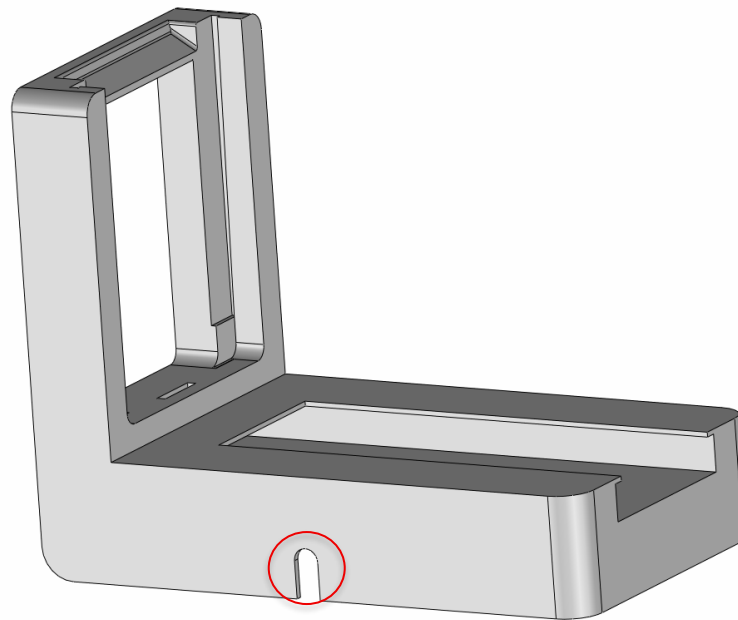


Figure 15: Redesign of Suction Assembly

A parametric redesign converting the closed guide hole in the main body to an open slot, allowing lateral insertion of the lever bar rather than threading it through multiple constrained openings. This change reduces the insertion metric from 2.43 to approximately 1.86—a 23% improvement—while maintaining structural integrity as the suction base provides adequate constraint for the lever bar during operation.

Redesign Change 3: Elimination of the push arm

User testing during our disassembly and operation analysis revealed that the push arm attached to the slider body sees minimal use during normal operation. Users demonstrated a strong preference for combining rotational and translational forces through the spiralizer handle rather

than using separate hands for rotation and pushing. Additionally, users frequently employed their free hand to stabilize the main body rather than operate the push arm, particularly when the suction base was used on less-than-ideal surfaces.

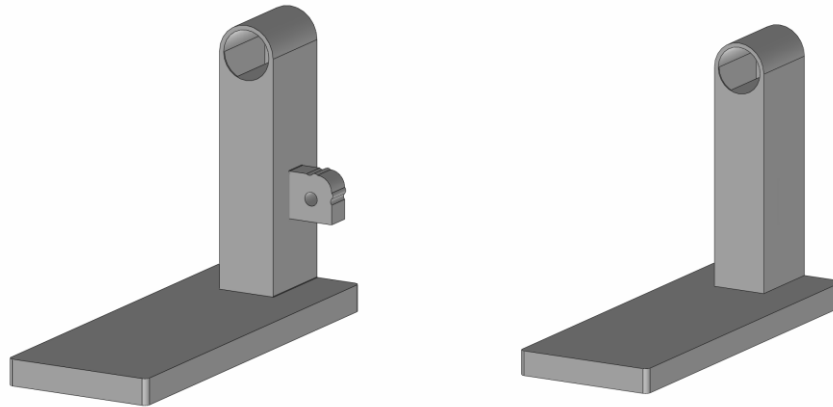


Figure 1616: Modified Slider Body

An adaptive redesign that eliminates the push arm, handle cap, pin, and slider base attachment point—components identified as non-essential through user behavior analysis showing they don't meaningfully enhance product functionality or user experience. Combined with Redesign 1, this improves part efficiency to 63.4% and provides secondary benefits including reduced material costs and simplified injection molding.

Adjusted Bill of Materials

This adjusted Bill of Materials reflects implementation of the proposed redesigns described in the prior section.

Item #	Part #	Qty	Name	Material
1	1M	1	Main Body, White	ABS
2	2	1	Main Body Backing, Gray	ABS
3	4	1	Suction base, White (screw secured)	ABS
4	5	1	Suction Lever Bar	Stainless Steel
5	6	1	Suction Lever Handle	ABS
6	7	1	Suction Pad	Silicone
7	8	1	Suction pad mount	ABS
8	9	4	Screws	Stainless Steel
9	10	3	Ribbon blade	Stainless Steel
10	11	1	Fettuccine blade	Stainless Steel

Item #	Part #	Qty	Name	Material
11	12	1	Spaghetti blade	Stainless Steel
12	13	3	Veg Core Remover/Holder	Stainless Steel
13	14	1	Single Blade Base	ABS
14	15	2	Dual Blade base	ABS
15	16	1	Cutting face holder	ABS
16	17M	6	Rivets	Stainless Steel
17	18	1	Slider Body	ABS
18	20	1	Pronged food holder	ABS
19	21	1	Spiralizer Arm	ABS
20	22	1	Spiralizer Handle	ABS
21	23	1	Handle End Cap	ABS

Table 7: Modified BOM After Redesign

Adjusted DFA

This DFA reflects implementation of the proposed redesigns.

DFA Analysis Worksheet																					
Assembly Name: <u>Spiralizer</u>				Team: <u>The Noodler</u>												Date: <u>10/5/2025</u>					
If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.																					
Part		DFA Complexity		Functional Analysis / Redesign Opportunity			Error Proofing		Handling			Insertion			Secondary Operations						
		Number of Parts (Np)	Number of Interfaces (Ni)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
Part Number	Part Name																				
Body and Suction Assembly																					
1	Main Body, White	1	19	1	0	H	1	0	0	0	0	0	0	1	0	0	1	0	0	0	0
2	Main Body Backing, Gray	1	5	0	0	M	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
4	Suction base, White (screw secured)	1	6	0	0	H	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
5	Suction Lever Bar	1	4	0	0	M	1	0	0	0	0	0	0	1	1	1	0	1	0	0	0
6	Suction Lever Handle	1	2	0	0	L	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
7	Suction Pad	1	1	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	Suction pad mount	1	2	0	0	L	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0
9	Screws	4	2	0	0	L	4	0	0	0	0	0	0	1	1	0	0	1	0	0	0
Cutting Faces																					
10	Ribbon blades	3	4	1	0	M	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0
11	Fettuccine blade	1	1	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
12	Spaghetti blade	1	1	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
13	Veg Core Remover/Holder	3	2	1	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	1	0
14	Single Blade Base	1	5	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	Dual Blade base	2	6	0	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	Cutting face holder	1	4	0	0	M	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
17	Rivets	6	2	0	0	L	6	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Slider Assembly																					
18	Slider Body	1	5	1	0	H	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0
20	Pronged food holder	1		1	0	H	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
21	Spiralizer Arm	1	3	0	0	L	0	0	0	1	0	0	0	1	1	0	0	0	0	0	0
22	Spiralizer Handle	1	2	0	0	L	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	Handle End Cap	1	1	0	0	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Totals		34	77	7	0	0	25	0	2	2	5	0	0	6	4	3	1	5	0	3	0
Design for Assembly Metrics		51.16639522		20.6%	←Theor. Effic. Pract. Effic.→		73.5%	0.29	1.00			1.86			1.29						
Targets																					

Table 8: Modified DFA After Redesign

Conclusion

This reverse engineering project deconstructed the OXO Good Grips Tabletop Spiralizer to understand its design decisions and implement changes that improve manufacturability, assembly efficiency, and cost-effectiveness. The objective was to improve design for assembly characteristics while maintaining functionality and user experience.

The spiralizer contains 44 parts with a practical minimum of 25 parts, yielding 56.8%-part efficiency below the 60% industry benchmark. DFA analysis identified key challenges: the insertion metric of 2.43 indicates moderate difficulty in assembling, specifically involving the suction lever bar through five alignment features with limited visibility, and the secondary operations metric of 1.71 reflects eight screws in the main body assembly. The handling metric of 1.00 shows acceptable performance, and the quality/error proofing metric of 0.29 reflects minimal assembly error risk.

Based on these metrics, we developed three targeted redesigns. First, a parametric redesign integrates custom fasteners into the main body during injection molding, eliminating eight parts and improving efficiency from 56.8% to 61.5%. Second, a parametric redesign converts the closed guide hole to an open slot, reducing the insertion metric from 2.43 to 1.86—a 23% improvement. Third, an adaptive redesign eliminates the underutilized push arm assembly, removing three parts based on user testing.

Combined, these redesigns reduce part count from 44 to 33 parts—a 25% reduction—and improve efficiency to 75.8%, exceeding the 60% benchmark. Assembly efficiency and reduction in parts also contributes to an overall reduction in costs for the manufacturing/assembly processes, increasing profit for the manufacturer. Secondary operations drop from 13 to 6 operations, a 54% reduction. Material analysis validated ABS plastic for 68% of parts and 304 stainless steel for 23% of blade components based on strength, manufacturability, and cost-effectiveness.

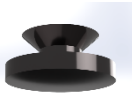
Appendices

Appendix A

Product Decomposition	
Design Organization: MCEN 5045	Date: September 1, 2025
Product Decomposed: OXO Tabletop Spiralizer	
Description: The tabletop spiralizer cuts vegetables into spiral noodle shapes.	
How it works: The tabletop spiralizer works by placing a prepared vegetable between the selected cutting face and food holder, then pressing and rotating the handles, respectively, to cut the vegetable into ribbons or ‘noodles’.	

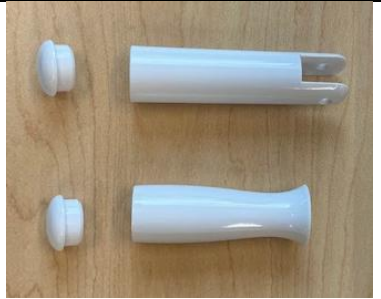
Parts:					
Part #	Part Name	# Req'd	Material	Mfg Process	Image
1	Main Body, White	1	ABS	Injection Molded	
2	Main Body Backing, Gray	1	ABS	Injection Molded	
3	Main Body Backing Fastener	4	ABS	Injection Molded	
4	Suction base, White (screw secured)	1	ABS	Injection Molded	
5	Suction Lever Bar	1	Stainless Steel		
6	Suction Lever Handle	1	ABS	Injection Molded	
7	Suction Pad	1	Silicone	Injection Molded	 *grey portion, silicone over molded
8	Suction pad mount	1	ABS	Injection Molded	 *beige portion

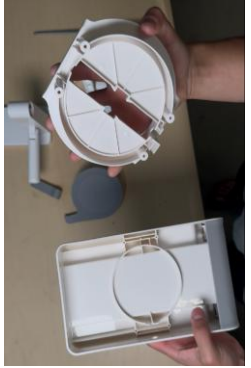
Part #	Part Name	# Req'd	Material	Mfg Process	Image
9	Screws	8	Stainless Steel	N/A	
10	Ribbon blades	3	Stainless Steel	Die Cut	
11	Fettuccine blade	1	Stainless Steel	Die Cut	
12	Spaghetti blade	1	Stainless Steel	Die Cut	
13	Veg Core Remover/Holder	3	Stainless Steel	Sheet metal Fabrication	
14	Single Blade Base	1	ABS	Injection Molded	
15	Dual Blade base	2	ABS	Injection Molded	
16	Cutting face holder	1	PP	Injection Molded	

Part #	Part Name	# Req'd	Material	Mfg Process	Image
17	Rivets	6	Stainless Steel	N/A	
18	Slider Body	1	ABS	Injection Molded	
19	Push Handle	1	ABS	Injection Molded	
20	Pronged food holder	1	ABS	Injection Molded	
21	Spiralizer Arm	1	ABS	Injection Molded	
22	Spiralizer Handle	1	ABS	Injection Molded	
23	Handle End Cap	2	ABS	Injection Molded	
24	Spiralizer Pin	1	Stainless Steel	N/A	

Disassembly:

Step #	Procedure	Part #s removed	Image
1	Remove slider body from main body	18	
2	Separate spiralizer arm from pronged food holder, and slide pronged food holder out of slider body	20, 21	

3	Remove spiralizer handle from spiralizer arm	22	
4	Bump pin out from push handle	24,19	
5	Pry handle end caps from spiralizer and push arms	23	
6	Remove suction lever handle from suction lever	6	
7	Wiggle and remove suction lever, and therefore suction pad/suction pad mount (suction pad is over molded on suction pad mount)	5,7, 8	

8	Unscrew (4) screws from suction base and remove from main body	9, 4	
9	Pry main body backing away from main body backing fastener	2	
10	Unscrew main body backing fastener from main body	1, 3, 9	
11	Apply heat to cutting faces, remove fettucine/spaghetti blade and vegetable holder	11, 12	

12	Drill out rivet to remove ribbon blade	10, 14, 15, 6	
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Table 9: Product Decomposition

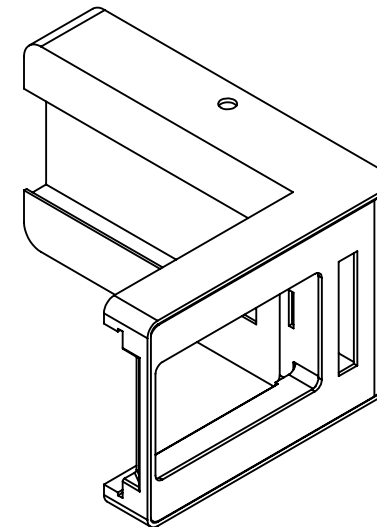
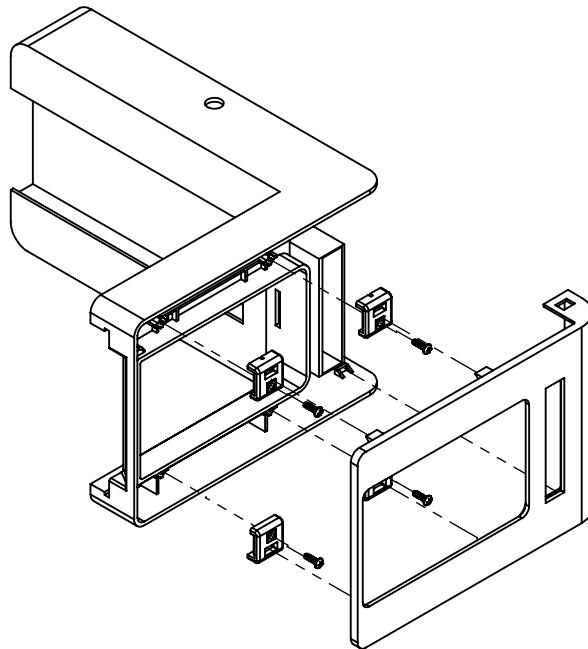
Appendix B

Definitions	Variables	Inputs	Equations	Solutions	*All boxes highlighted pink should be filled in	
Material Cost	C_M			\$0.51		
weight of material	m (lb/unit)	0.5481			Break Even Price (for 5000 parts)	\$2.43
cost of material	c_m (\$/lb)	0.83			Total Price	\$4.50
fraction of scrap material	f	0.1				
Labor Cost	C_L			\$0.004		
hourly cost of wages (\$/hr)	c_w (\$/hr)	0.22				
production rate (unit/hr)	n' (units/hr)	60				
Tooling Cost	C_T			\$0.40		
cost of making tooling	c_t (\$)	1000				
# of parts	n	5000				
tooling wear factor (k=n/ life of toolin k		2				
Equipment Cost	C_E			\$0.21		
cost of capital equipment	c_e (\$)	50000				
production rate	n' (units/hr)	60				
load factor, fractional time equipmen L		0.15				
capitol write off-time	t_w (yrs)	5				
fraction of equipment sharing between q		0.375				
Overhead Cost	C_OH			\$0.33		
Overheader Hourly	c_oh (\$/hr)	20				
Production Rate	n' (units/hr)	60				
Unit Cost	C_U			\$1.45		
Break Even Point	Q_B			\$9,833.33	Additional variables (monthly)	
fixed costs (\$)	f	9833.33			Labor costs (\$/unit)	0.003666666667
sales price (\$/unit)	P	1.458589667			Material costs (\$/unit)	0.454923
variable costs (\$/unit)	v	0.4585896667			G&A expenses	3000
					Depreciation	833.33
Break Even Price (for 5000 parts)	P			\$0.66	Factory Expenses	2000
fixed costs (\$)	f	9833.33			Sales and OH	4000
variable costs (\$/unit)	v	0.4585896667			Profit	1 3* cost of manufacturing - cost of the material
# of units (5000)	u	50000				
Order of Magnitude Estimate (OME)						
Cost of material	c_m (\$/lb)	0.83	Material	\$0.50		
part weight	m (lbs/part)	0.5481	Manufactuing	\$1.50		
total material multiplier (waste + part	1.1	1.1	Price	\$4.50		
*assumptions	small sized plant (25,000 sqft) operating in China, buying bulk materials at market price					
ABS Price Index	https://businessanalytiq.com/procurementanalytics/index/acrylonitrile-butadiene-styrene-abs-usa-price-index-2/?utm_source=chatgpt.com					
Injection Mold operator salary	https://www.erieri.com/salary/job/injection-molding-machine-operator/china?utm_source=chatgpt.com					

Table 10: Cost Analysis Spreadsheet Example: Part #1 (Slider Body)

Appendix C

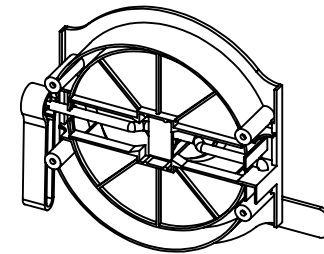
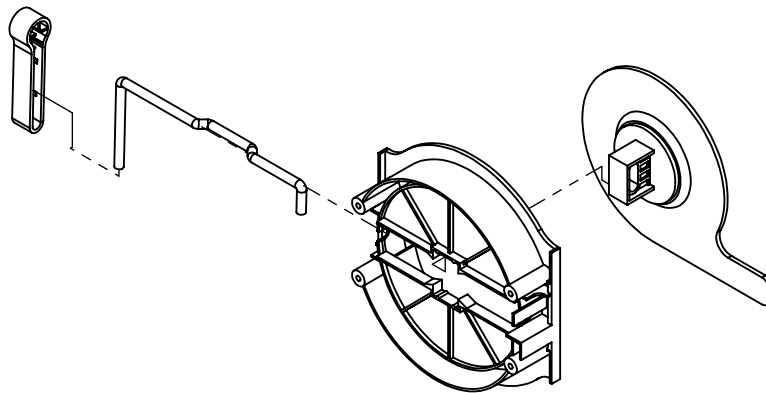
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	1. MAIN BODY WHITE	MAIN BODY	1
2	2. MAIN BODY BACKING FASTENER GREY	MAIN BODY BACKING	1
3	3. MAIN BODY BACKING FASTENER	BODY BACKING FASTENER	4
4	91772A108	PASSIVATED 18-8 STAINLESS STEEL PAN HEAD PHILLIPS SCREW	4



NOTE: EXPLODED VIEW

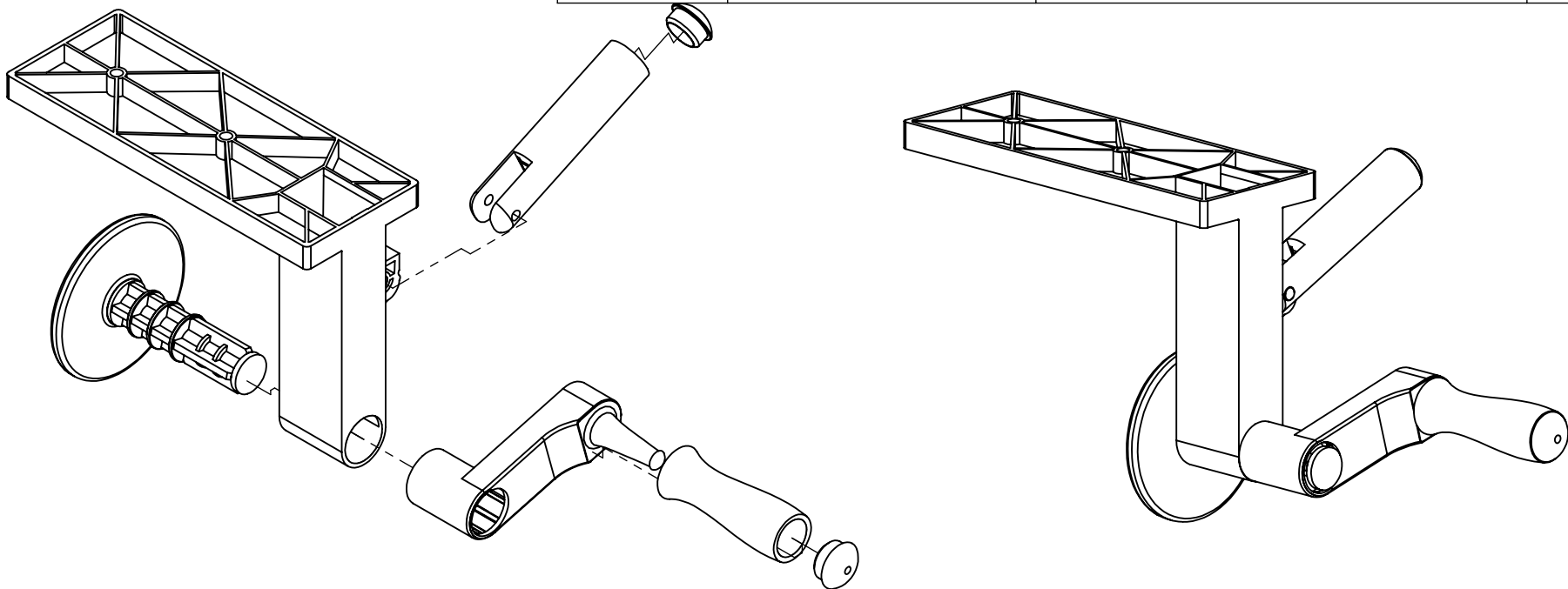
TOLERANCES UNLESS NOTED: X.X + X.XX +- X.XXX +- UNITS: INCHES X.X°		UNIVERSITY OF COLORADO 1876	
MATERIAL	AS REQUIRED	DESCRIPTION MAIN BODY ASSEMBLY	
FINISH	N/A	PN ASSEM 1	REV 1 of 1
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	4. SUCTION BASE (white) SCREW SECURED	BASE (WHITE) WITH SCREW HOLES	1
2	5. SUCTION LEVER BAR	SUCTION LEVER BAR	1
3	6. SUCTION LEVER HANDLE	LEVER HANDLE	1
4	8. SUCTION PAD MOUNT	PAD MOUNT - PERMANENTLY ATTACHED TO PAD	1
5	7. SUCTION PAD	SUCTION PAD	1



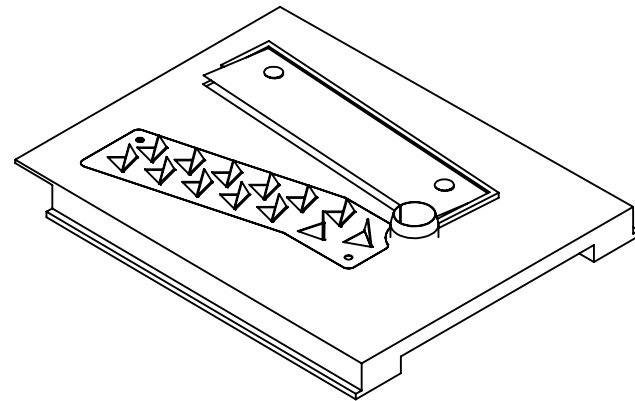
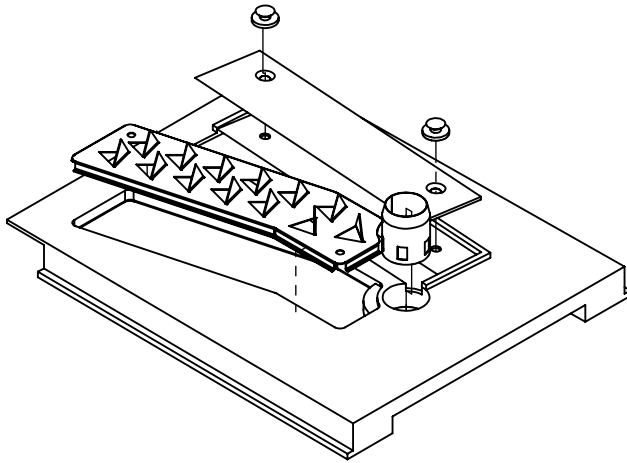
TOLERANCES UNLESS NOTED: X.X + X.XX ++ X.XXX +++ UNITS: INCHES X.X° °			
MATERIAL AS REQUIRED		DESCRIPTION SUCTION BASE	
FINISH N/A		PN ASSEM 2	REV   SHEET 1 of 1
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	18. SLIDER BODY	SLIDER BODY	1
2	20. PRONGED FOOD HOLDER	FOOD HOLDER	1
3	21. SPIRALIZER ARM	SPIRALIZER ARM	1
4	22. SPIRALIZER HANDLE	SPIRALIZER HANDLE	1
5	96008A180	CARBON STEEL CENTER-KNUTED DOWEL PINS	1
6	22. HANDLE END CAP	HANDLE END CAP	2
7	19. PUSH ARM	PUSH ARM	1



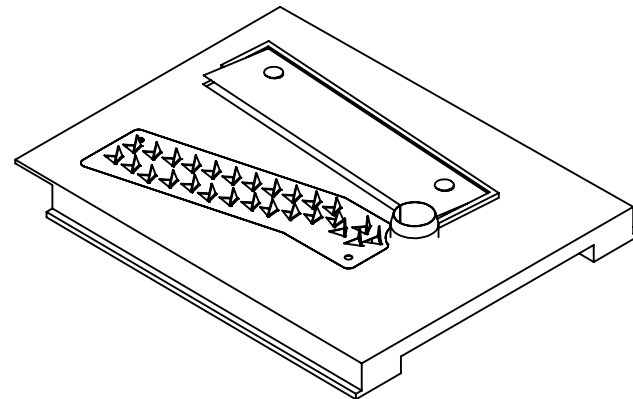
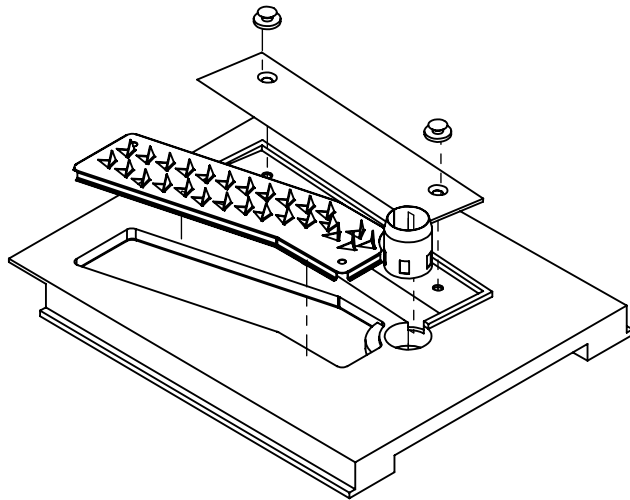
TOLERANCES UNLESS NOTED: X.X ± X.XX ± ± X.XXX ± ± ± UNITS: INCHES X.X°			
MATERIAL AS REQUIRED		DESCRIPTION SLIDER ASSEMBLY	
FINISH N/A		PN ASSEM 3	REV   SHEET 1 of 1
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			

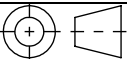
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	15.2. DUAL BLADE BASE (ORANGE)	BLADE HOLDER	1
2	10. RIBBON BLADE	RIBBON BLADE	1
3	11. FETTUCINE BLADE	FETTUCINI BLADE CUTTER	1
4	MS20470AD 3-16	RIVETS	2
5	13. VEGGIE HOLDER PART	VEGETABLE CORER	1



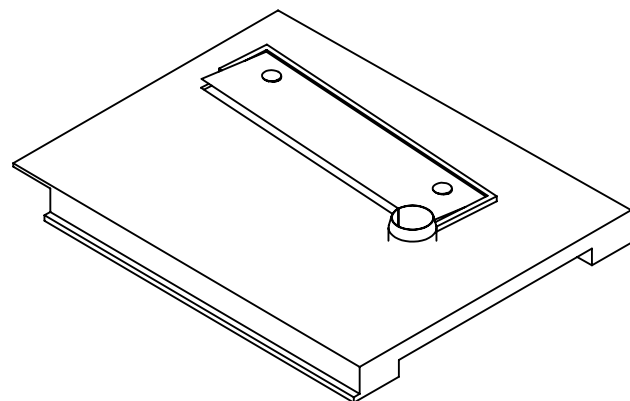
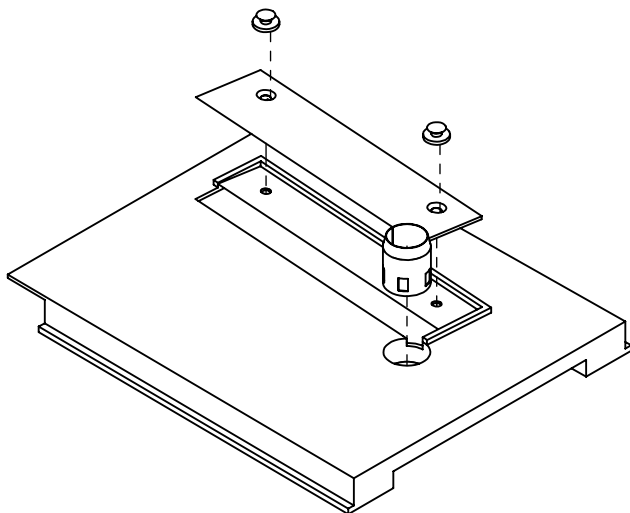
TOLERANCES UNLESS NOTED: X.X + X.XX ++ X.XXX +++ UNITS: INCHES X.X°			
MATERIAL AS REQUIRED		DESCRIPTION DUAL BLADE HOLDER - FETTUCINI	
FINISH N/A		PN ASSEM 4	REV 
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	
			SHEET 1 of 1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	12. SPAGHETTI BLADE	SPAGHETTI BLADE	1
2	10. RIBBON BLADE	RIBBON BLADE	1
3	15.1. DUAL BLADE BASE (GREEN)	BLADE HOLDER	1
4	13. VEGGIE HOLDER PART	VEGETABLE HOLDER	1
5	MS20470AD 3-16	RIVET	2



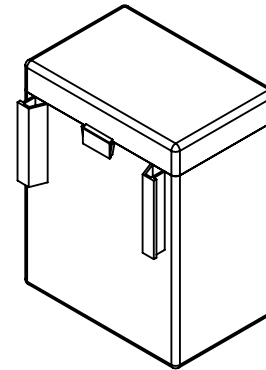
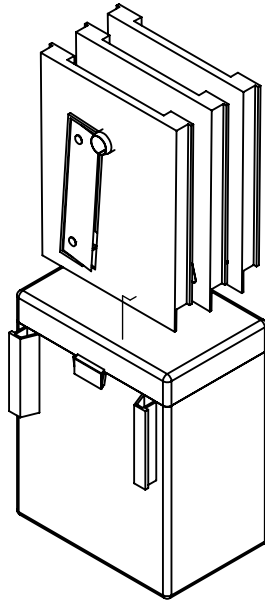
TOLERANCES UNLESS NOTED: X.X + X.XX ++ X.XXX +++ X.X° +° UNITS: INCHES			
MATERIAL AS REQUIRED		DESCRIPTION DUAL BLADE HOLDER - SPAGHETTI	
FINISH N/A		PN ASSEM 5	REV 
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	
		SHEET 1 of 1	

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	14. SINGLE BLADE BASE (RED)	BLADE HOLDER	1
2	10. RIBBON BLADE	RIBBON BLADE	1
3	13. VEGGIE HOLDER PART	VEGETABLE HOLDER	1
4	MS20470AD 3-16	RIVETS	2

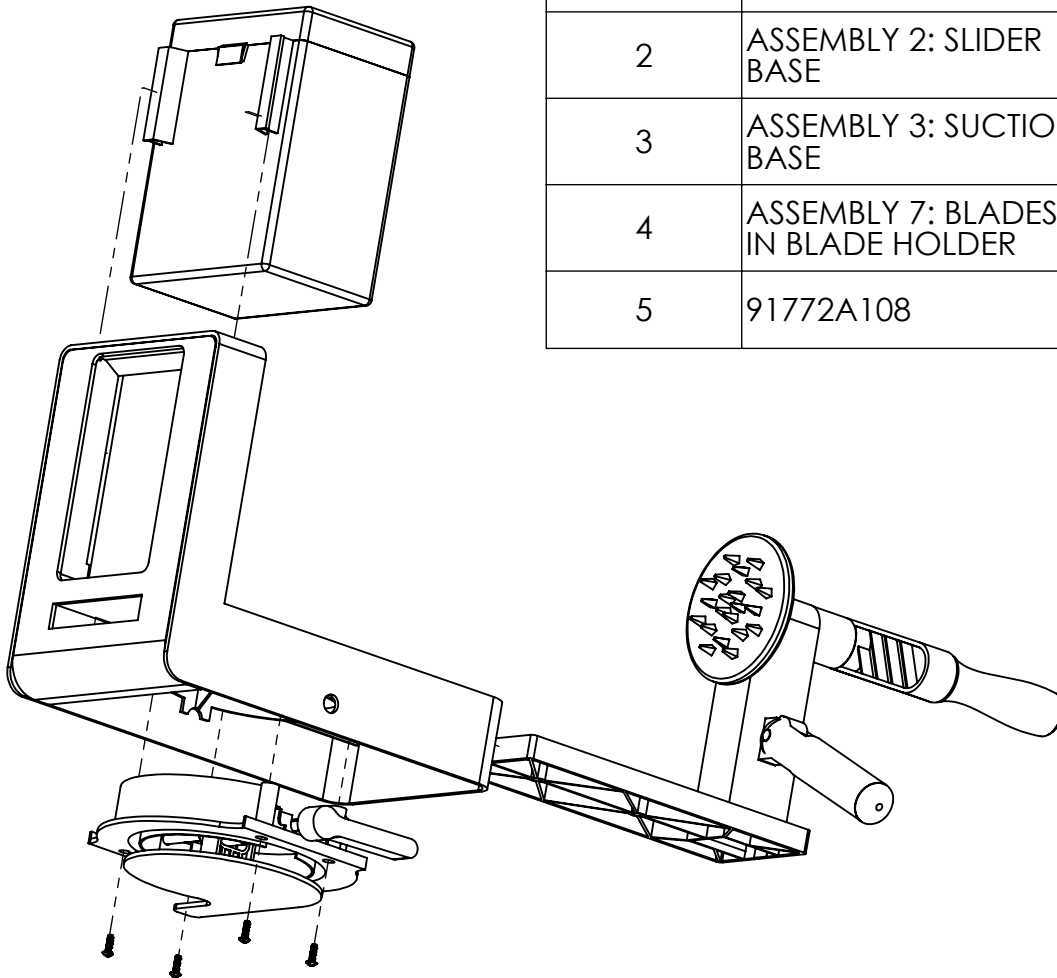


TOLERANCES UNLESS NOTED: UNITS: INCHES		X.X X.XX X.XXX X.X°					
MATERIAL		AS REQUIRED		DESCRIPTION SINGLE BLADE HOLDER			
FINISH		N/A		PN ASSEM 6	REV		SHEET 1 of 1
PROPRIETARY AND CONFIDENTIAL							
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.							

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	16. CUTTING FACE HOLDER	CUTTING FACE HOLDER (TRANSPARENT)	1
2	ASSEMBLY 6: SINGLE BLADE HOLDER (RED)	RED BLADE (SPIRALIZER)	1
3	ASSEMBLY 5: DUAL BLADE HOLDER (GREEN)	GREEN BLADE (SPAGHETTI)	1
4	ASSEMBLY 4: DUAL BLADE HOLDER (ORANGE)	ORANGE BLADE (FETTUCINI)	1

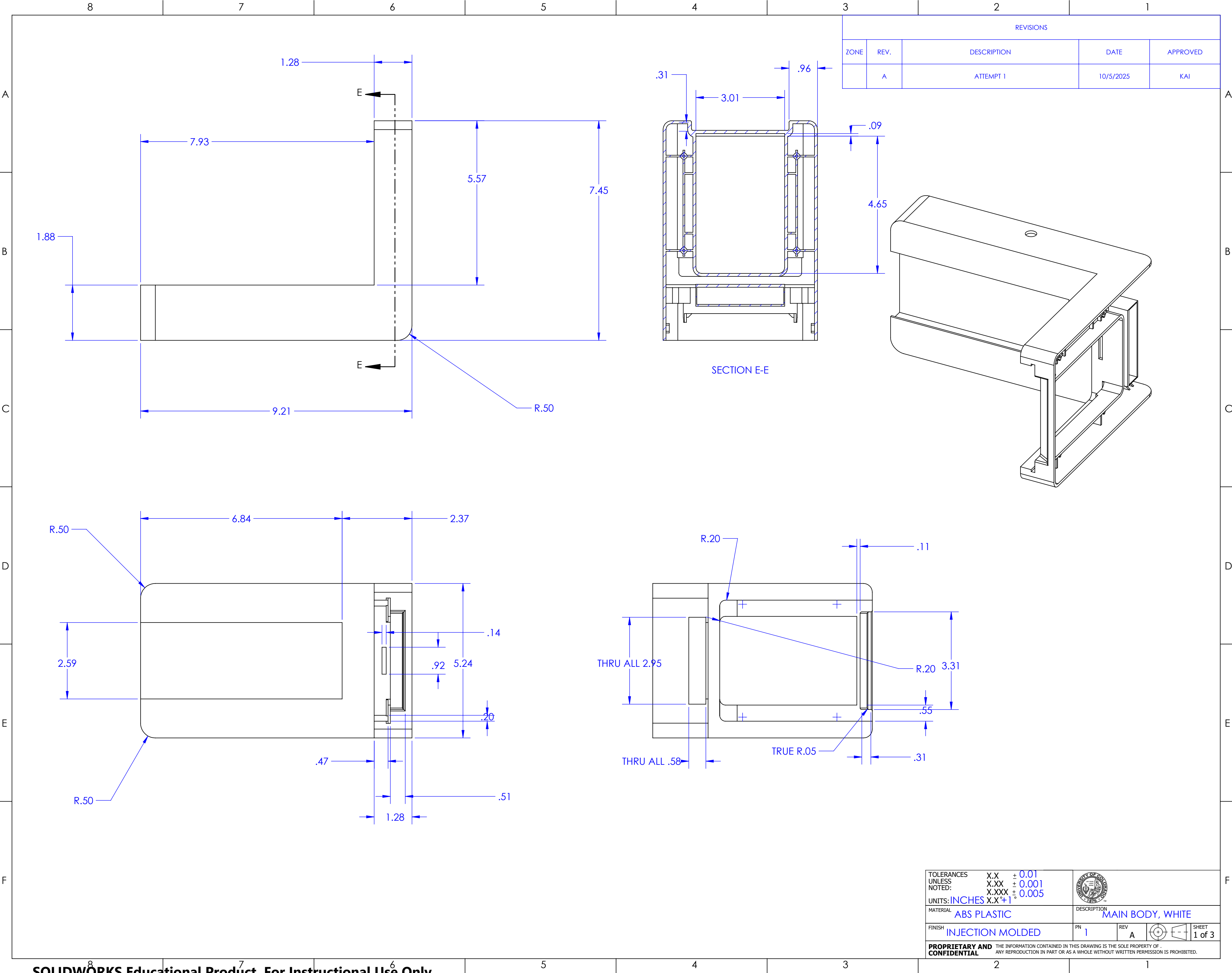


TOLERANCES UNLESS NOTED: X.X ± X.XX ± X.XXX ± UNITS: INCHES X.X° °			
MATERIAL AS REQUIRED		DESCRIPTION BLADES IN BLADE HOLDER	
FINISH N/A		PN ASSEM 7	REV 
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	
		SHEET 1 of 1	



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	ASSEMBLY 1: MAIN BODY ASSEMBLY	MAIN BODY	1
2	ASSEMBLY 2: SLIDER BASE	SLIDER BASE	1
3	ASSEMBLY 3: SUCTION BASE	SUCTION BASE	1
4	ASSEMBLY 7: BLADES IN BLADE HOLDER	BLADES IN BLADE HOLDER	1
5	91772A108	PASSIVATED 18-8 STAINLESS STEEL PAN HEAD PHILLIPS SCREW	4

TOLERANCES UNLESS NOTED: X.X + X.XX ++ X.XXX +++ UNITS: INCHES X.X°			
MATERIAL AS REQUIRED		DESCRIPTION FULL ASSEMBLED SLIDER	
FINISH N/A		PN ASSEM 8	REV  
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	
		SHEET 1 of 1	



REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	ATTEMPT 1	10/5/2025	KAI

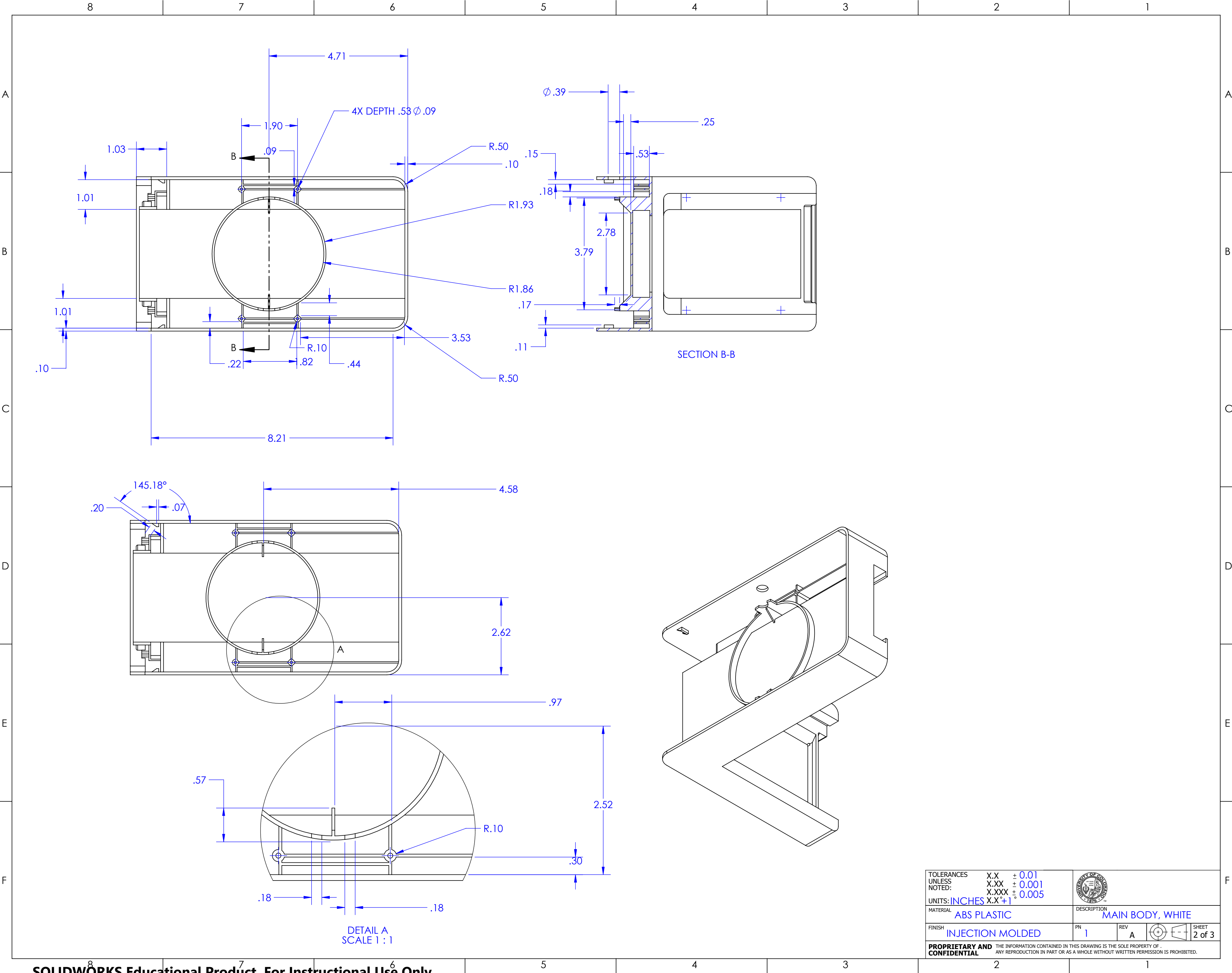
SECTION E-E

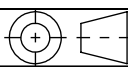
TOLERANCES
UNLESS
NOTED:
X.X ± 0.01
X.XX ± 0.001
X.XXX ± 0.005
UNITS: INCHES X.X⁺1

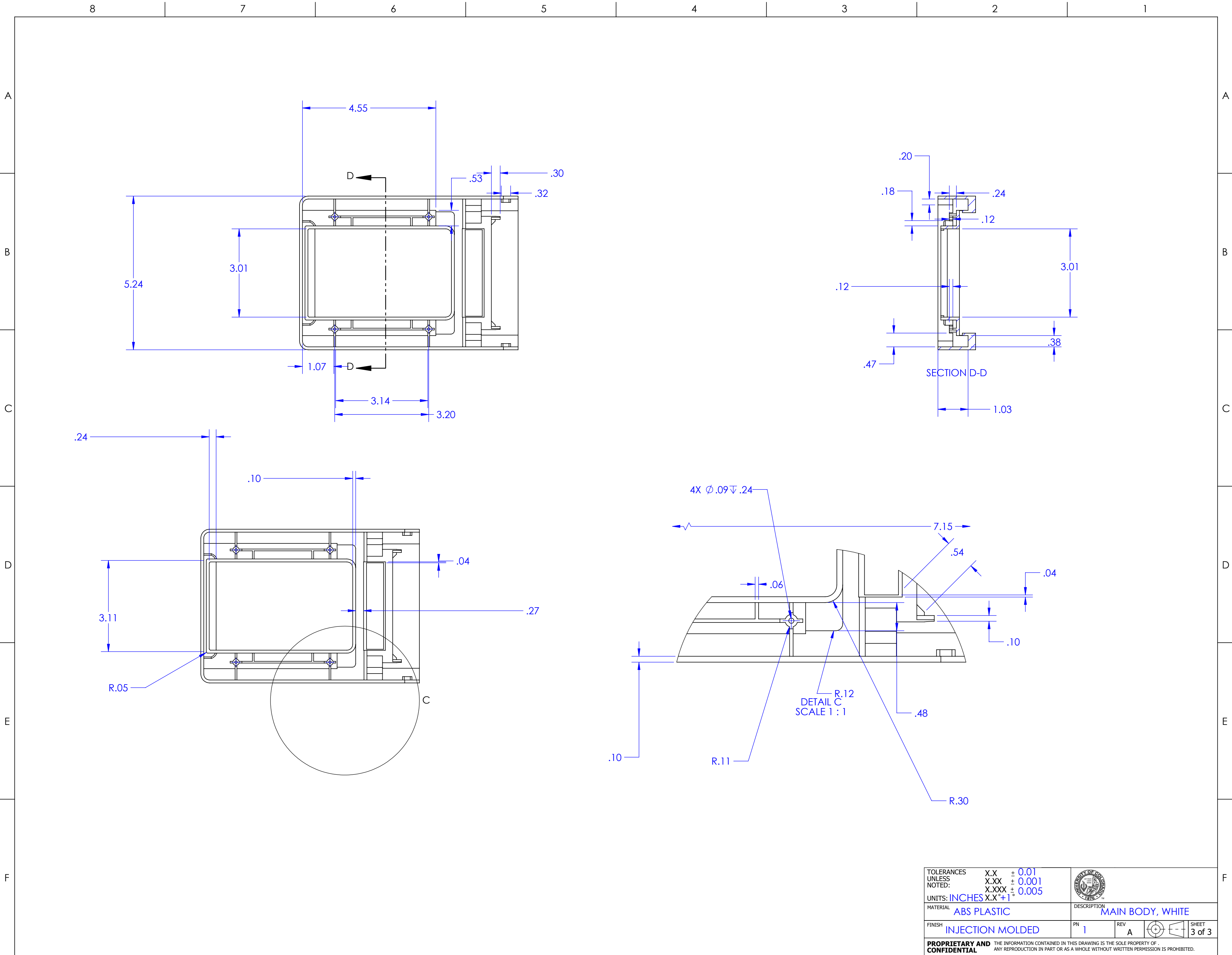


MATERIAL ABS PLASTIC		DESCRIPTION MAIN BODY, WHITE		
FINISH INJECTION MOLDED	PN 1	REV A		SHEET 1 of 3

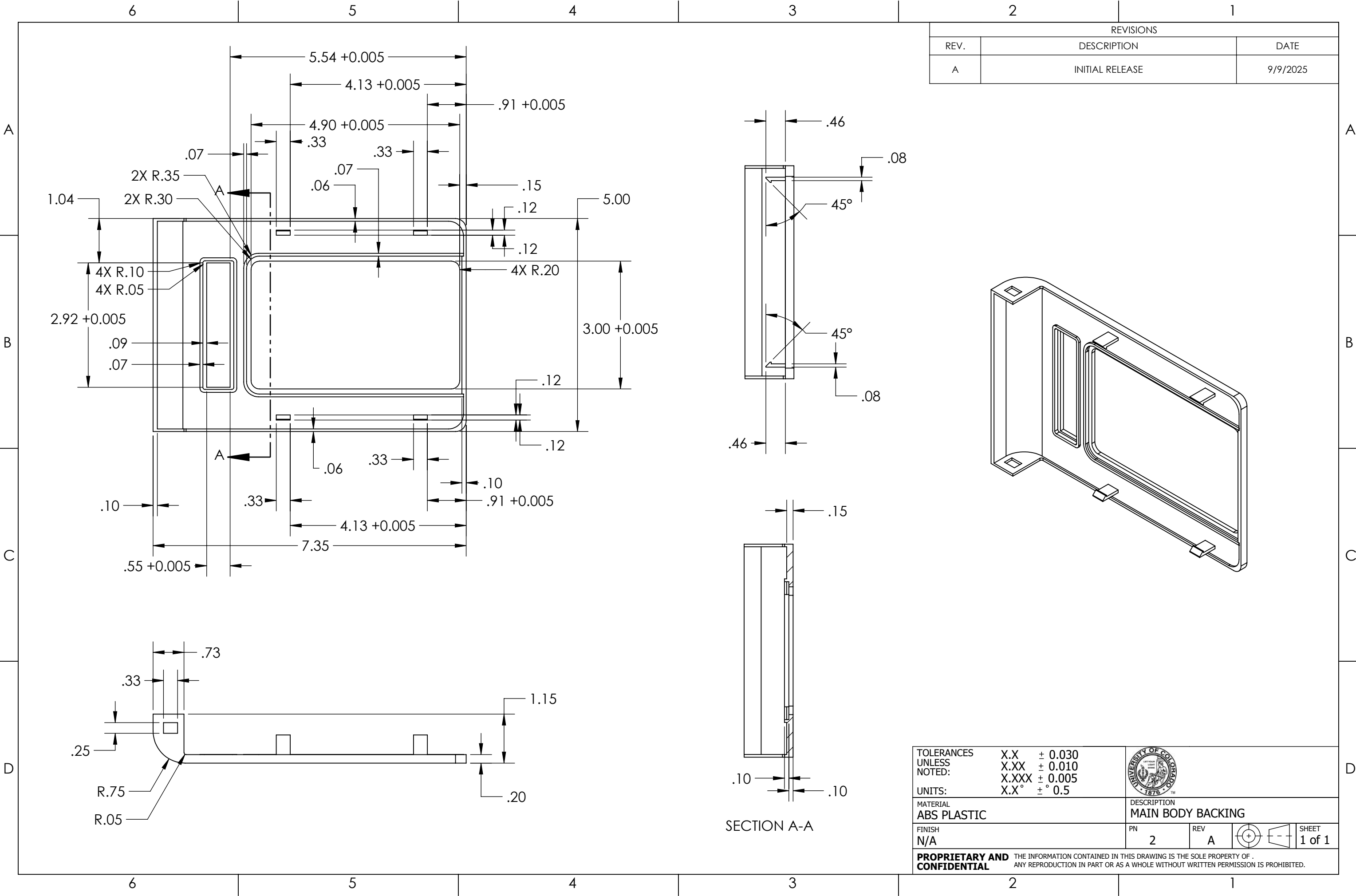
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.



TOLERANCES UNLESS NOTED:		X.X ± 0.01 X.XX ± 0.001 X.XXX ± 0.005	
UNITS: INCHES		X.X [+]	
MATERIAL		ABS PLASTIC	DESCRIPTION MAIN BODY, WHITE
FINISH	INJECTION MOLDED	PN 1	REV A
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.	
			
		SHEET 2 of 3	

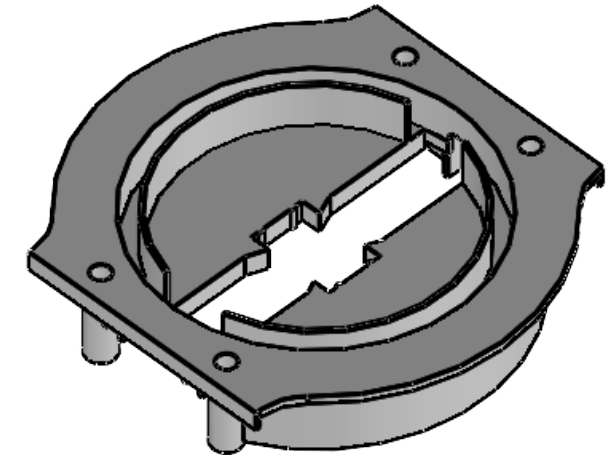
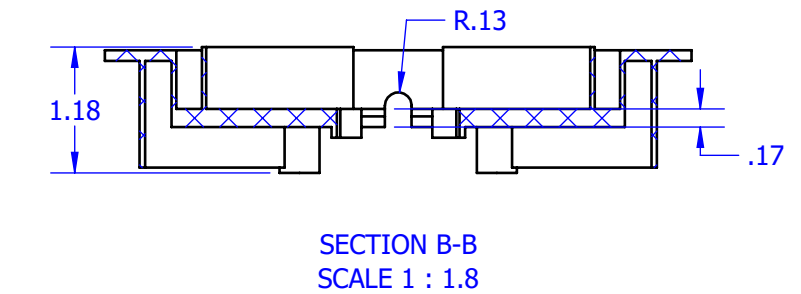
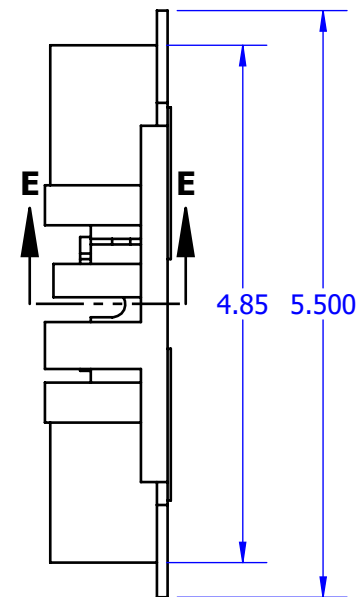
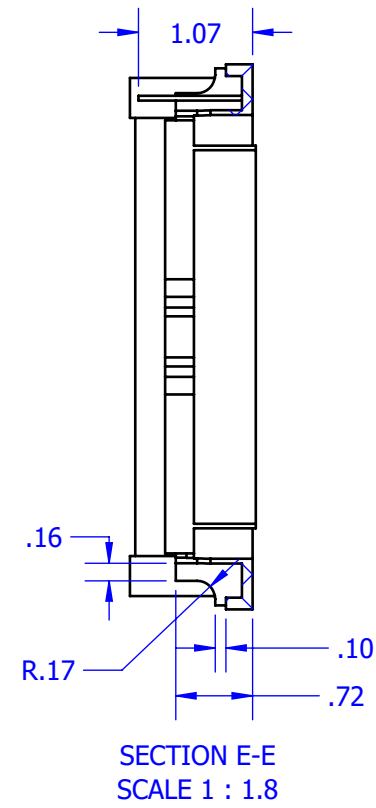
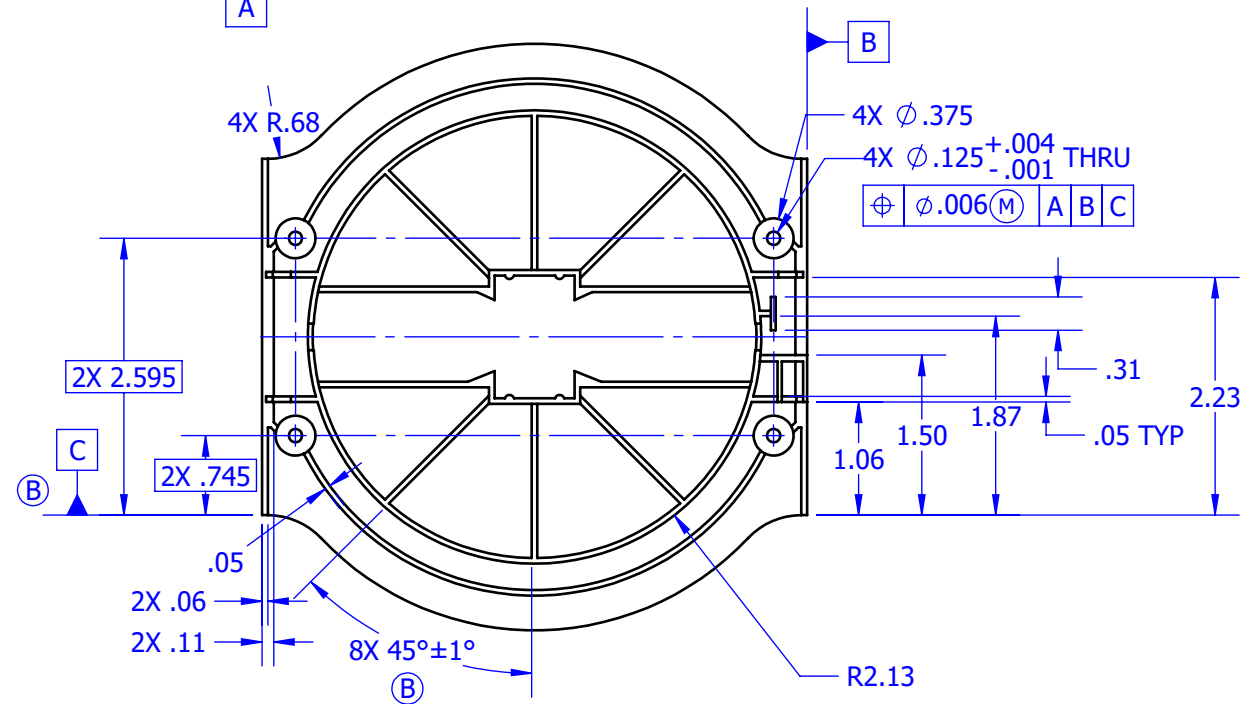
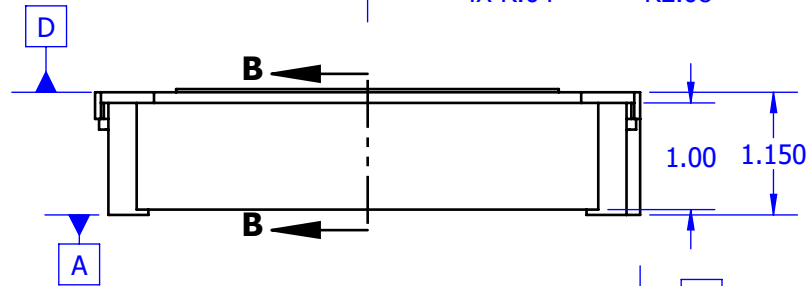
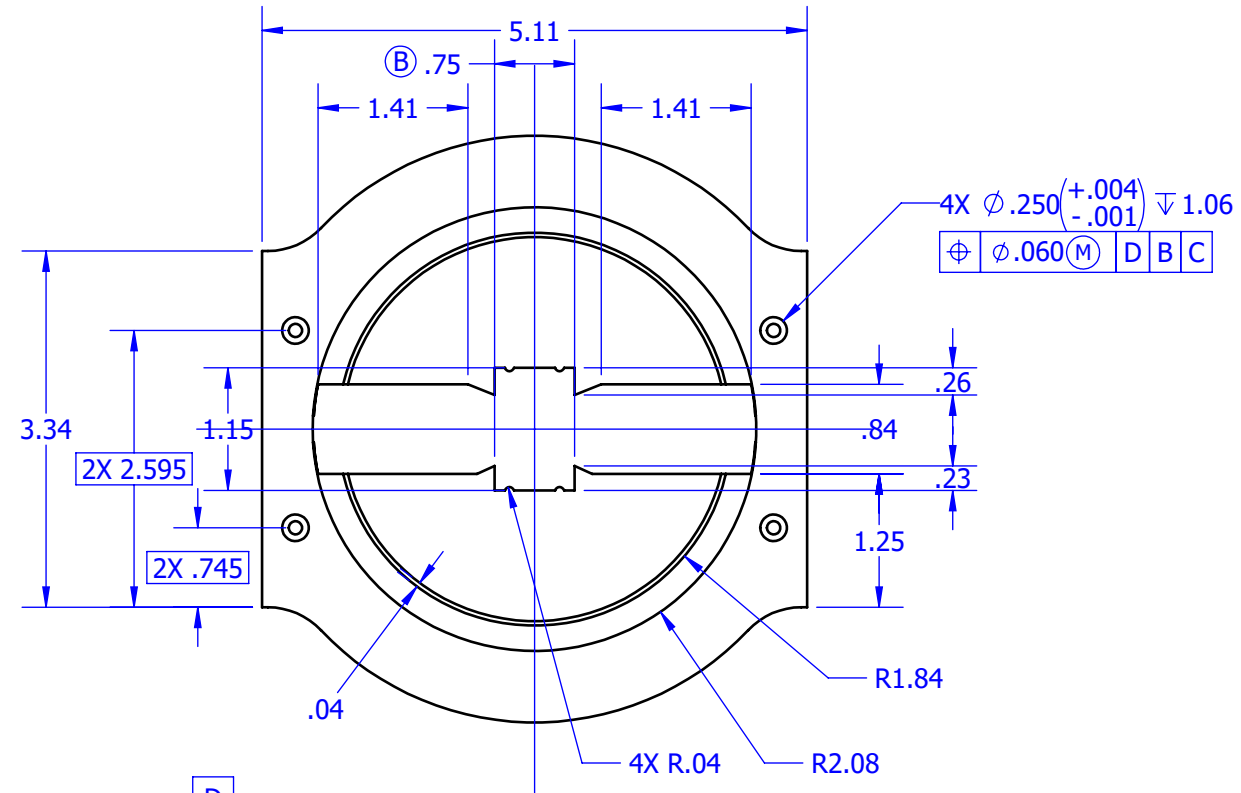


TOLERANCES UNLESS NOTED: X.X ± 0.01 X.XX ± 0.001 X.XXX ± 0.005			
UNITS: INCHES X.X" ± 1°		DESCRIPTION MAIN BODY, WHITE	
MATERIAL ABS PLASTIC		FINISH INJECTION MOLDED	
PN 1		REV A	
			
SHEET 3 of 3			
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			



NOTES:

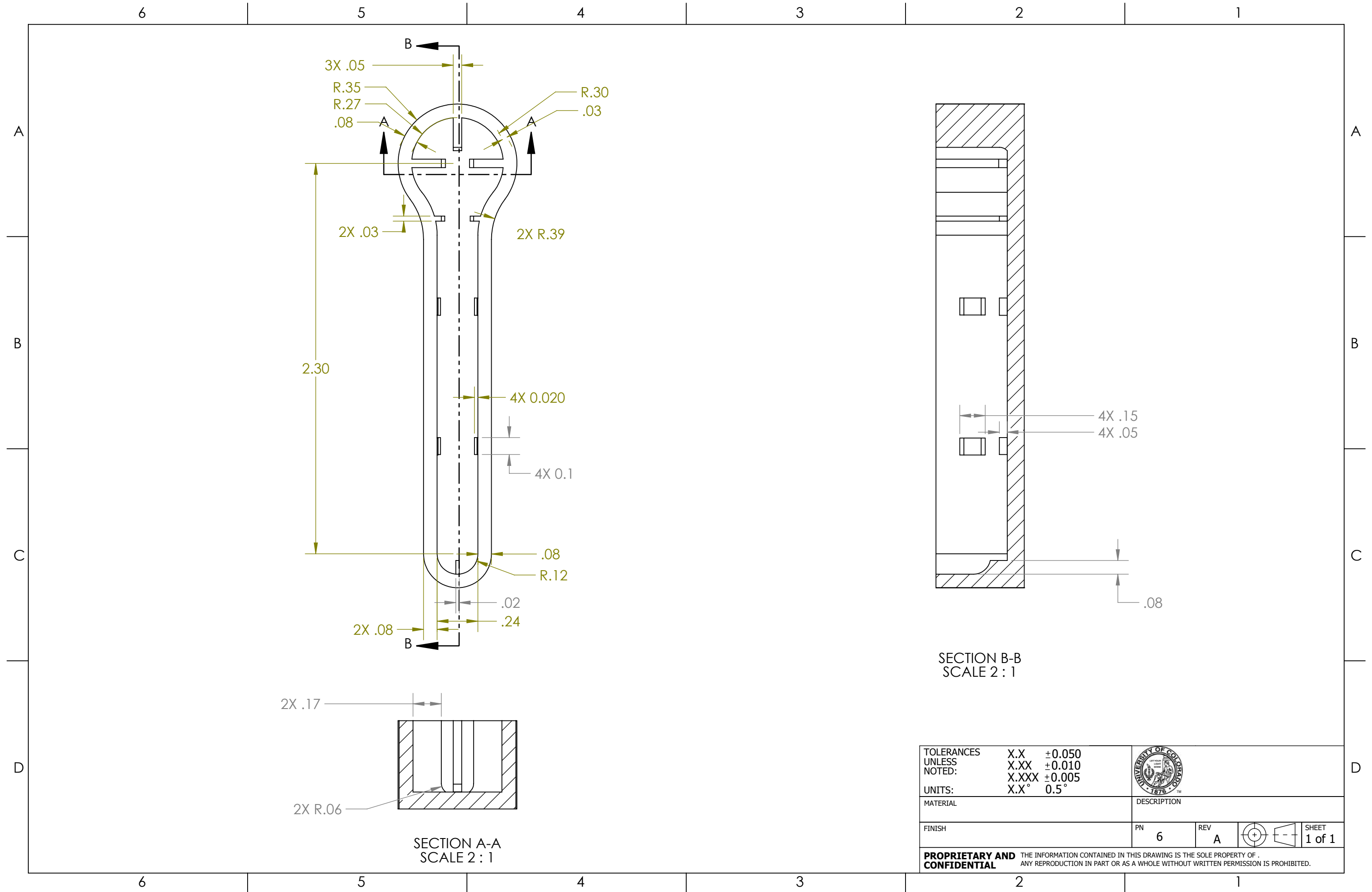
1. FOR ANY UNDIMENSIONED FEATURE, PLEASE REFER TO THE SOLID MODEL 4. SUCTION BASE WHITE V1. (B)

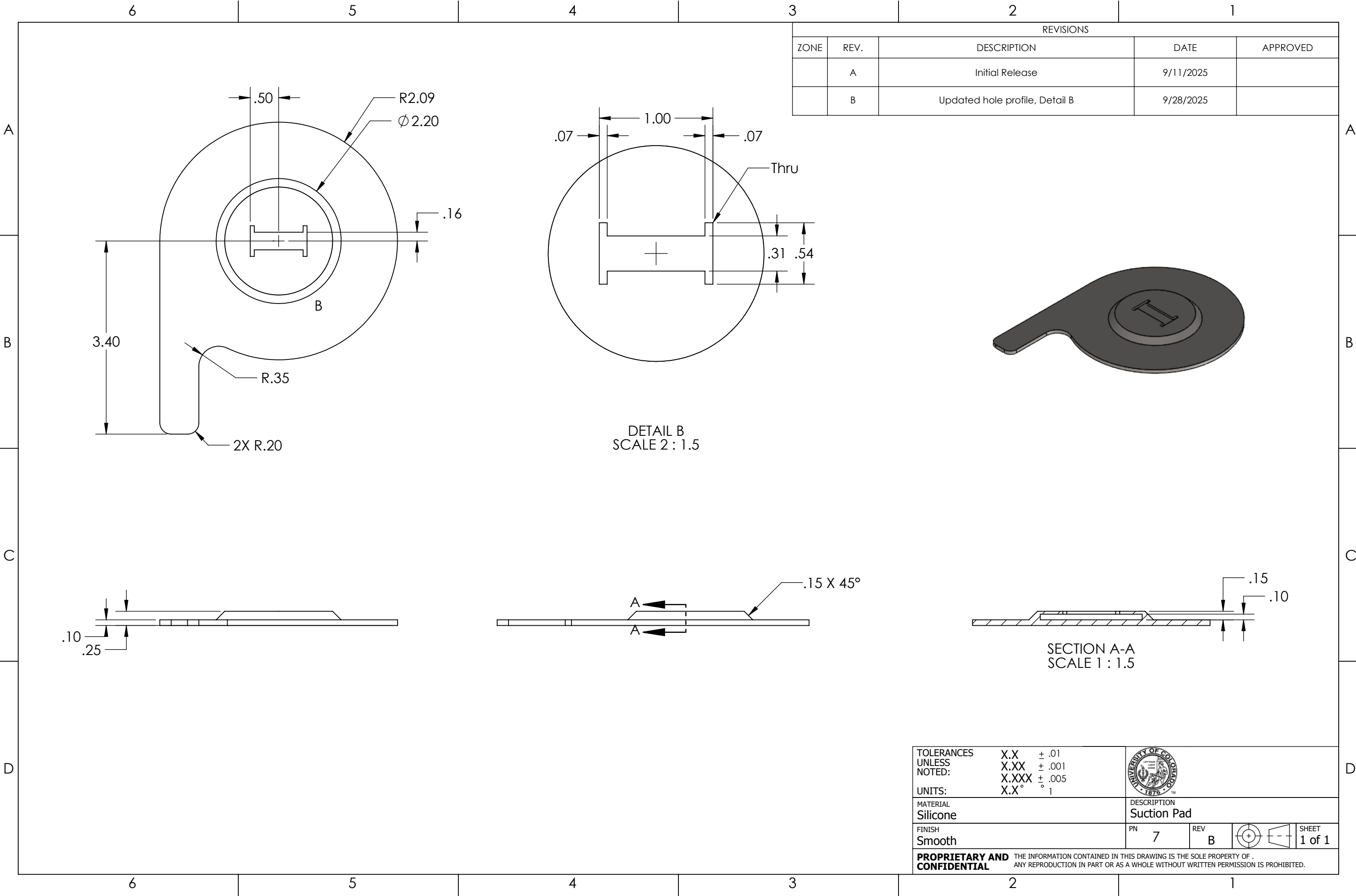


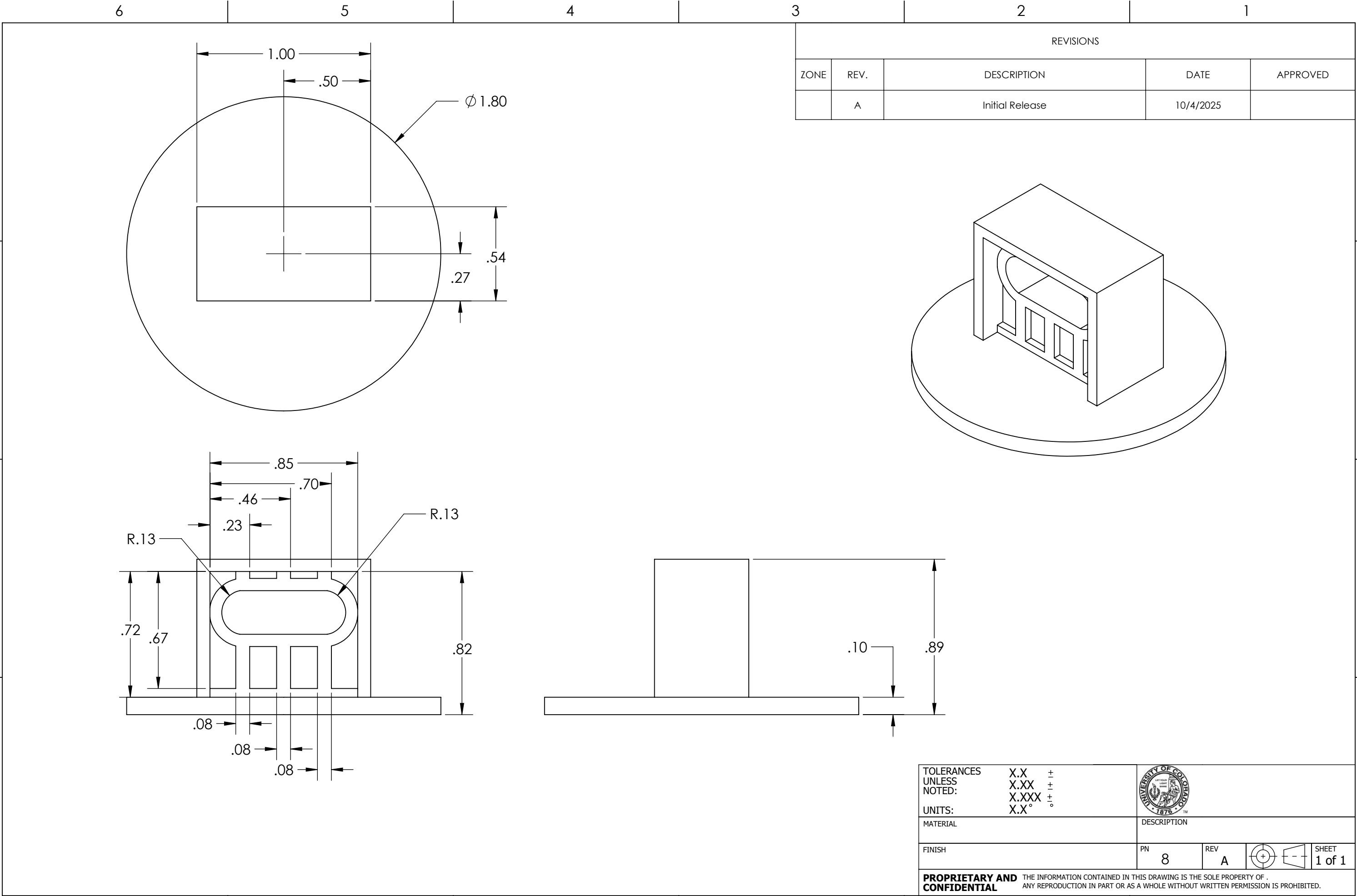
ISOMETRIC VIEW
FOR REFERENCE ONLY

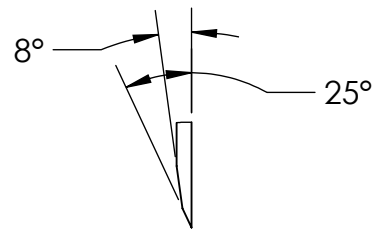
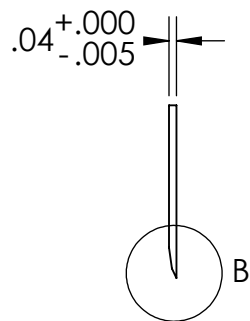
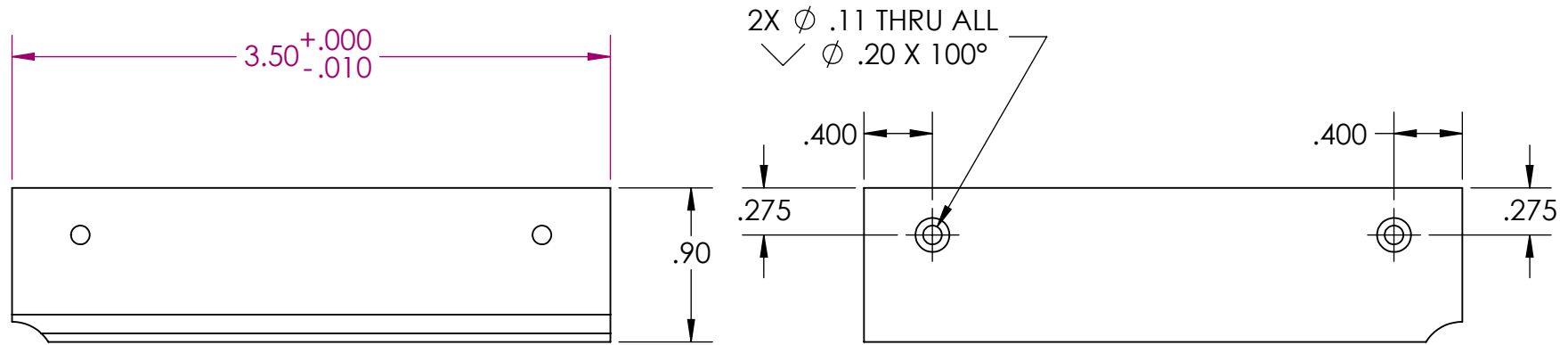
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	9/4/2025
B	ADJUSTED CROSSING LINES, DATUM C, TOLERANCE ON INNER ANGLE, ADDED SOLID MODEL NAME AND REV	10/5/2025

SPONSOR		dc DESIGN CENTER COLORADO	
PROJECT		FILE NAME	4. Suction Base (White) Screw Secured
TEAM NUMBER		APPROVAL	
TOLERANCES UNLESS NOTED:	X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005		
UNITS: INCHES	X.X° ± 0.5		
DRAWN BY	J. LAFYATIS	DRAWN DATE	9-4-25
MATERIAL	ABS PLASTIC	MFD BY	
STOCK SIZE	N/A	DESCRIPTION	NOODLER BASE FRAME
PROPRIETARY AND CONFIDENTIAL		PN	1
		REV	B
		SHEET	1 of 1

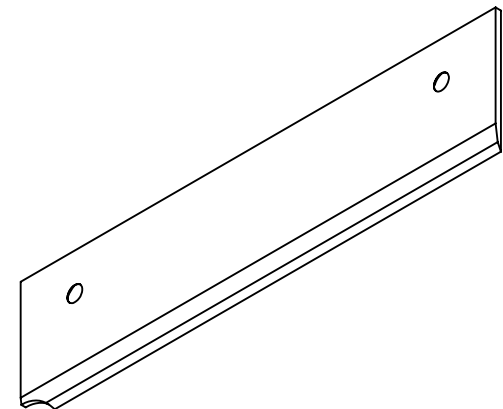








DETAIL B
SCALE 2 : 1

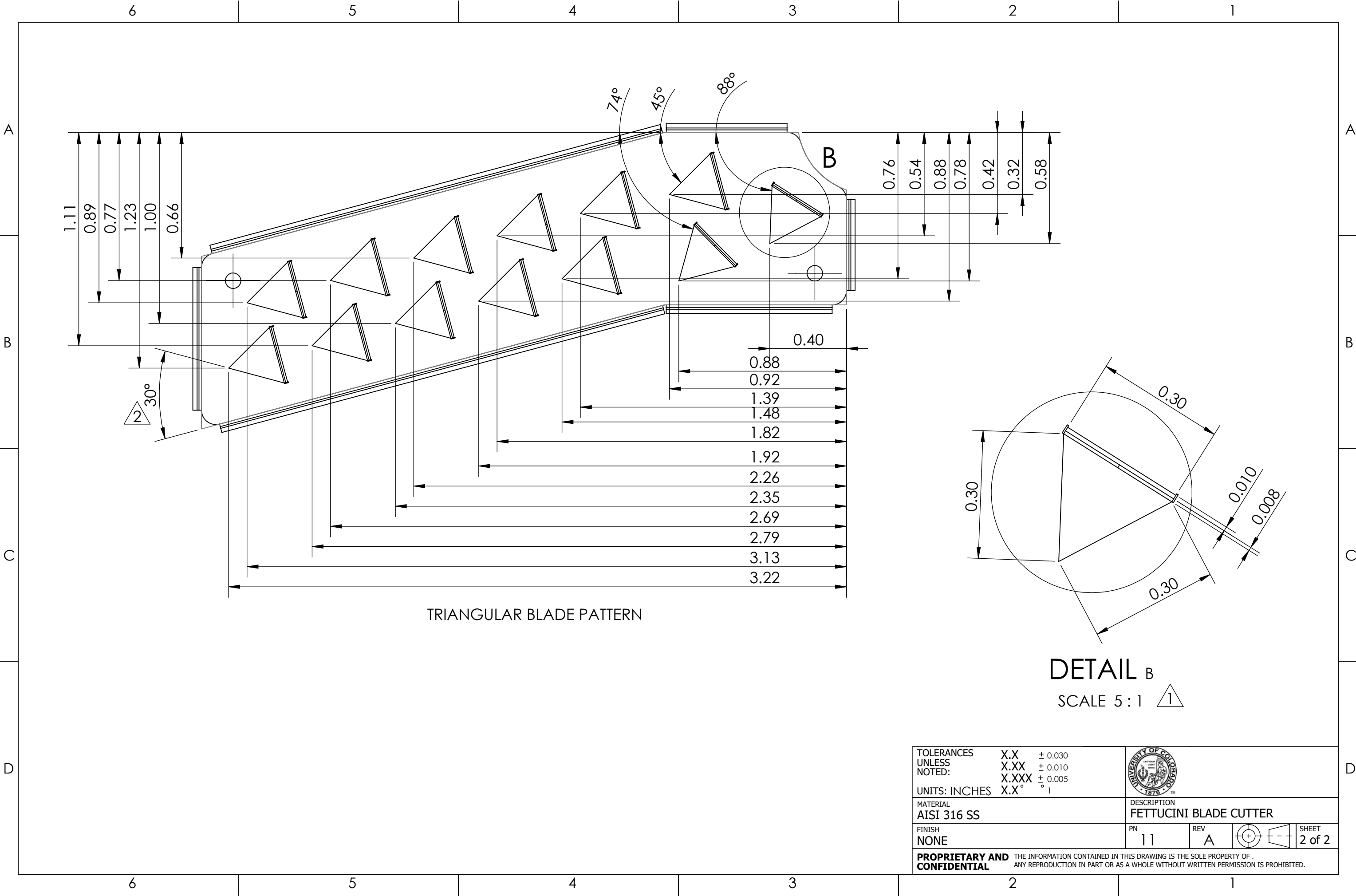


TOLERANCES UNLESS NOTED:		X.X ± .01 X.XX ± .001 X.XXX ± .005 X.X° ± 1			DESCRIPTION Spiralizer Blade			
UNITS:INCHES								
MATERIAL		STAINLESS STEEL, 20 GAUGE						
FINISH		BRUSHED						
		PN	10	REV	A			SHEET 1 of 1
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.						

- 1 ALL TRIANGULAR BLADES ARE OF EQUAL SIZE AND SHAPE AS DEFINED IN SECTION VIEW A AND DETAIL VIEW A.
- 2 ASSUME THIS ANGLE IN TRIANGULAR BLADE PATTERN VIEW UNLESS OTHERWISE SPECIFIED.



TOLERANCES UNLESS NOTED: UNITS: INCHES		X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005 X.X° ± .1			
MATERIAL AISI 316 SS		DESCRIPTION FETTUCINI BLADE CUTTER			
FINISH NONE		PN 11	REV A		SHEET 1 of 2
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			



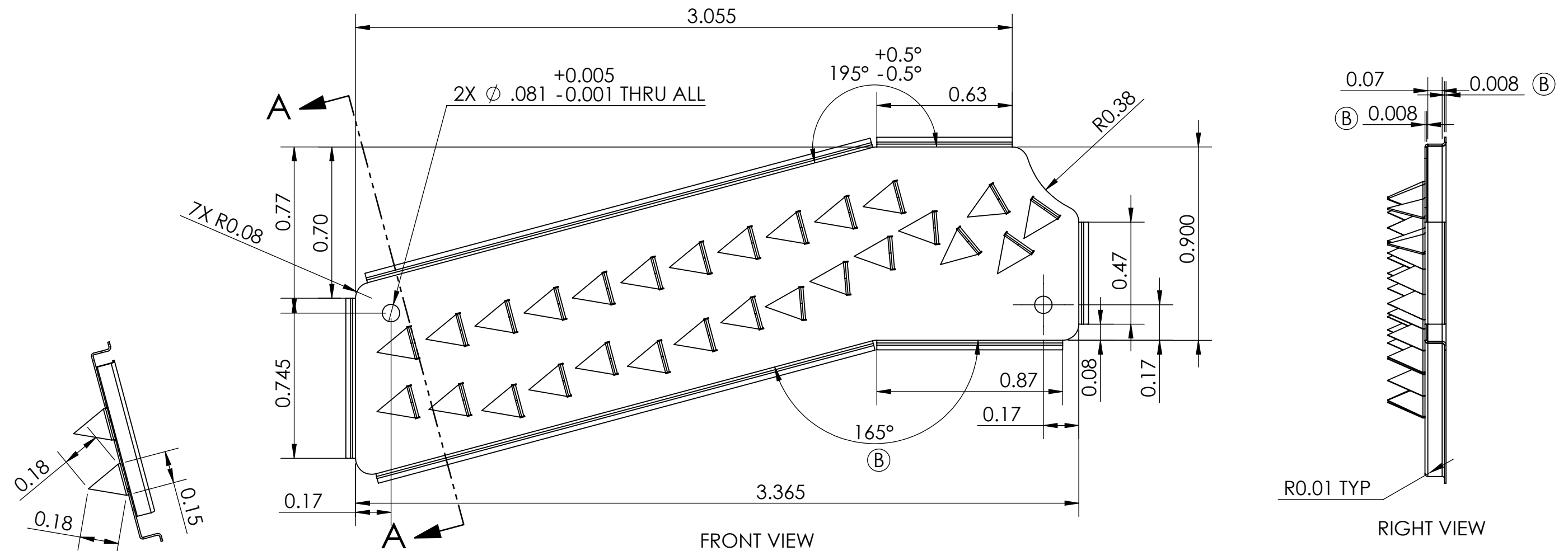
TOLERANCES UNLESS NOTED:		X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005 X.X° 1		
UNITS: INCHES				
MATERIAL AISI 316 SS		DESCRIPTION FETTUCINI BLADE CUTTER		
FINISH NONE		PN 11	REV A	 SHEET 2 of 2
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.				

NOTES:

- 1 ALL TRIANGULAR BLADES ARE OF EQUAL SIZE AND SHAPE AS DEFINED IN SECTION VIEW A AND DETAIL VIEW A.
- 2 ASSUME THIS ANGLE IN TRIANGULAR BLADE PATTERN VIEW UNLESS OTHERWISE SPECIFIED

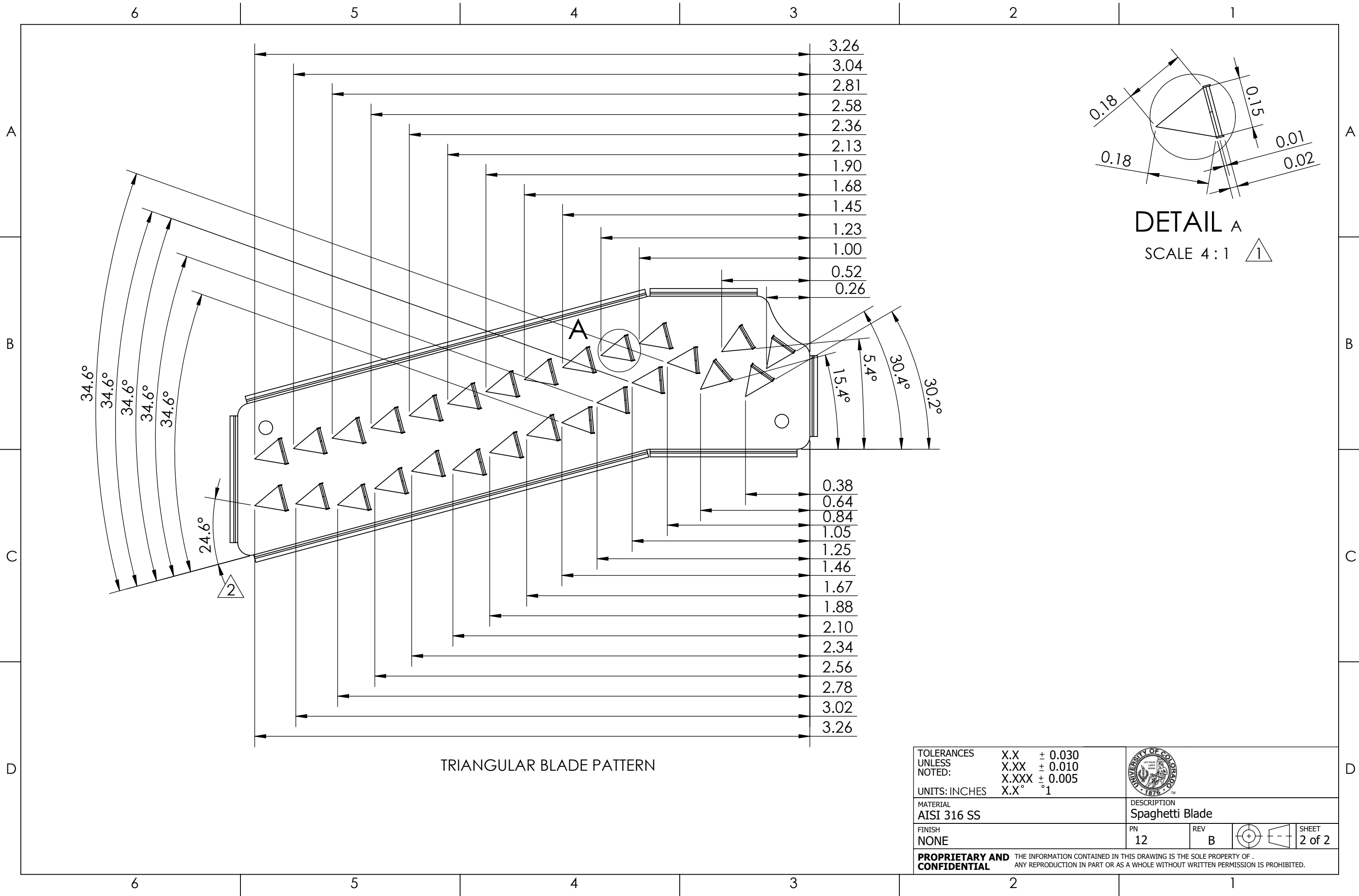
REVISIONS

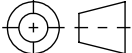
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	9/7/2025
B	UPDATED ANGLE TOLERANCE, FIXED IMPLIED TOLERANCES	10/5/2025



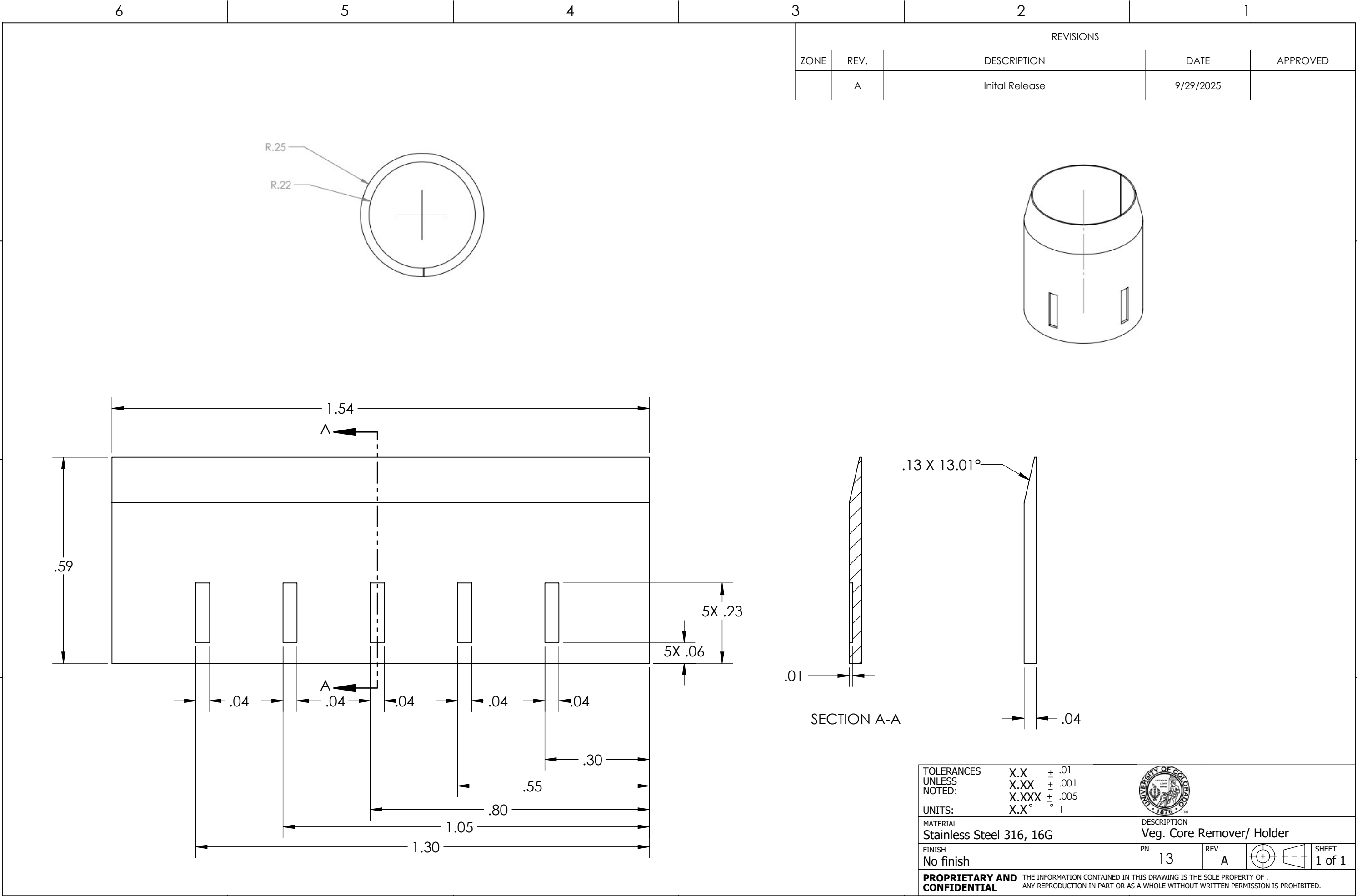
SECTION A-A 1

TOLERANCES UNLESS NOTED:	X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005 X.X° ± 1°	UNIVERSITY OF COLORADO 1876
UNITS: INCHES		
MATERIAL AISI 316 SS	DESCRIPTION Spaghetti Blade	
FINISH NONE	PN 12	REV B
PROPRIETARY AND CONFIDENTIAL		SHEET 1 of 2
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.		



TOLERANCES UNLESS NOTED:		X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005		
UNITS: INCHES		X.X° ± 1		
MATERIAL AISI 316 SS			DESCRIPTION Spaghetti Blade	
FINISH NONE			PN 12	REV B
PROPRIETARY AND CONFIDENTIAL				
			SHEET 2 of 2	

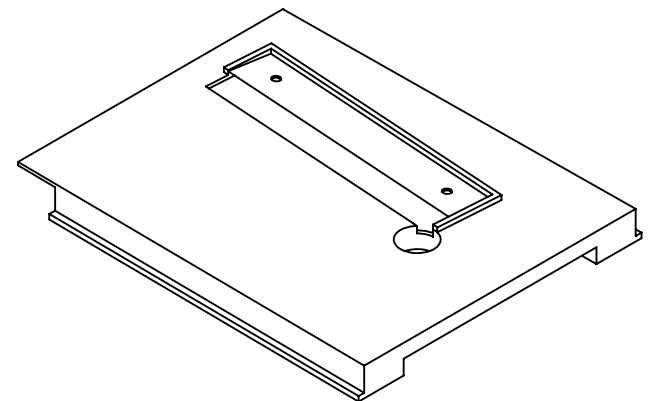
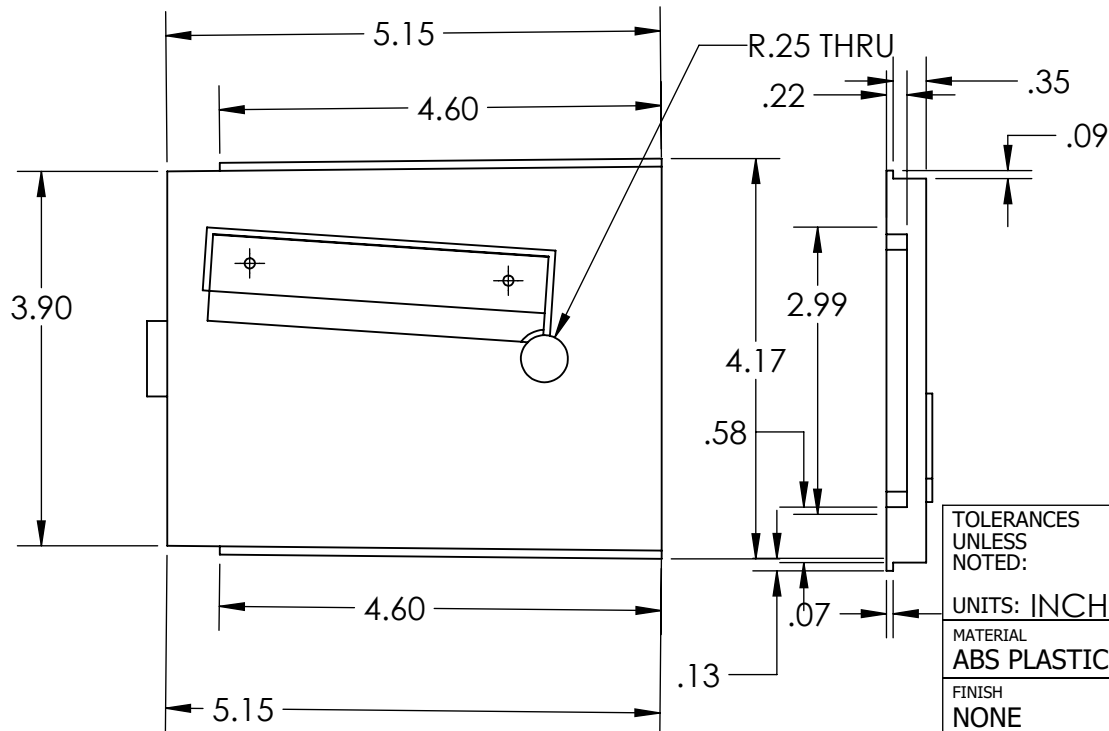
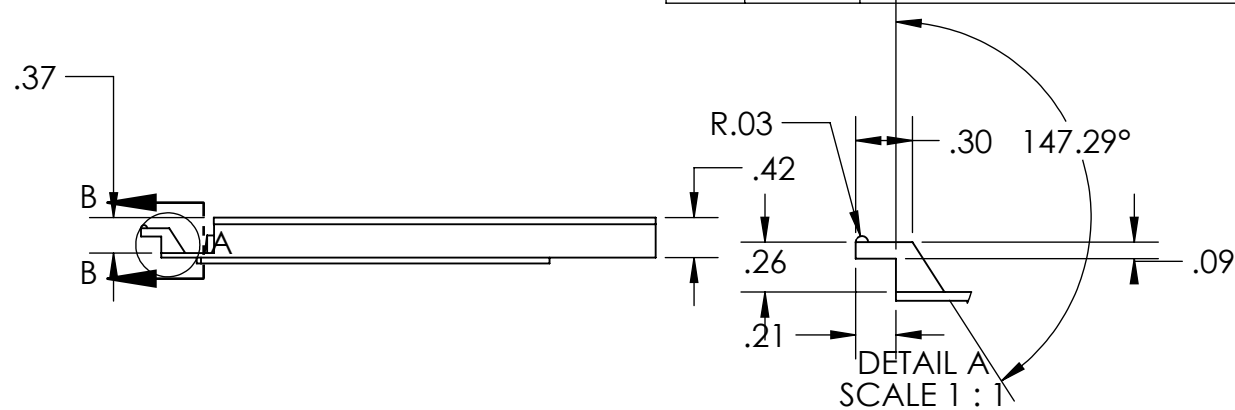
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF .
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REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	Initial Release	9/29/2025	

TOLERANCES UNLESS NOTED: UNITS:	X.X ± .01 X.XX ± .001 X.XXX ± .005 X.X° 1				
MATERIAL Stainless Steel 316, 16G	DESCRIPTION Veg. Core Remover/ Holder				
FINISH No finish	PN 13	REV A		SHEET 1 of 1	
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.					

REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL RELEASE	10/5/2025	KAI



TOLERANCES
UNLESS
NOTED:
X.X ± .030
X.XX ± .010
X.XXX ± .005
UNITS: INCHES X.X° .5



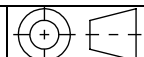
MATERIAL
ABS PLASTIC

DESCRIPTION
BLADE HOLDER

FINISH
NONE

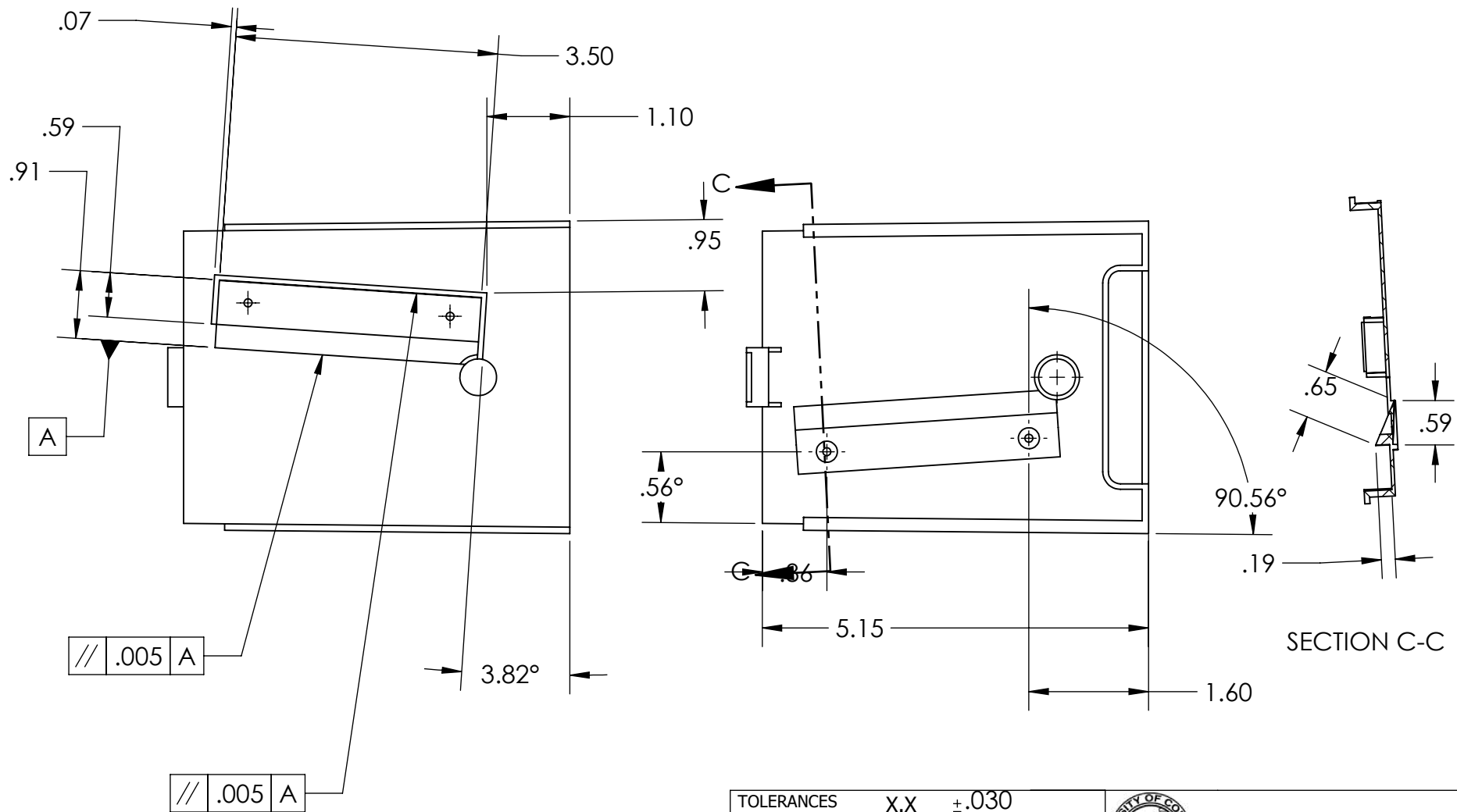
PN
14

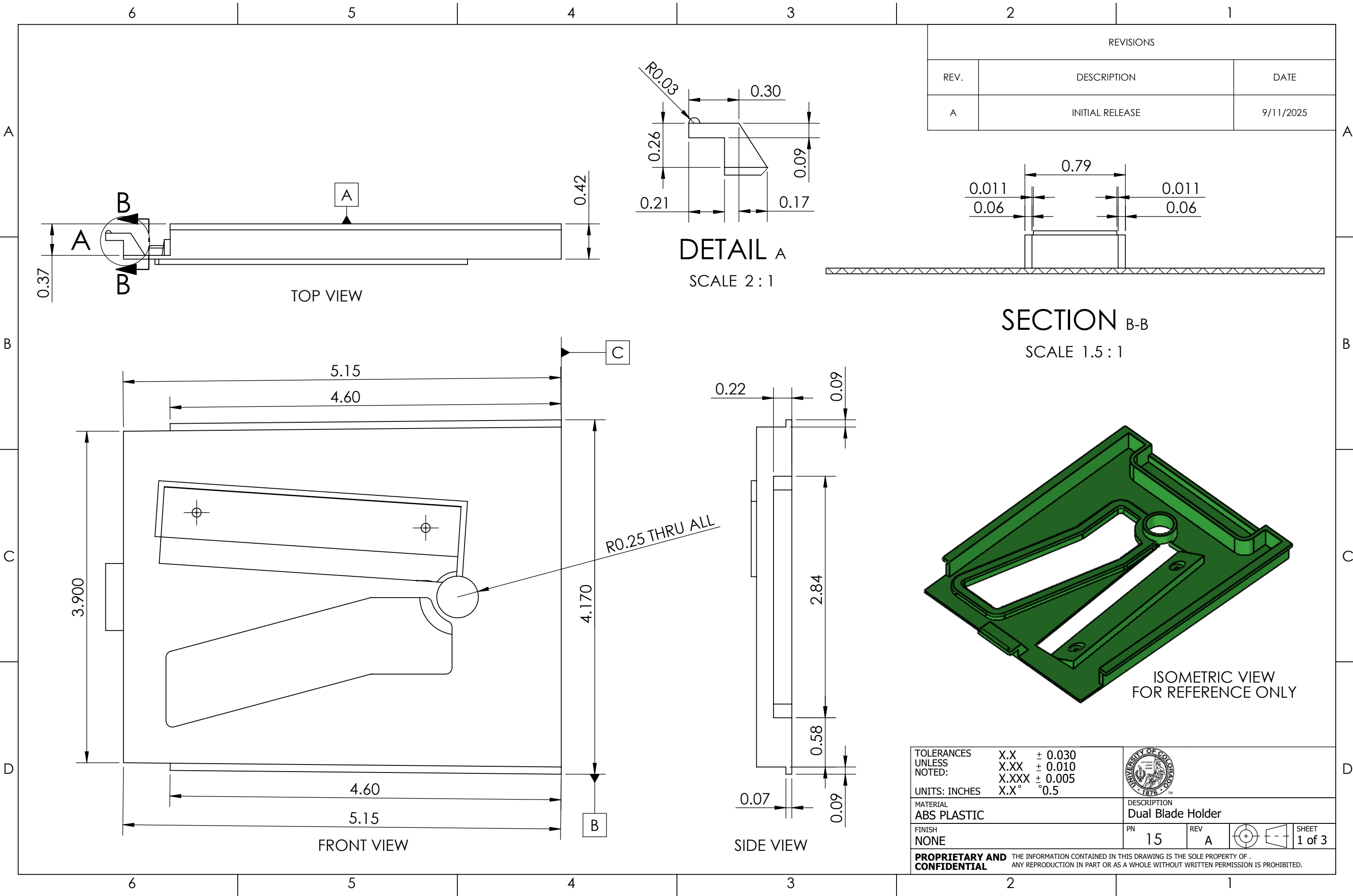
REV
A



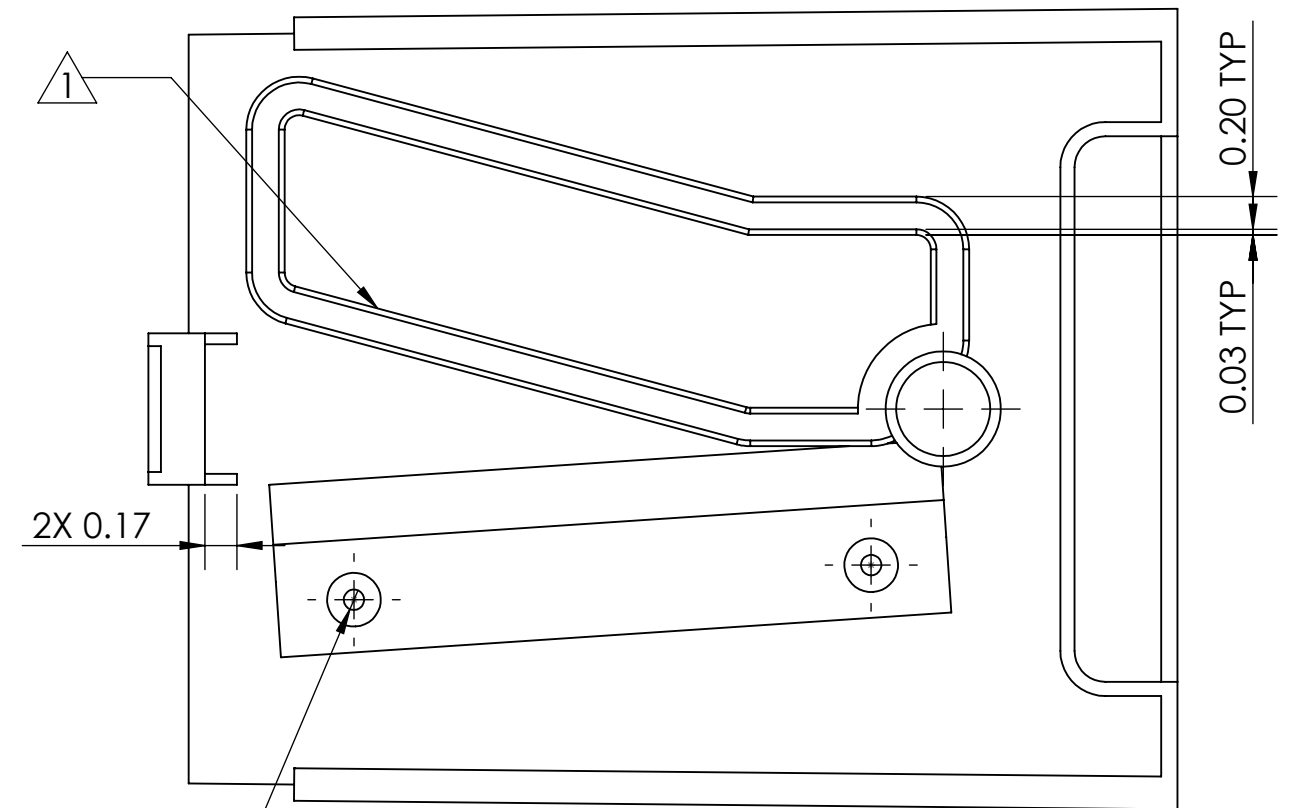
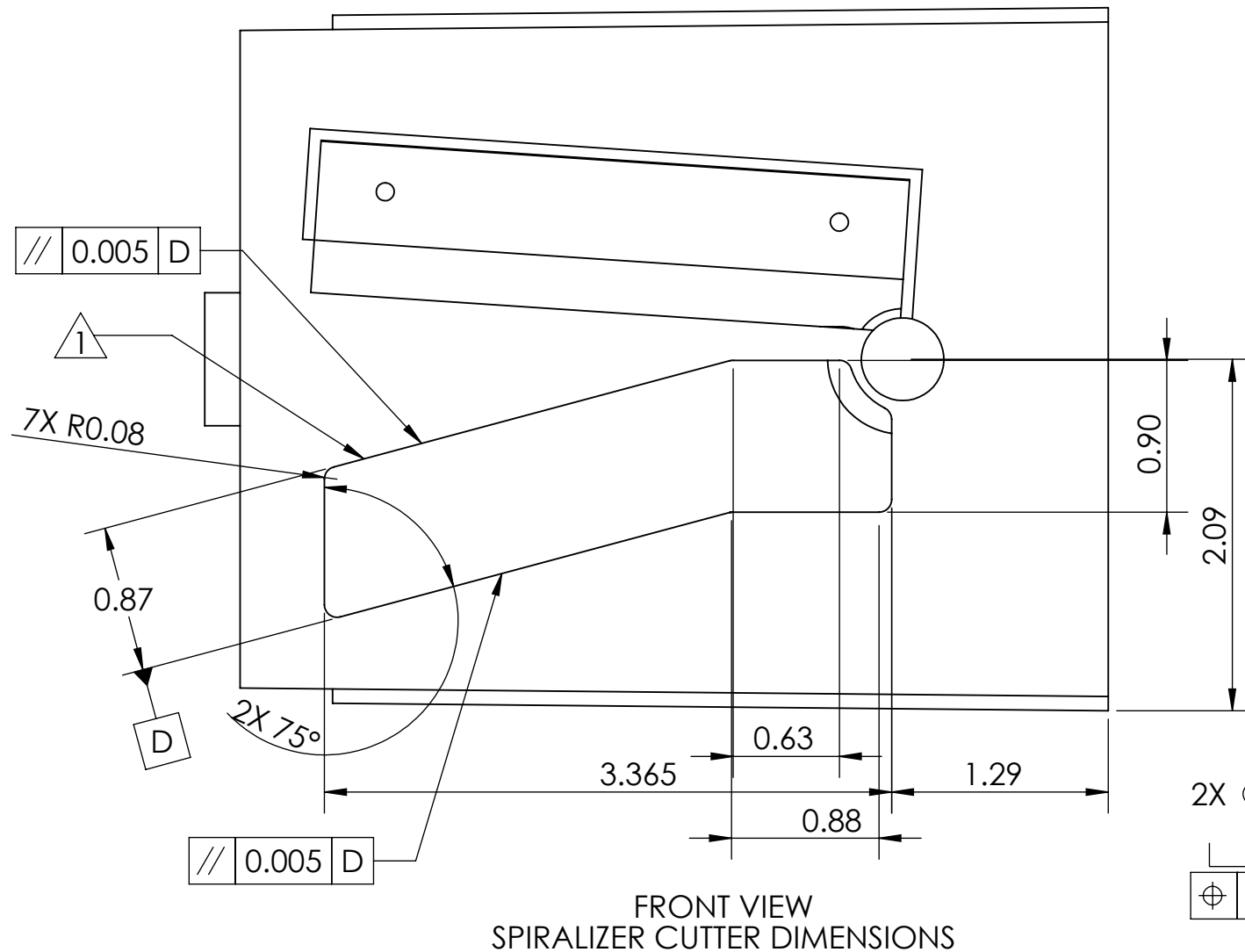
SHEET
1 of 2

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

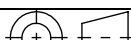


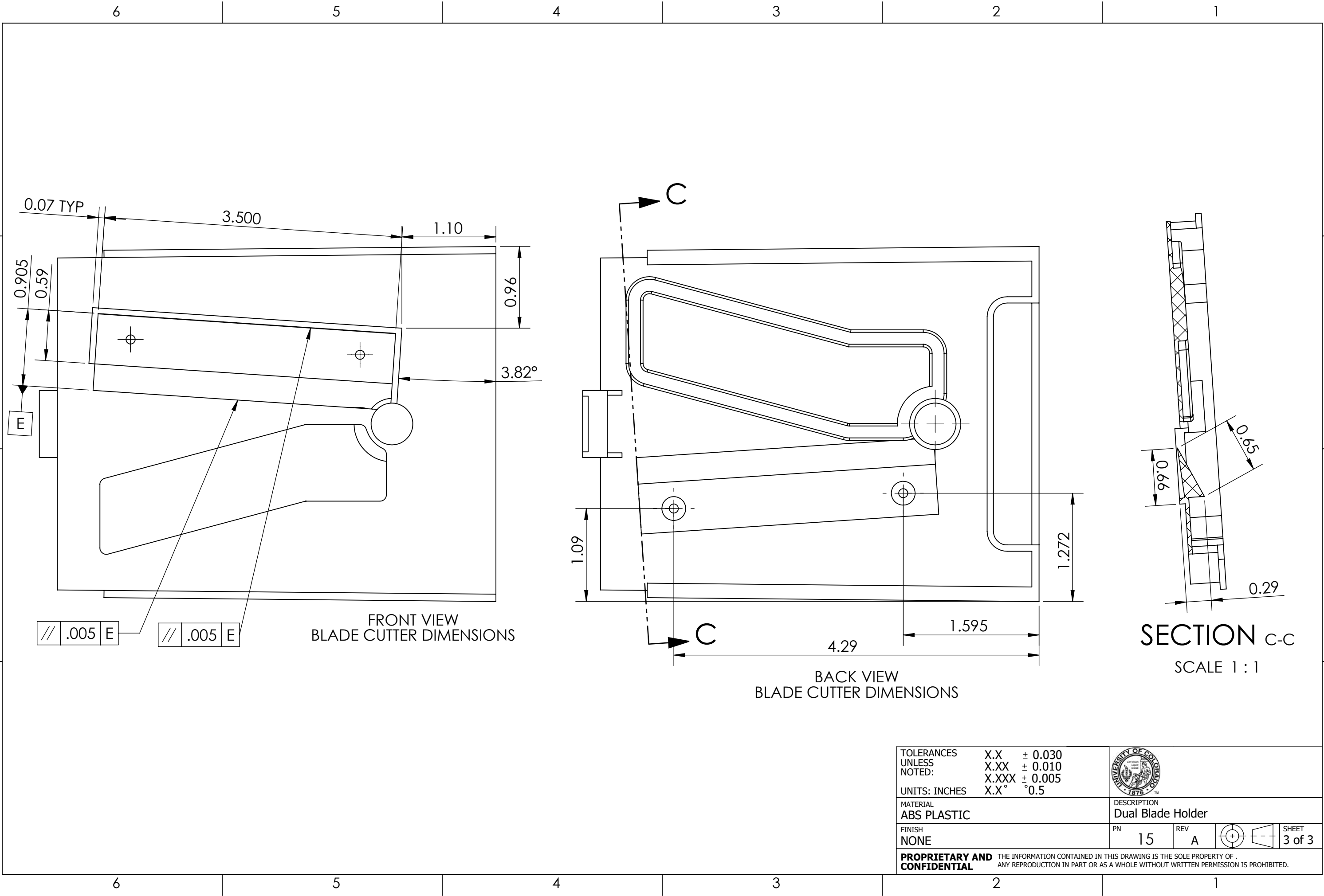


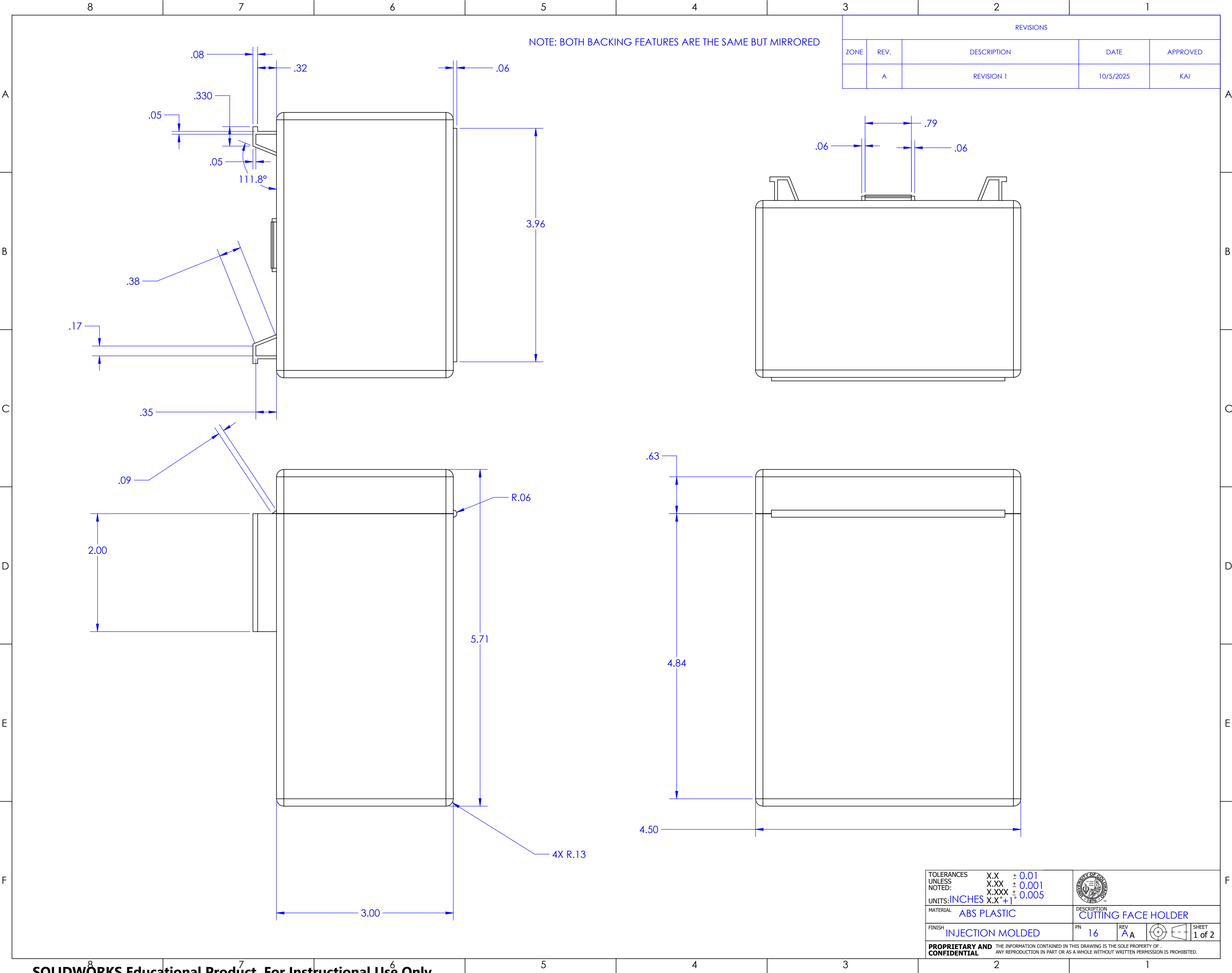
1 DIMENSIONS OF THE SPRIALIZER CUTTER IN THE FRONT VIEW MATCH THE INNER DIMENSIONS OF THE SPRIALIZER CUTTER IN THE BACK VIEW

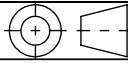


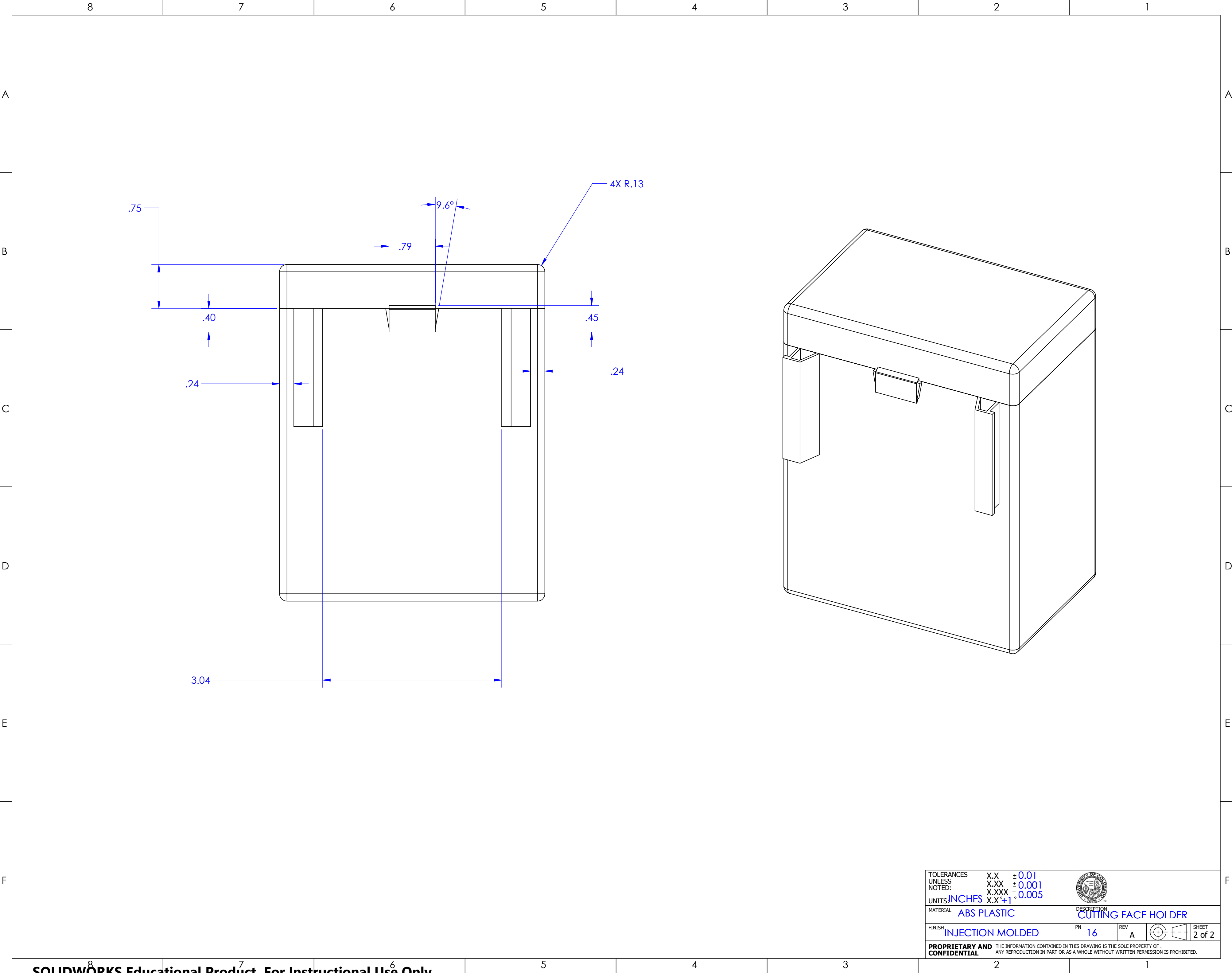
2X $\phi .107$ $\begin{matrix} +0.005 \\ -0.001 \end{matrix}$ THRU ALL
 $\begin{matrix} +0.005 \\ -0.001 \end{matrix}$ $\phi .280$ $\downarrow .23$
 $\oplus \phi .010$ (M) A B C

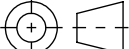
TOLERANCES UNLESS NOTED:		X.X ± 0.030 X.XX ± 0.010 X.XXX ± 0.005			
UNITS: INCHES		X.X° ± 0.5			
MATERIAL ABS PLASTIC			DESCRIPTION Dual Blade Holder		
FINISH NONE		PN 15	REV A		SHEET 2 of 3
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			

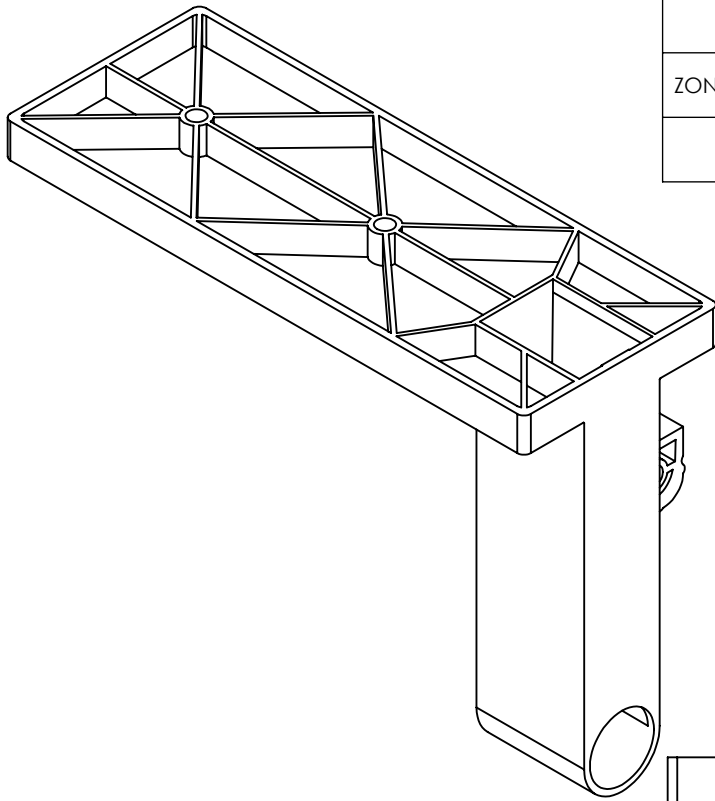




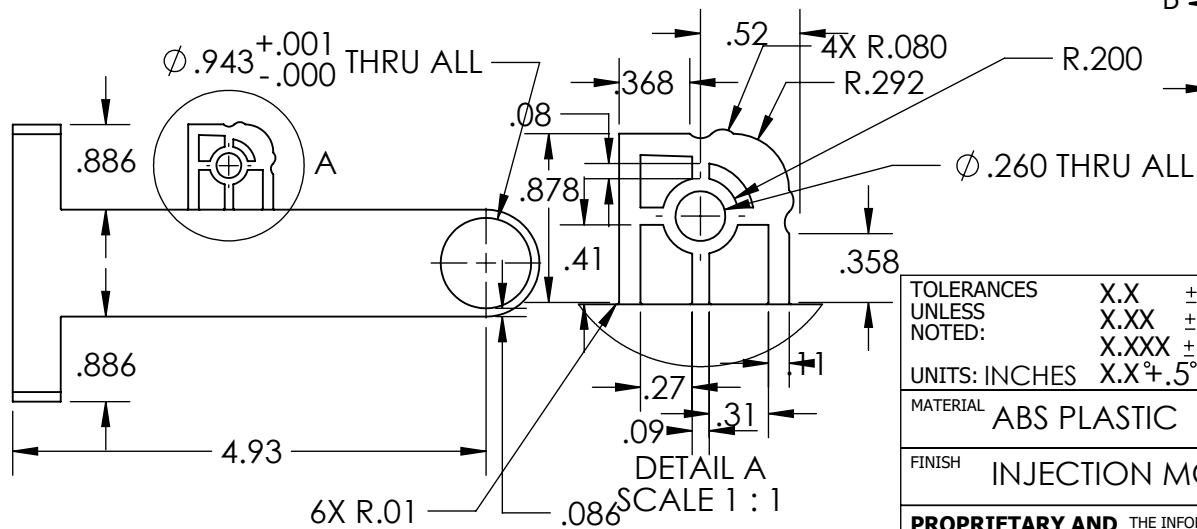
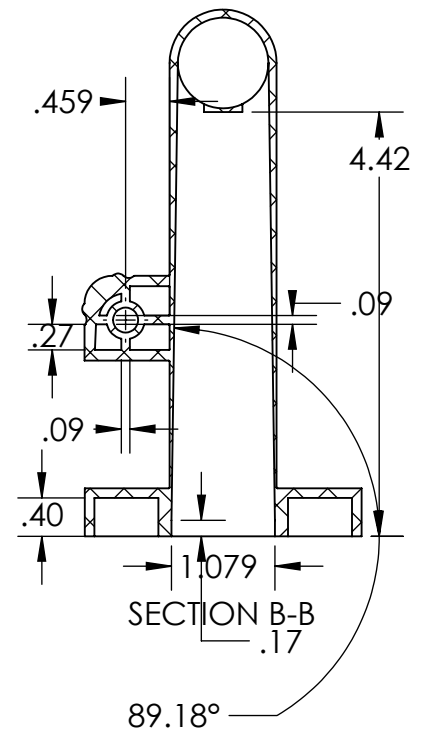
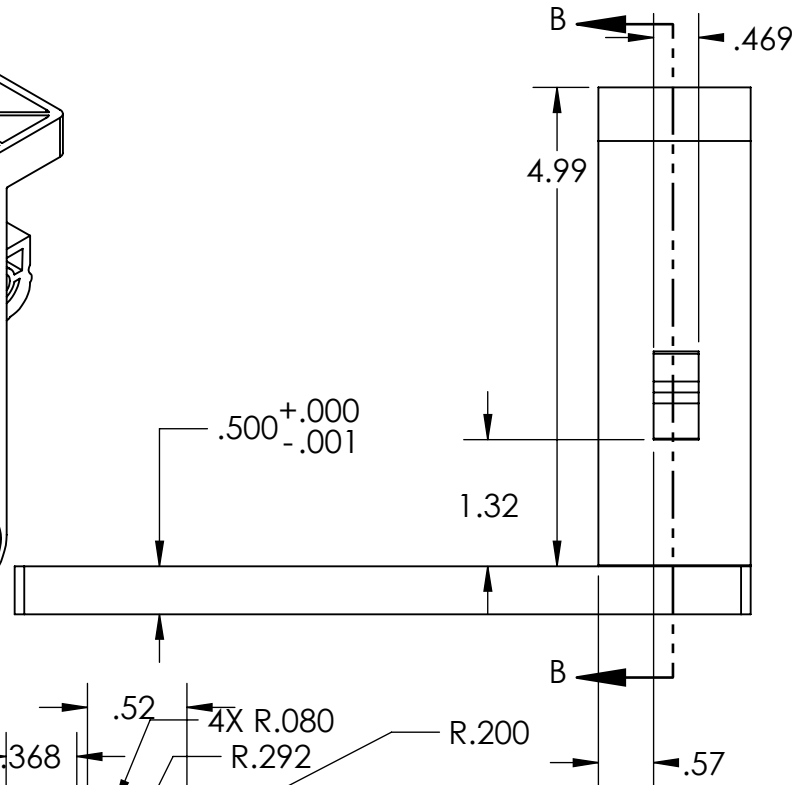
TOLERANCES UNLESS NOTED: X.X ± 0.01 X.XX ± 0.001 X.XXX ± 0.005 UNITS: INCHES X.X" ± .01			
MATERIAL ABS PLASTIC		DESCRIPTION CUTTING FACE HOLDER	
FINISH INJECTION MOLDED	PN 16	REV A A	 SHEET 1 of 2
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			



TOLERANCES UNLESS NOTED: X.X ± 0.01 X.XX ± 0.001 X.XXX ± 0.005			
UNITS: INCHES	X.X ± .1		
MATERIAL: ABS PLASTIC		DESCRIPTION: CUTTING FACE HOLDER	
FINISH: INJECTION MOLDED	PN: 16	REV: A	 SHEET 2 of 2
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			



REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A		9/8/2025	



TOLERANCES
UNLESS
NOTED:
X.X ± 0.0
X.XX ± 0.01
X.XXX ± 0.005
UNITS: INCHES X.X° ± .5°



MATERIAL ABS PLASTIC

DESCRIPTION SLIDER BODY

FINISH INJECTION MOLDED

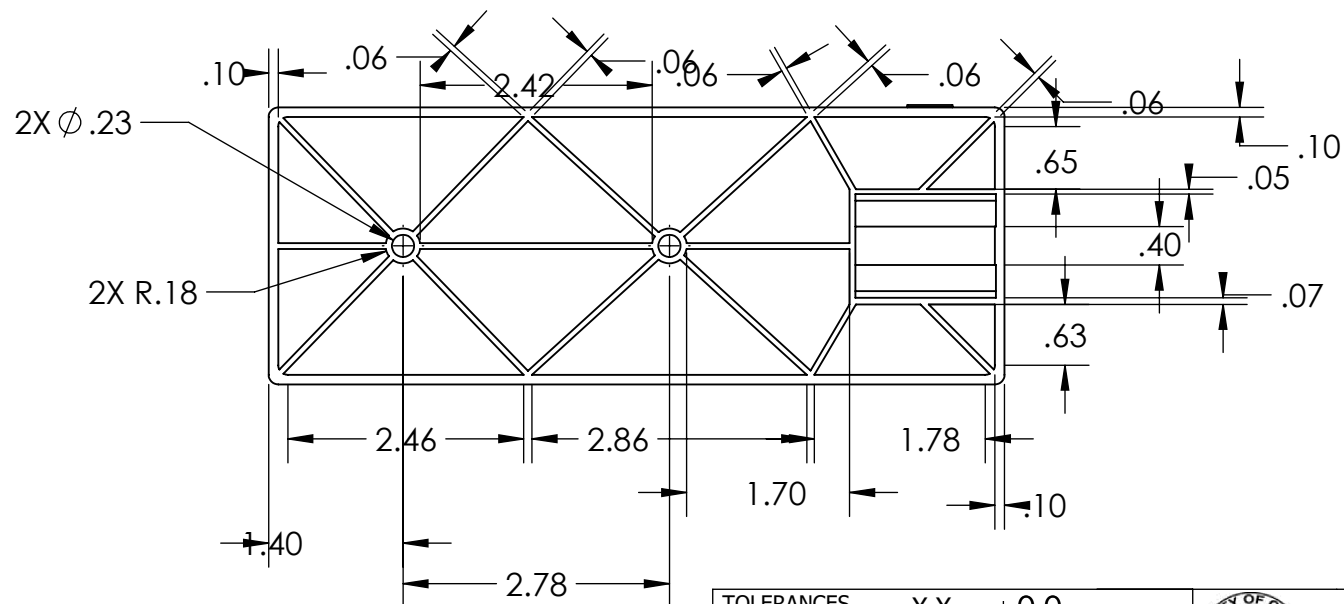
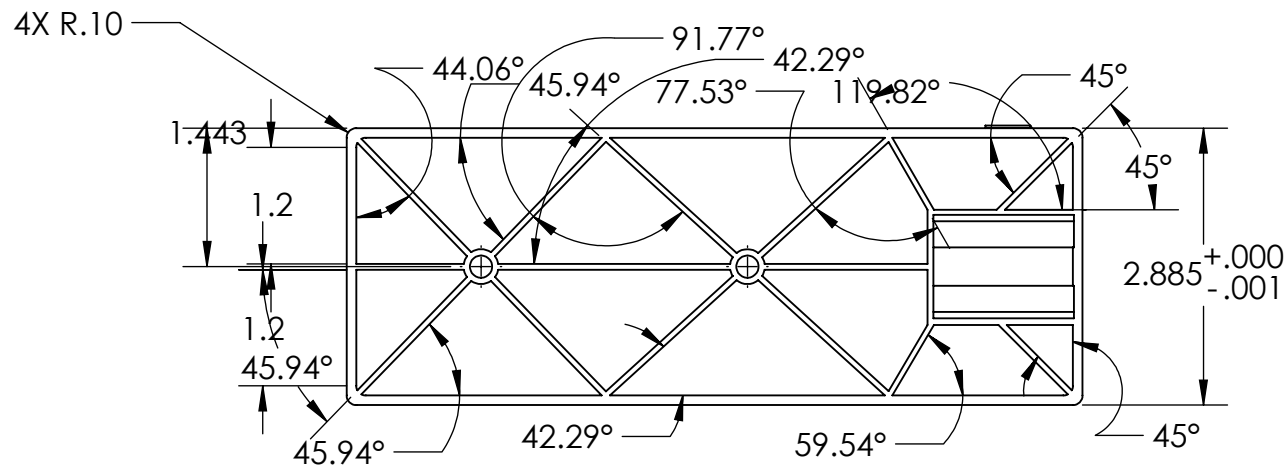
PN 17

REV A



SHEET 1 of 2

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.



TOLERANCES UNLESS NOTED: X.X ± 0.0 X.XX ± 0.01 X.XXX ± 0.005 UNITS: INCHES X.X°.5°			
MATERIAL ABS PLASTIC		DESCRIPTION SLIDER BODY	
FINISH INJECTION MOLDED		PN 17	REV A
			SHEET 2 of 2
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.			

6

5

4

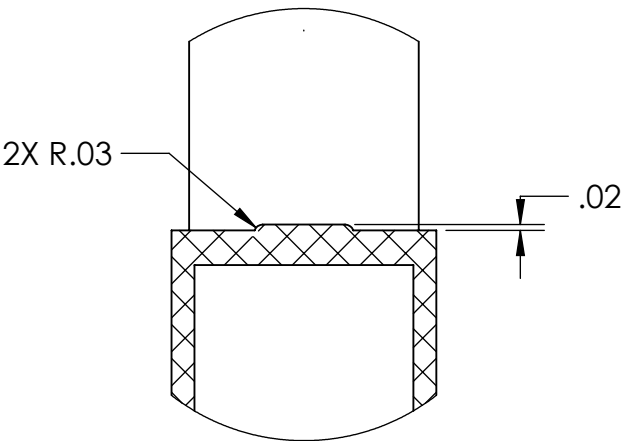
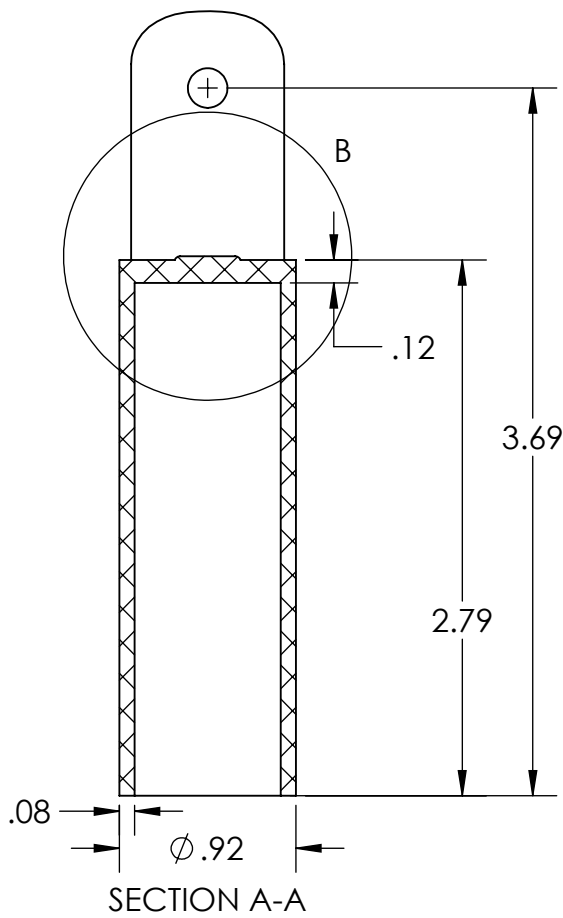
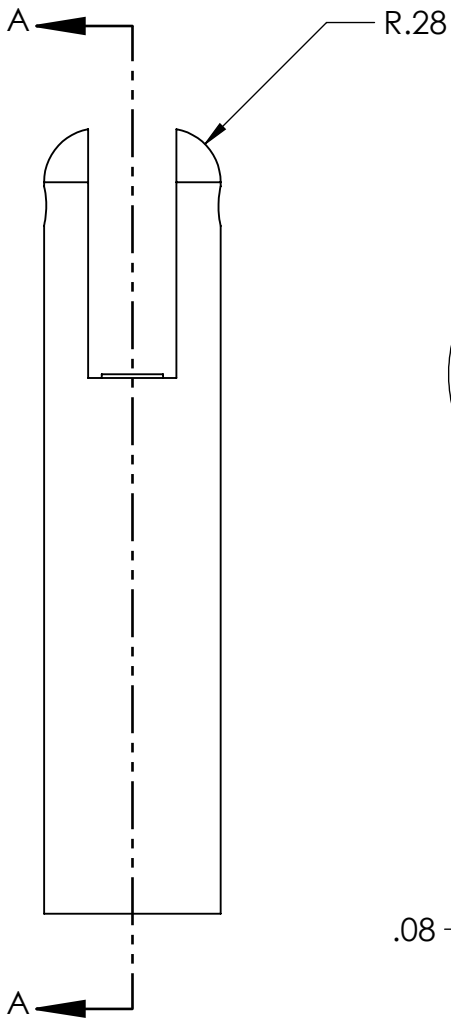
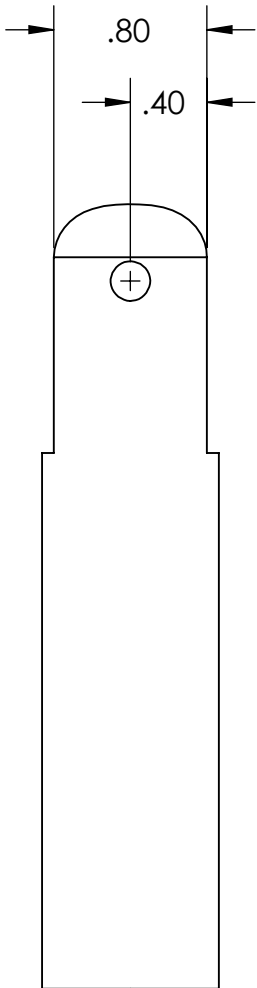
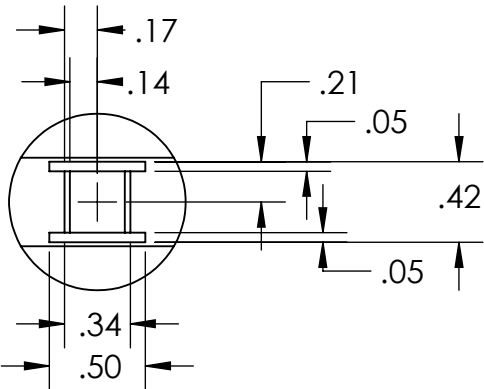
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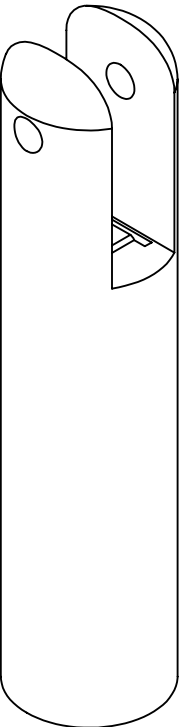
1

REVISIONS

ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	Initial Release	10/4/2025	
	B	Clarified Dimensions	10/4/2025	



DETAIL B
SCALE 1.5 : 1



TOLERANCES UNLESS NOTED:	X.X ± .01 X.XX ± .001 X.XXX ± .005				
UNITS:	X.X° ± 1				
MATERIAL ABS	DESCRIPTION Push Handle				
FINISH Smooth	PN 19	REV B		SHEET 1 of 1	
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6

5

4

3

2

1

6

5

4

3

2

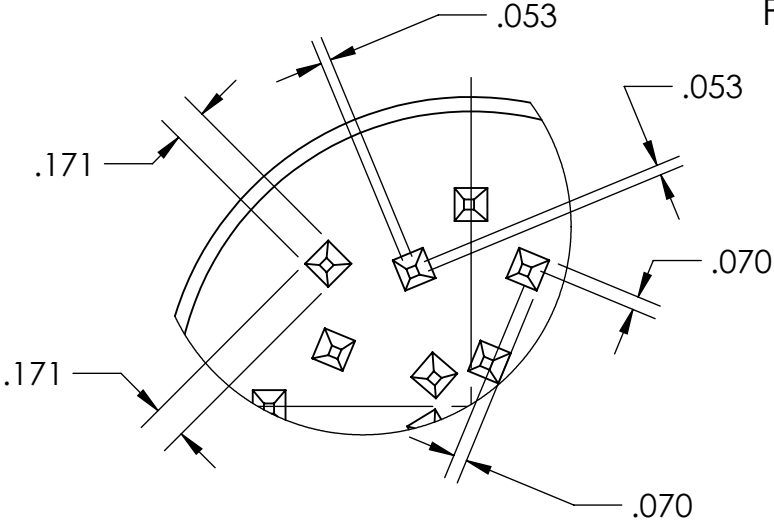
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REVISIONS

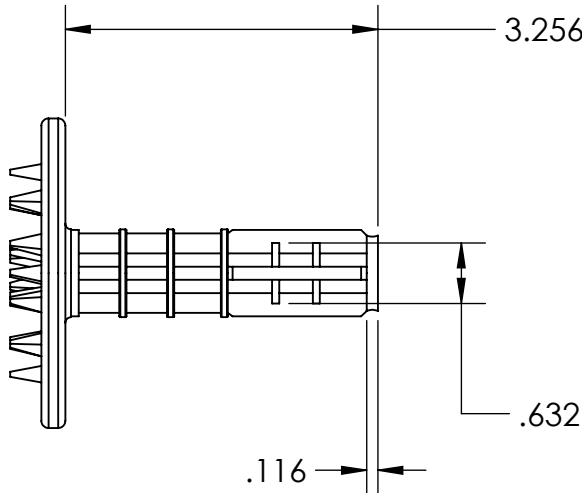
ZONE	REV.	DESCRIPTION	DATE	APPROVED
A4	A	INITIAL DESIGN WITH WRITTEN TOLERANCES	9/10/2025	KAI

NOTE:
X21 (8+8+5) IDENTICAL FEATURES

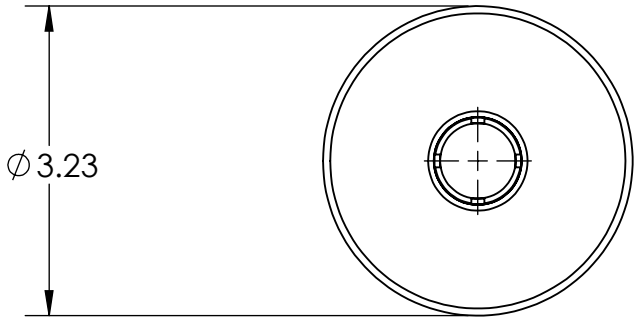
PATTERN HAS 3 LAYERS TO ROTATE VEGETABLES
OFF OF. THESE PATTERNS MUST BE PRECISE TO
PLACE IN X AND Y DIRECTION BY .001. OTHERWISE
FOOD COULD SLIP OUT.



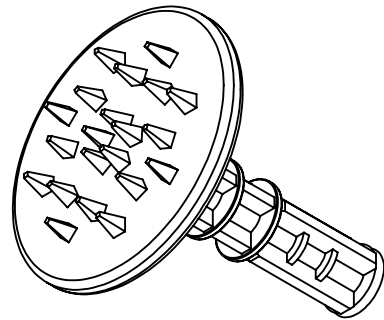
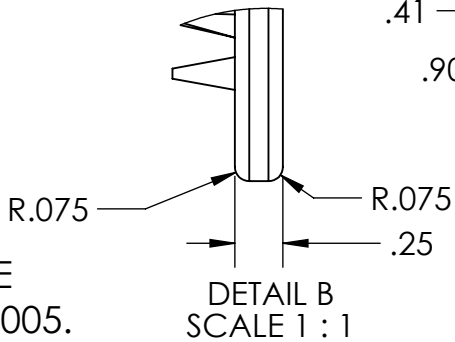
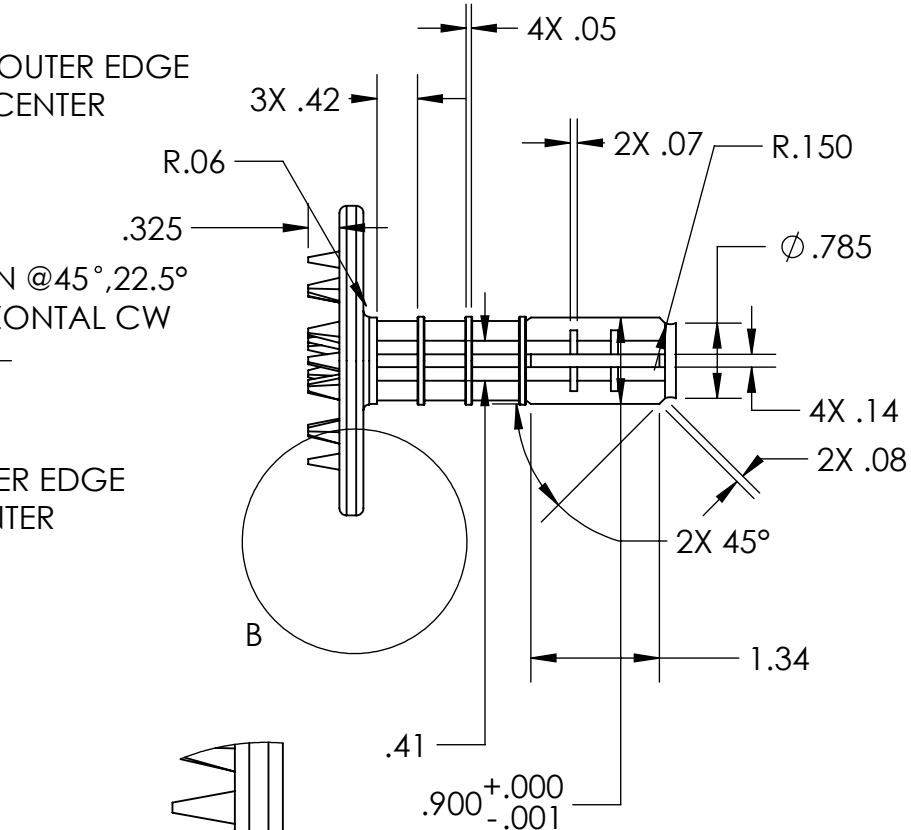
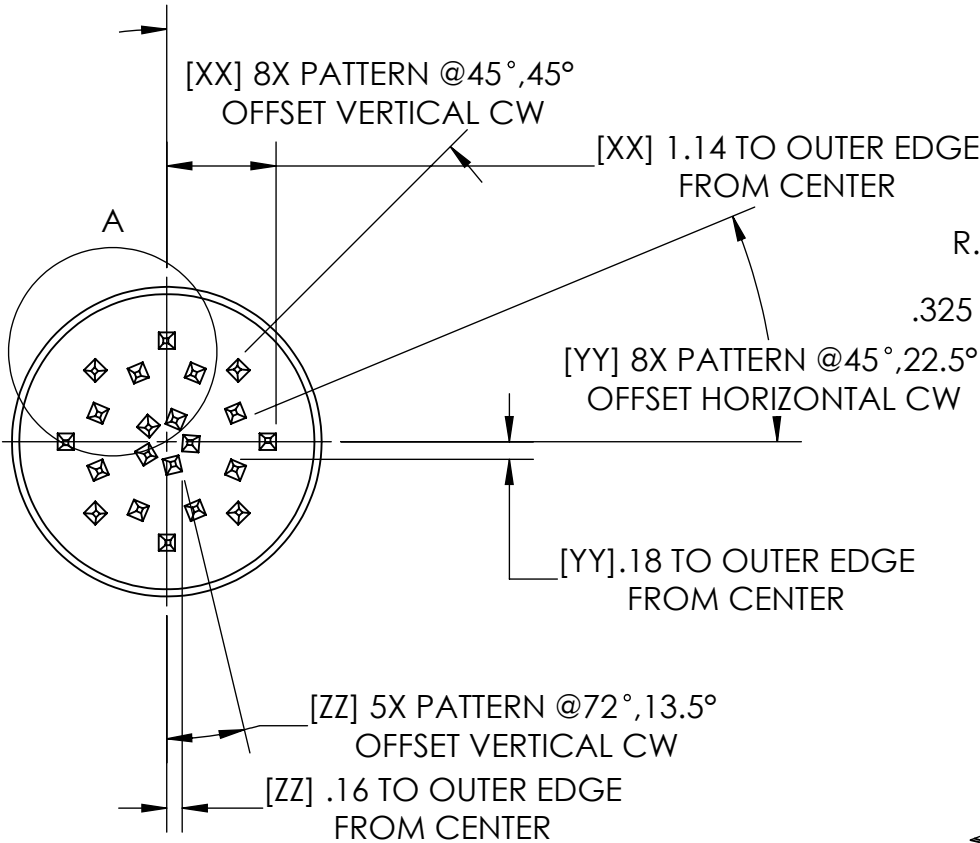
DETAIL A
SCALE 1 : 1



NOTE:
- CYLINDER TO BE PERPENDICULAR
TO THE FLAT FACE BY +/- .001. THIS IS
EXTREMELY IMPORTANT AS IT IS
NEEDED TO SPIN THE VEGETABLE IN
A PERFECT CIRCLE. THIS MUST ALSO
BE FLAT TO .001 SO IT SPINS
SMOOTHLY.



NOTE: CIRCLES MUST BE
CONCENTRIC TO +/- .0005.
OTHERWISE THIS WILL NOT
SPIN EVENLY AND WILL MAKE
THE MACHINE FAIL.



TOLERANCES
UNLESS
NOTED:
X.X ± 0.1
X.XX ± 0.005
X.XXX ± 0.0005
UNITS: INCHES X.X° .5°

MATERIAL
ABS PLASTIC

FINISH
INJECTION MOLDED

PROPRIETARY AND
CONFIDENTIAL



DESCRIPTION

PN 19

REV AA



SHEET 1 of 1

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6

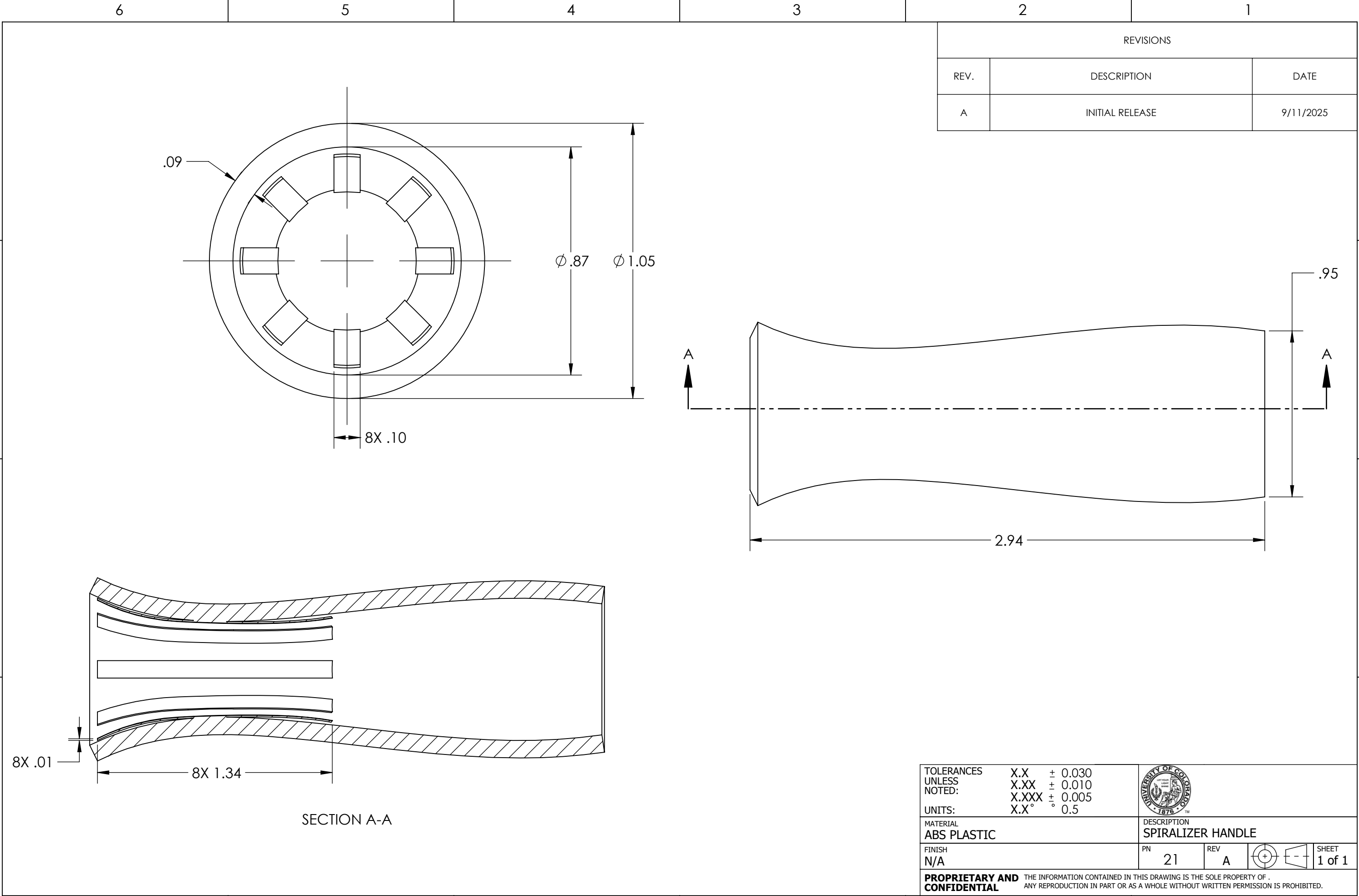
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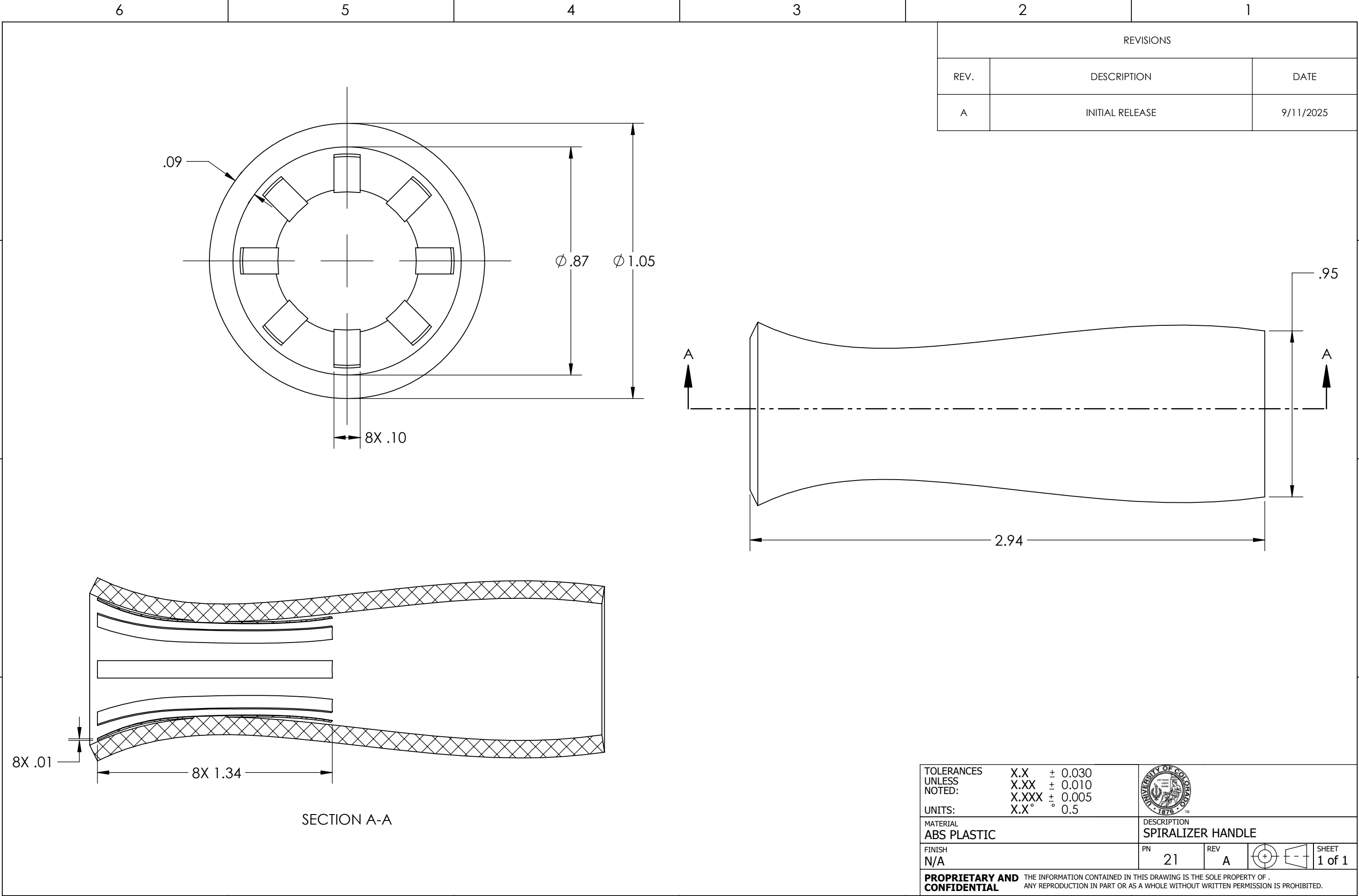
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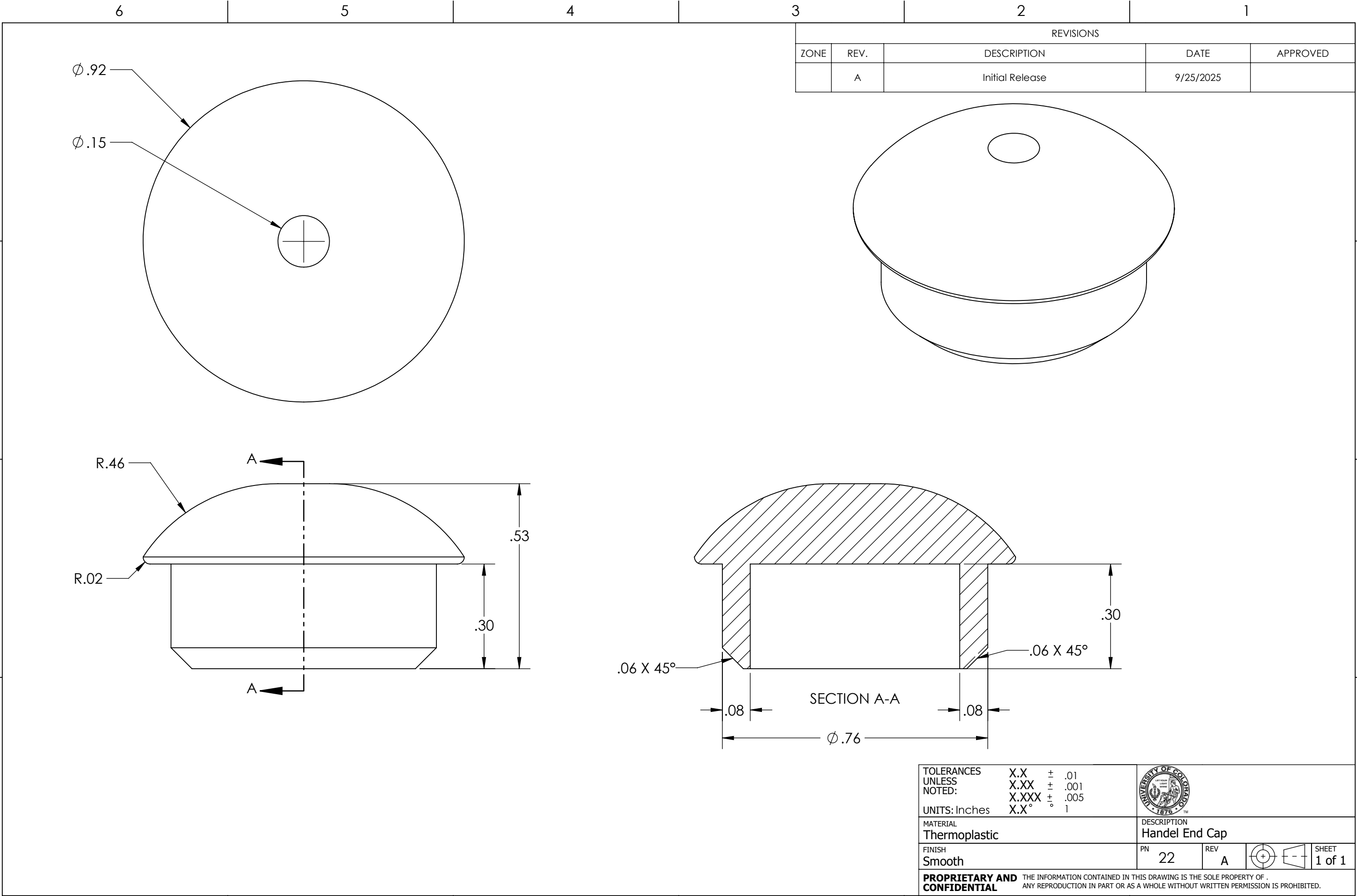
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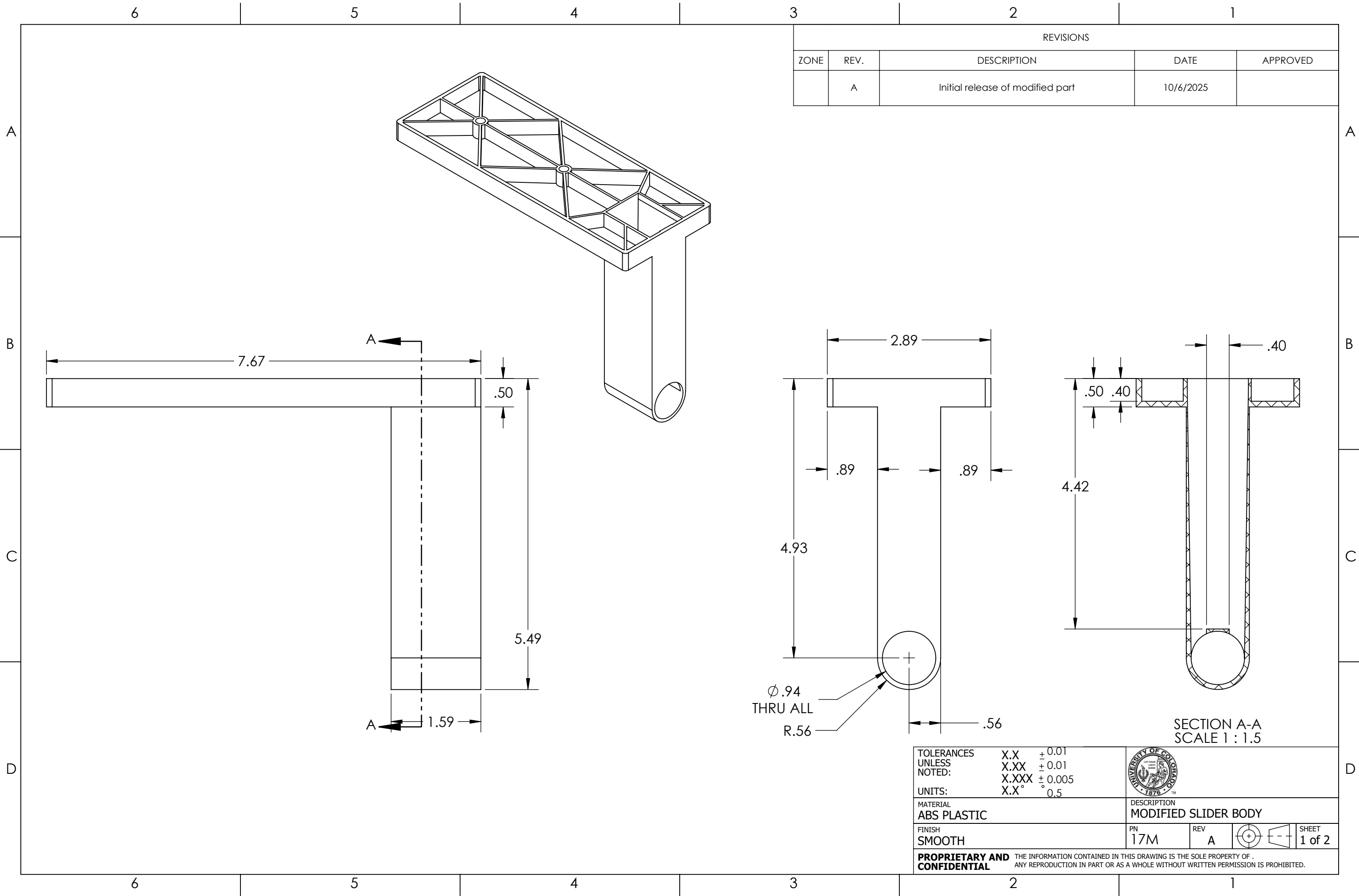
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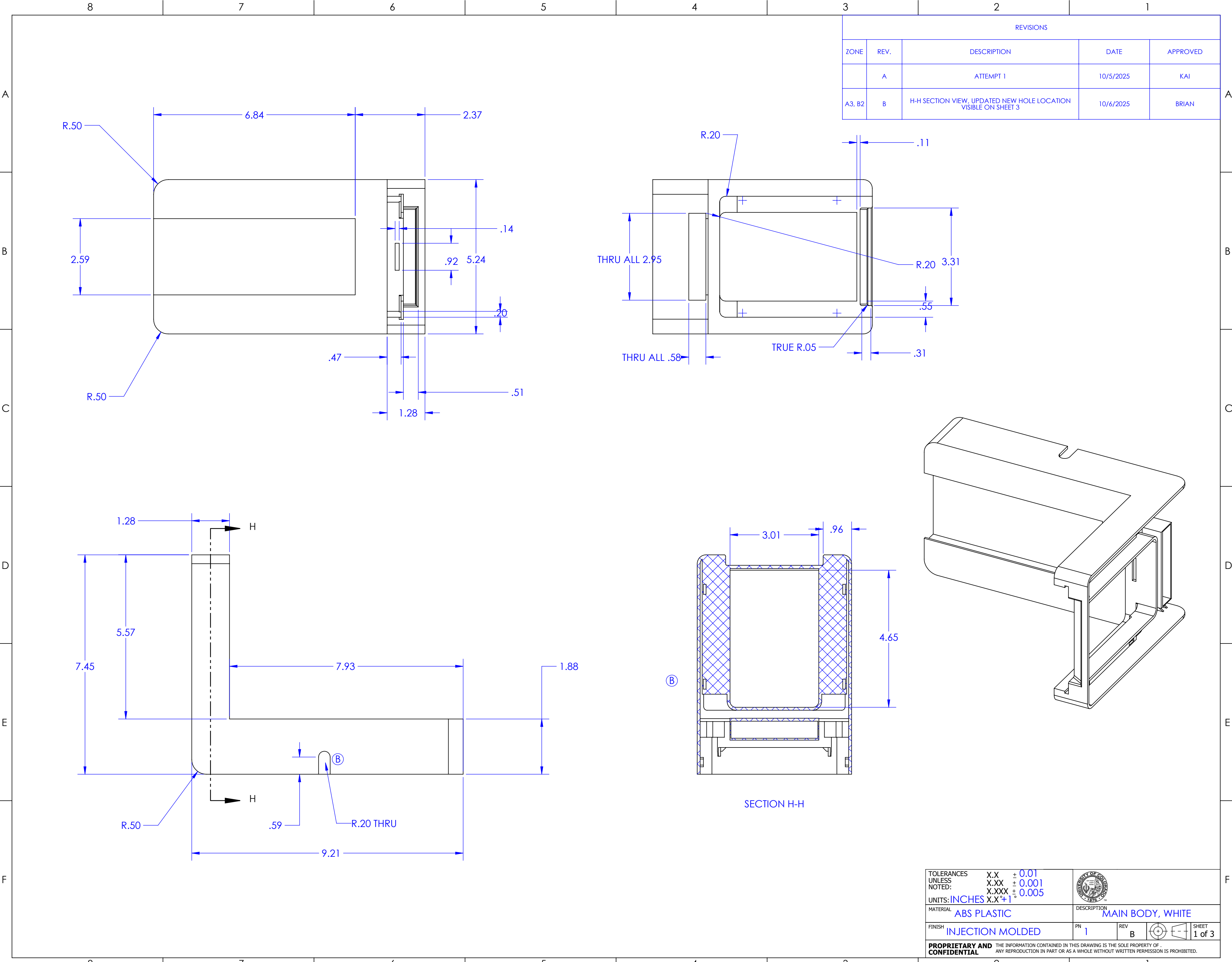






Appendix D

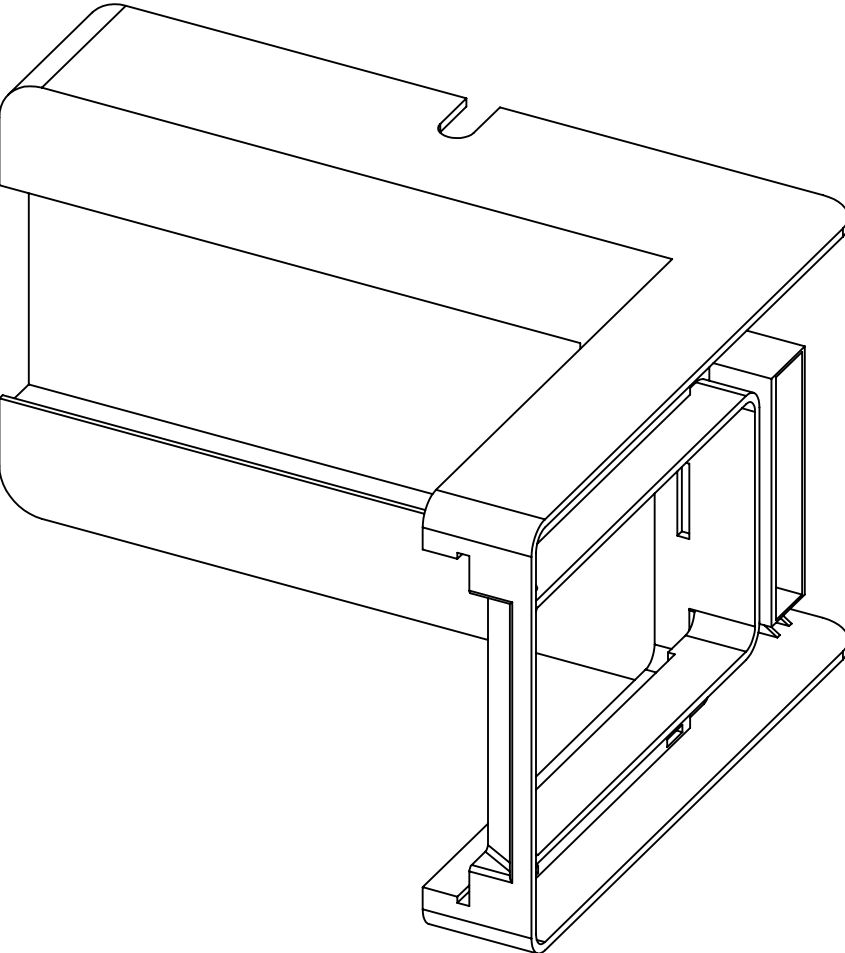




REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	ATTEMPT 1	10/5/2025	KAI
A3, B2	B	H-H SECTION VIEW, UPDATED NEW HOLE LOCATION VISIBLE ON SHEET 3	10/6/2025	BRIAN

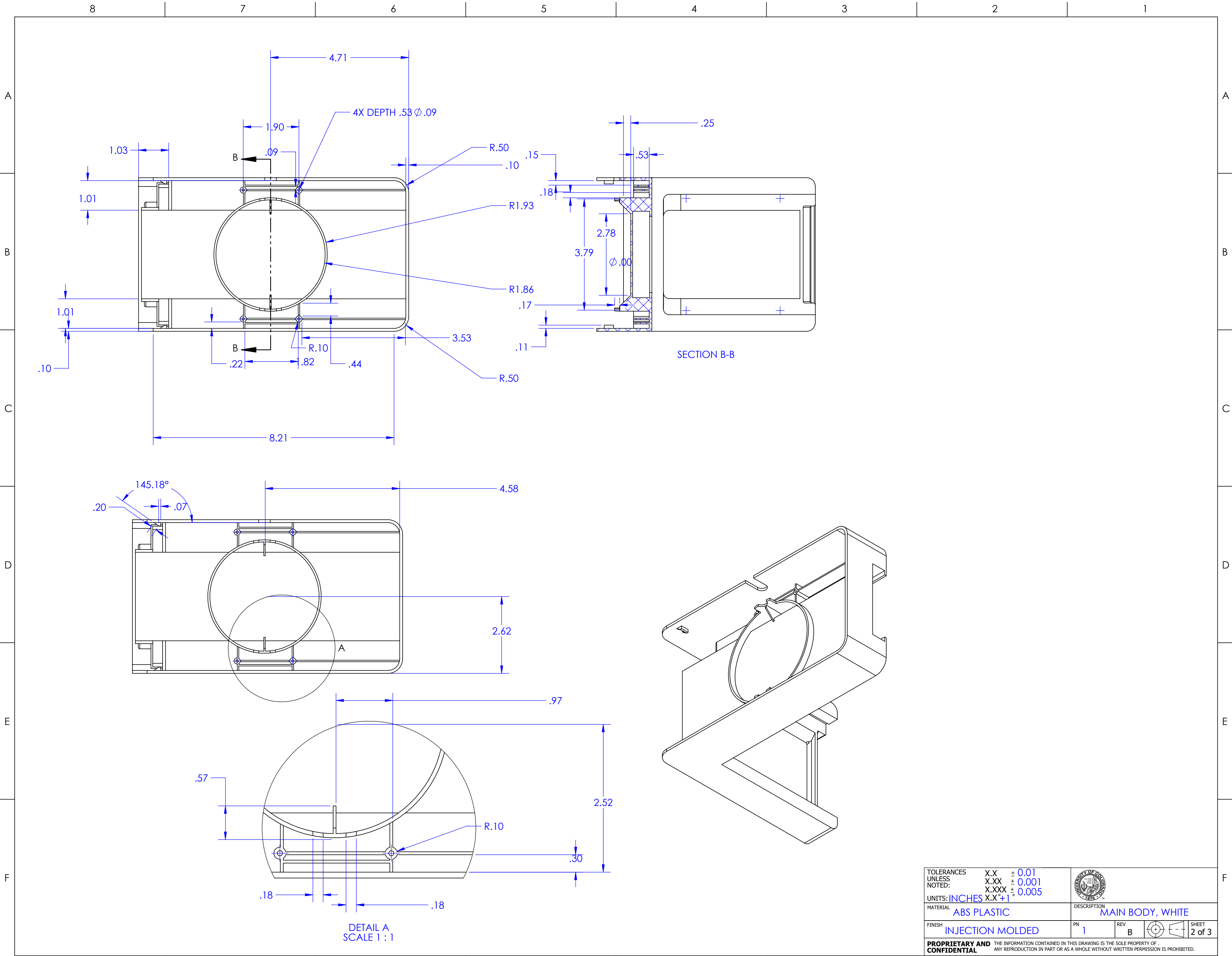
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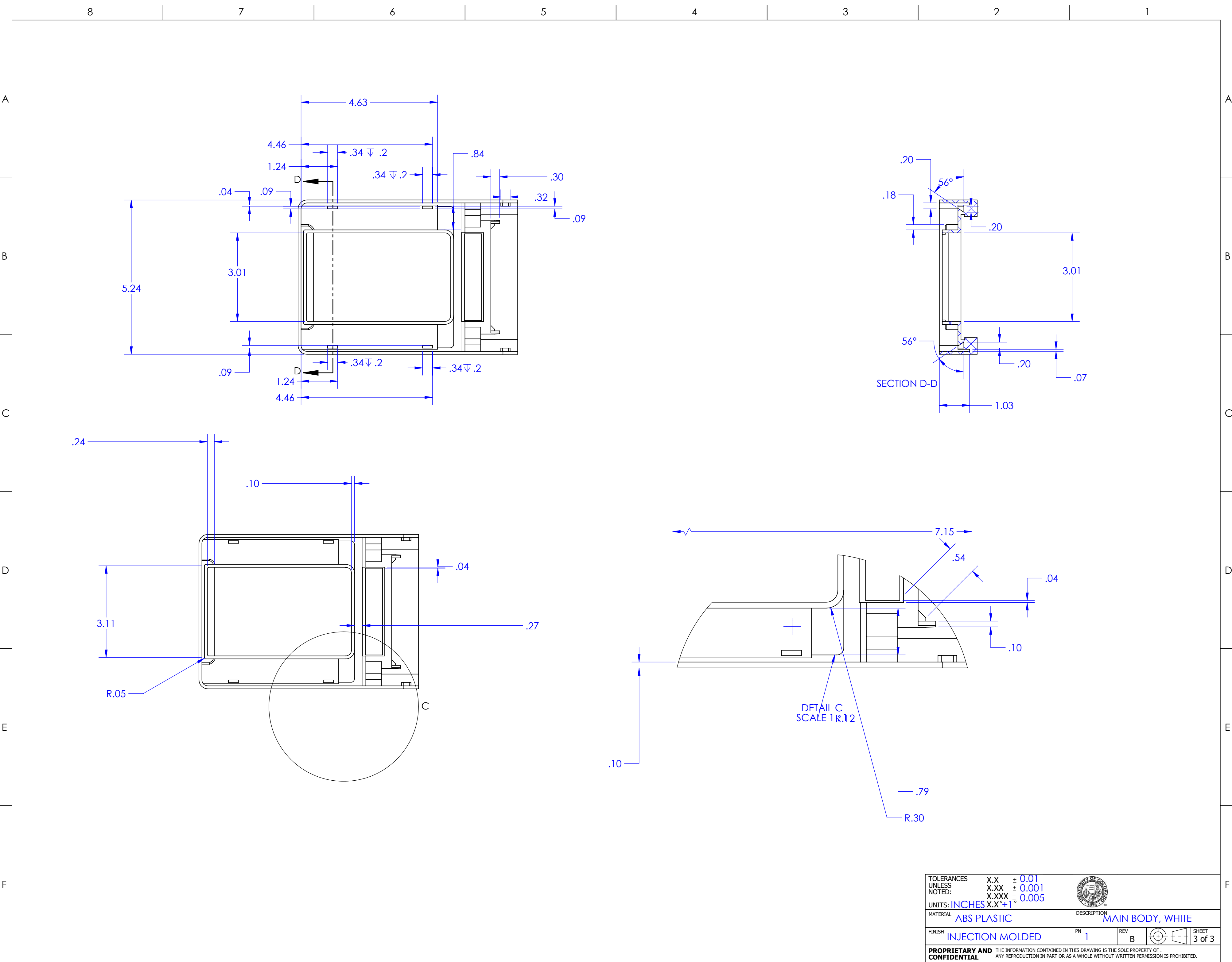
SECTION H-H



TOLERANCES UNLESS NOTED: X.X ± 0.01 X.XX ± 0.001 X.XXX ± 0.005	
UNITS: INCHES X.X ± .01	
MATERIAL ABS PLASTIC	DESCRIPTION MAIN BODY, WHITE
FINISH INJECTION MOLDED	PN 1 REV B
SHEET 1 of 3	

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TOLERANCES UNLESS NOTED: X.X $\pm$ 0.01 X.XX $\pm$ 0.001 X.XXX $\pm$ 0.005		UNITS: INCHES X.X $\pm$ 1		DESCRIPTION MAIN BODY, WHITE	
MATERIAL ABS PLASTIC		FINISH INJECTION MOLDED		PN 1	REV B
PROPRIETARY AND CONFIDENTIAL		THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.		SHEET 3 of 3	